



IEC 61439-5

Edition 3.0 2023-05
REDLINE VERSION

INTERNATIONAL STANDARD



**Low-voltage switchgear and controlgear assemblies –
Part 5: Assemblies for power distribution in public networks**



THIS PUBLICATION IS COPYRIGHT PROTECTED

Copyright © 2023 IEC, Geneva, Switzerland

All rights reserved. Unless otherwise specified, no part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from either IEC or IEC's member National Committee in the country of the requester. If you have any questions about IEC copyright or have an enquiry about obtaining additional rights to this publication, please contact the address below or your local IEC member National Committee for further information.

IEC Secretariat
3, rue de Varembe
CH-1211 Geneva 20
Switzerland

Tel.: +41 22 919 02 11
info@iec.ch
www.iec.ch

About the IEC

The International Electrotechnical Commission (IEC) is the leading global organization that prepares and publishes International Standards for all electrical, electronic and related technologies.

About IEC publications

The technical content of IEC publications is kept under constant review by the IEC. Please make sure that you have the latest edition, a corrigendum or an amendment might have been published.

IEC publications search - webstore.iec.ch/advsearchform

The advanced search enables to find IEC publications by a variety of criteria (reference number, text, technical committee, ...). It also gives information on projects, replaced and withdrawn publications.

IEC Just Published - webstore.iec.ch/justpublished

Stay up to date on all new IEC publications. Just Published details all new publications released. Available online and once a month by email.

IEC Customer Service Centre - webstore.iec.ch/csc

If you wish to give us your feedback on this publication or need further assistance, please contact the Customer Service Centre: sales@iec.ch.

IEC Products & Services Portal - products.iec.ch

Discover our powerful search engine and read freely all the publications previews. With a subscription you will always have access to up to date content tailored to your needs.

Electropedia - www.electropedia.org

The world's leading online dictionary on electrotechnology, containing more than 22 300 terminological entries in English and French, with equivalent terms in 19 additional languages. Also known as the International Electrotechnical Vocabulary (IEV) online.



IEC 61439-5

Edition 3.0 2023-05
REDLINE VERSION

INTERNATIONAL STANDARD



Low-voltage switchgear and controlgear assemblies – Part 5: Assemblies for power distribution in public networks

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

ICS 29.130.20

ISBN 978-2-8322-6987-9

Warning! Make sure that you obtained this publication from an authorized distributor.

CONTENTS

FOREWORD	3
1 Scope	6
2 Normative references	7
3 Terms and definitions	8
4 Symbols and abbreviations	9
5 Interface characteristics	9
6 Information	10
7 Service conditions	10
8 Constructional requirements	10
9 Performance requirements	14
10 Design verification	14
11 Routine verification	30
Annexes	31
Annex Q L (informative) Guidance on verification of temperature rise verification	32
Annex AA (normative) Cross-section of conductors	33
Annex BB (informative) Items subject to agreement between the assembly manufacturer and the user	35
Annex CC (informative) Design verification	39
Annex DD (informative) List of notes concerning certain countries	41
Bibliography	42
Figure 101 – Typical distribution network	7
Figure 102 – Diagram of test to verify resistance to shock load of a PENDA-O	18
Figure 103 – Diagram of test to verify impact force withstand of a PENDA-O	19
Figure 104 – Diagram of test to verify the resistance to static load	20
Figure 105 – Sandbag for test to verify the resistance to shock load	21
Figure 106 – Diagram of test to verify resistance to torsional stress of a PENDA-O	23
Figure 107 – Diagram of test to verify the mechanical strength of doors	26
Figure 108 – Striker element for test of resistance to mechanical shock impacts induced by sharp-edged objects	27
Figure 109 – Typical test arrangement for mechanical strength of base	28
Table 101 – Values of assumed loading	9
Table 102 – Axial load to be applied to the inserts	27
Table AA.1 – Minimum and maximum cross-section of copper and aluminium conductors, suitable for connection (see 8.8)	33
Table AA.2 – Standard cross-sections of round copper conductors and approximate relationship between mm ² and AWG/kcmil sizes (see 8.8 of IEC 61439-1:2020)	34
Table BB.1 – Items subject to agreement between the ASSEMBLY manufacturer and the user	35
Table CC.1 – List of design verifications to be performed	39

INTERNATIONAL ELECTROTECHNICAL COMMISSION

LOW-VOLTAGE SWITCHGEAR AND CONTROLGEAR ASSEMBLIES –

Part 5: Assemblies for power distribution in public networks

FOREWORD

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
- 2) The formal decisions or agreements of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees.
- 3) IEC Publications have the form of recommendations for international use and are accepted by IEC National Committees in that sense. While all reasonable efforts are made to ensure that the technical content of IEC Publications is accurate, IEC cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.
- 4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC Publication and the corresponding national or regional publication shall be clearly indicated in the latter.
- 5) IEC itself does not provide any attestation of conformity. Independent certification bodies provide conformity assessment services and, in some areas, access to IEC marks of conformity. IEC is not responsible for any services carried out by independent certification bodies.
- 6) All users should ensure that they have the latest edition of this publication.
- 7) No liability shall attach to IEC or its directors, employees, servants or agents including individual experts and members of its technical committees and IEC National Committees for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC Publication or any other IEC Publications.
- 8) Attention is drawn to the Normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.
- 9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

This redline version of the official IEC Standard allows the user to identify the changes made to the previous edition IEC 61439-5:2014. A vertical bar appears in the margin wherever a change has been made. Additions are in green text, deletions are in strikethrough red text.

IEC 61439-5 has been prepared by subcommittee 121B: Low-voltage switchgear and controlgear assemblies, of IEC technical committee 121: Switchgear and controlgear and their assemblies for low voltage. It is an International Standard.

This third edition cancels and replaces the second edition published in 2014. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- a) omission of the requirement to conduct mechanical tests at -25 °C when enclosures are made of a metallic material;
- b) addition of assumed loading factors generation supplies and electric vehicle charging applications;
- c) additional dielectric tests when a PENDA is used in a distribution substation with separate HV and LV earths;
- d) further clarification of representative samples for design verification.

The text of this International Standard is based on the following documents:

Draft	Report on voting
121B/173/FDIS	121B/178/RVD

Full information on the voting for its approval can be found in the report on voting indicated in the above table.

The language used for the development of this International Standard is English.

This document was drafted in accordance with ISO/IEC Directives, Part 2, and developed in accordance with ISO/IEC Directives, Part 1 and ISO/IEC Directives, IEC Supplement, available at www.iec.ch/members_experts/refdocs. The main document types developed by IEC are described in greater detail at www.iec.ch/publications.

This document is to be read in conjunction with IEC 61439-1. The provisions of the general rules dealt with in IEC 61439-1 are only applicable to this document insofar as they are specifically cited. When this document states "addition", "modification" or "replacement", the relevant text in IEC 61439-1:2020 is to be adapted accordingly. Subclauses that are numbered with a 101 (102, 103 etc.) suffix are additional to the same subclause in IEC 61439-1:2020.

Tables and figures in IEC 61439-5:2023 that are new are numbered starting with 101.

New annexes in IEC 61439-5:2023 are lettered AA, BB, etc.

The reader's attention is drawn to the fact that Annex DD lists all of the "in-some-country" clauses on differing practices of a less permanent nature relating to the subject of this document.

A list of all parts of the IEC 61439 series, under the general title *Low-voltage switchgear and controlgear assemblies* can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under webstore.iec.ch in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

IMPORTANT – The "colour inside" logo on the cover page of this document indicates that it contains colours which are considered to be useful for the correct understanding of its contents. Users should therefore print this document using a colour printer.

LOW-VOLTAGE SWITCHGEAR AND CONTROLGEAR ASSEMBLIES –

Part 5: Assemblies for power distribution in public networks

1 Scope

This document defines the specific requirements for public electricity network distribution assemblies (PENDAs).

PENDAs have the following criteria:

- used for the distribution of electrical energy in three phase systems for which the rated voltage does not exceed 1 000 V AC (see Figure 101 for a typical distribution network) and DC systems not exceeding 1 500 V DC;
- stationary;
- open type assemblies are not covered by this document;
- suitable for installation in places where only skilled persons have access for their use, however, outdoor types ~~may~~ can be installed in situations that are accessible to ordinary persons
 - intended for use in energy distribution in public power grids;
 - indoor use: assemblies for installation inside of electric power substations;
 - outdoor use: assemblies containing an enclosure suitable for open air installation.

The object of this document is to state the definitions and to specify the service conditions, construction requirements, technical characteristics and tests for PENDAs. ~~Network parameters may require~~ Tests at higher performance level can be applicable with some network parameters.

PENDAs ~~may~~ can also include control and or signalling devices associated with the distribution of electrical energy.

NOTE 1 Control and monitoring devices can be used in smart grid applications or the transmission of smart grid data.

This document applies to all PENDAs whether they are designed, manufactured on a one-off basis or fully standardised and manufactured in quantity.

The manufacture and/or assembly ~~may~~ can be carried out other than by the original manufacturer (see 3.10.1 of IEC 61439-1:2014/2020).

This document does not apply to individual devices and self-contained components, such as motor starters, fuse switches, electronic equipment, etc. which comply with the relevant product standards.

If the substation is owned or operated by a public distribution system operator (DSO), PENDA's which are used as LV distribution panels in transformer substations are within the scope of this document,

This document does not apply to specific types of assemblies covered by other parts of IEC 61439 series.

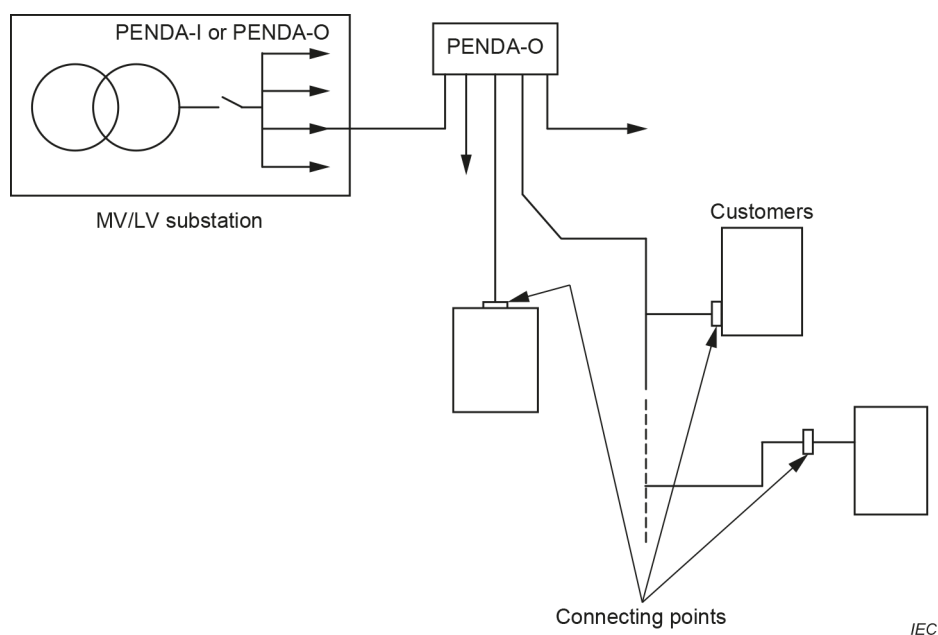


Figure 101 – Typical distribution network

NOTE 2 If a PENDA is equipped with additional equipment (for example meters), in such a way that the main function is changed considerably, then other standards can also apply as agreed between user and manufacturer (see 8.5 of IEC 61439-1:2014/2020).

NOTE 3 Where local regulations and practices permit, a PENDA according to this document can be used in other than public networks.

NOTE 4 DSO's can define additional requirements for their PENDA's.

2 Normative references

This clause of IEC 61439-1:2020 applies except as follows.

Addition:

IEC 60695-11-10:2013, *Fire hazard testing – Part 11-10: Test flames – 50 W horizontal and vertical flame test methods*

IEC 61439-1:2014/2020, *Low-voltage switchgear and controlgear assemblies – Part 1: General rules*

IEC 62262, *Degrees of protection provided by enclosures for electrical equipment against external mechanical impacts (IK code)*

ISO 9223:2012, *Corrosion of metals and alloys – Corrosivity of atmospheres – Classification, determination and estimation*

ISO 6506-1:2014, *Metallic materials – Brinell hardness test – Part 1: Test method*

3 Terms and definitions

This clause of IEC 61439-1:2020 applies except as follows.

3.1 General terms

Additional terms:

3.1.101

public electricity network distribution assembly PENDA

assembly, generally for installation in a public electricity network which in use, receives electrical energy from one or more supplies and distributes that energy through one or more cables to other equipment

Note 1 to entry: A PENDA is installed, operated and maintained solely by skilled persons.

Note 2 to entry: Some types of a PENDA were previously known as a cable distribution cabinet (CDC).

3.1.101.1

outdoor public electricity network distribution assembly PENDA-O

cubicle type public electricity network distribution assembly that is suitable for outdoor installation in places that ~~may~~ can, or ~~may not~~ cannot, be accessible to the public

3.1.101.2

indoor public electricity network distribution assembly PENDA-I

public electricity network distribution assembly suitable for installation indoors, generally without an enclosure, but including all structural parts necessary to support busbars, functional units and other ancillary devices, necessary to complete the assembly

3.1.102

design life

minimum duration for which specified performance characteristics of equipment are expected when the equipment is operated as intended and regularly maintained by instructed persons in accordance with the manufacturer's instructions

[SOURCE: IEC 60050-395:2014, 395-07-93, modified – Note to entry omitted and 'granted' replaced by 'expected when instructions'.]

3.3 External design of assemblies

3.3.1

open-type assembly

This term of IEC 61439-1:2020 does not apply.

3.9 Verification

Modifications:

3.9.1

design verification

Delete the note.

3.9.1.2

verification comparison

This term of IEC 61439-1:2020 does not apply.

3.9.1.3**verification assessment**

This term of IEC 61439-1:2020 does not apply.

4 Symbols and abbreviations

This clause of IEC 61439-1:2020 applies.

5 Interface characteristics

This clause of IEC 61439-1:2020 applies, except as follows.

5.4 Rated diversity factor (RDF)

Addition:

In the absence of an agreement between the assembly manufacturer and user concerning the actual load currents, the assumed loading of the outgoing circuits of the assembly or group of outgoing circuits ~~may~~ can be based on the values given in Table 101.

For distribution and final circuits, it is assumed that the load current is the rated current of the protective device, I_n , as required by the user, multiplied with the loading factor of Table 101.

Table 101 – Values of assumed loading

Number of main circuits	Assumed loading factor
2 and 3	0,9
4 and 5	0,8
6 to 9 inclusive	0,7
10 (and above)	0,6

Application	Assumed loading factor
Regular distribution grid connections 2 to 3 circuits	0,9
Regular distribution grid connections 4 to 5 circuits	0,8
Regular distribution grid connections 6 to 9 circuits	0,7
Regular distribution grid connections ≥ 10 circuits	0,6
Generation supply (e.g. PV, wind farm, biomass)	1,0
Charging infrastructure for EV	1,0

6 Information

This clause of IEC 61439-1:2020 applies except as follows.

6.1 Assembly designation marking

Addition to first paragraph:

Designation plates ~~may~~ can be placed inside an enclosure of an assembly provided their intended place ensures good legibility and visibility when the door(s) is(are) open or the cover is removed.

Replacement of item dg):

dg) IEC 61439-5.

6.3 Device and/or component identification

Additional paragraph:

In the case of removable fuse-carriers which are specific to a fuseway, a label shall be placed on the fuse carrier as well as on the fuse base, to avoid incorrect interchangeability of the fuse-carrier.

Additional subclause:

6.101 Circuit identification

It shall be possible to identify each functional unit in a clearly visible manner.

7 Service conditions

This clause of IEC 61439-1:2020 applies except as follows.

7.1 Normal service conditions

7.1.1 Climatic conditions

Addition to the first paragraph:

Unless the user specifies that a PENDA shall be suitable for use in an arctic climate, the lower limit of ambient air temperature is -25 °C as specified in Table 15 of IEC 61439-1:2020. For an arctic climate the lower limit of ambient temperature is -50 °C .

~~7.1.1.2 Ambient air temperature for outdoor installations~~

~~Replacement of last paragraph with:~~

~~Unless the user specifies a PENDA shall be suitable for use in an arctic climate, the lower limit of ambient air temperature is -25 °C . For an arctic climate the lower limit of ambient temperature is -50 °C .~~

7.1.2 Pollution degree

Replace the first paragraph with the following:

The pollution degree referred to in Annex BB is the macro-environmental condition for which the assembly is intended.

7.2 Special service conditions

Addition of the following note to item h):

~~NOTE—Exposure to vibration arising from traffic and/or occasional ground excavation is a normal service condition for PENDAs.~~

Additional paragraph:

Additional requirements for a PENDA-O, to be installed where heavy snowfalls occur and where they are adjacent to areas where there is snow clearance by ploughing, are subject to agreement between manufacturer and user.

8 Constructional requirements

This clause of IEC 61439-1:2020 applies except as follows.

8.1 Strength of materials and parts

8.1.1 General

Change the reference to Annex C to Annex BB in paragraph 5 of 8.1.1 of IEC 61439-1:2020.

Addition:

A PENDA-O shall be arranged for ground mounting, transformer mounting, pole mounting, surface wall mounting or mounting within a recess within a wall, as agreed between user and manufacturer.

A PENDA ~~may~~ can be directly coupled to a transformer by means of a flange coupling or it ~~may~~ can connect to its supply by means of cable or via busbars as agreed between user and manufacturer. Outgoing circuits shall be suitable for connection by means of cables.

A ~~reliable~~ locking device shall be provided on outdoor enclosures which prevents access by unauthorized persons. ~~Doors, lids and covers shall be so designed that, after they are locked, they do not open due to subsequent moderate ground settlement, nor due to exposure to vibration arising from traffic and/or ground excavation and reinstatement works.~~ The fixings of any covers etc. which are removable for installation or maintenance operations shall only be accessible while the door(s) is (are) open.

Any auxiliary equipment, e.g. meters, relays, instruments, circuit breaker trip units, communications equipment, that can be readily replaced, are excluded from the minimum design life of a PENDA.

NOTE When applicable, a design life can be agreed between user and manufacturer, assuming it is operated as intended and regularly maintained by instructed personnel in accordance with the manufacturer's instructions.

8.1.3.2 Resistance of insulating materials to heat and fire

Additional subclause:

8.1.3.2.101 Verification of category of flammability

The insulating materials used for enclosures, barriers and other insulating parts shall have ~~flame-retardant~~ flammability properties in accordance with ~~10.2.3.102 of this standard~~ 10.2.3.101.

8.1.5 Mechanical strength

Additional subclause:

8.1.5.101 Verification of mechanical strength

The mechanical properties of a PENDA-O shall comply with 10.2.101.

Parts of the PENDA-O intended to be embedded in the ground shall withstand the stresses imposed on them during installation and normal service and comply with 10.2.101.9.

~~Additional subclause:~~

~~8.1.101 Thermal stability~~

~~The thermal stability of a PENDA shall be verified according to 10.2.3.101.~~

8.2 Degree of protection provided by an assembly enclosure

8.2.1 Protection against mechanical impact (IK code)

Subclause 8.2.1 of IEC 61439-1:2020 does not apply.

The mechanical impact tests required by this document are at least equal to IK10 in accordance with IEC 62262 (see 8.1.5.101).

8.2.2 Protection against contact with live parts, ingress of solid foreign bodies and water (IP code)

Addition:

Open type assemblies (IP00) are not covered by this document.

When a PENDA-O is intended to be installed in places accessible to the public, its enclosure shall, when fully-installed in accordance with the manufacturer's instructions, provide a degree of protection of at least IP34D according to IEC 60529. In other locations, the minimum level of protection shall be at least IP33.

PENDA-O's that are intended to be installed in places accessible to the public shall, unless otherwise stated by the user, be designed such that when any temporary cables are connected, the enclosure shall provide a degree of protection of at least IP23C according to IEC 60529. See 8.8.

~~8.4 Protection against electric shock~~

8.4.2.1 General

The third paragraph does not apply.

Additional subclause:

8.4.2.101 Earthing and short-circuiting means

When specified by the user, the outgoing units in an assembly shall be so constructed that they can be earthed and short-circuited in a secure manner by means of a device(s) recommended by the manufacturer, which ensures the ~~manufacturer's indicated~~ required degree of protection (IP code) is maintained for all parts of the assembly. ~~This requirement is not applicable if it could cause a safety hazard arising from the system conditions and/or operational practice.~~

8.4.3.1 Installation conditions

Additional paragraph:

~~For an assembly that is expected~~ When a user specifies the assembly is to feed overhead cable lines, outgoing units shall be designed in such a way that an attached cable(s) can be earthed at the termination(s).

8.8 Terminals for external ~~conductors~~ cables

Replacement of the first three paragraphs with the following:

In the absence of a special agreement between user and manufacturer, terminals shall be capable of accommodating cables having copper or aluminium conductors from the smallest to the largest cross-sectional area corresponding to the appropriate rated current (see Table AA.1).

The terminations for outgoing circuits shall be located so that adequate spacing is provided and to facilitate terminating the phase conductors of a cable irrespective of their lay.

When specified by the user the phase cable terminals of each outgoing circuit shall be separated from all other hazardous live parts. When the terminals of an outgoing circuit are exposed protection shall be provided against accidental contact with other hazardous live parts. Separation and protection shall be from the normal direction of access and in accordance with IPXXB of IEC 60529.

Where specified by the user, the incoming circuit shall be suitable for connection by means of either bare or insulated bars.

Additional subclauses:

8.101 Marking as an obstacle to snow clearance

When specified by the user, ~~Where~~ a PENDA-O that is intended for use in regions where heavy snowfalls occur in accordance with 7.2, ~~or alternatively, if required by the user, it shall be possible to mark it~~ marked as an obstacle to snow clearance. Holders shall be provided, attached to the PENDA-O, to accommodate marking rods and it shall be possible to install and make adjustments to the position of the marking rod from outside the PENDA-O. ~~The holders shall be constructed in a manner which ensures that the holder or marking rod will give way to a mechanical force before the transmitted force to the PENDA-O's enclosure reaches the value which would adversely affect the degree of protection (IP code).~~

8.102 Ease of operation and maintenance

All parts of the assembly shall, as far as practicable, be readily accessible and replaceable without excessive dismantling. ~~The conditions for interchangeability of parts of the ASSEMBLY may be subject to an agreement between the user and the manufacturer.~~

The design shall be such that the cables can be readily connected from the front.

When a PENDA does not have a means of measurement incorporated, it shall be possible, by the use of a portable instrument, to readily and safely measure voltages in all phases of incoming units and on both sides of all current breaking and/or switch devices of outgoing units, also the current in one phase of all outgoing units. During this operation all live parts of the PENDA shall be protected sufficiently to retain the required degree of protection in accordance with 8.2. Instructions concerning the procedure to be adopted shall be provided by the manufacturer.

If the assembly is intended to be connected to a live reserve power, for example a standby generator, the switchgear connecting device shall be designed so that connection can be made with the live parts having a degree of protection of ~~IP10~~ IPXXB according to IEC 60529.

~~Locking arrangements shall be provided on a PENDA to secure the door(s) and prevent unauthorised access. The fixings of any covers etc. which are removable for installation or maintenance operations shall only be accessible while the door(s) are open.~~

9 Performance requirements

This clause of IEC 61439-1:2020 applies.

10 Design verification

This clause of IEC 61439-1:2020 applies except as follows.

10.1 General

Replace ~~4th, 5th, and 6th~~ 6th, 7th, and 8th paragraph with:

Design verification shall only be achieved by the application of tests in accordance with Clause 10. The alternatives methods of verification by assessment or comparison with a reference design shall not be used (see Table CC.1).

It is not necessary to test every arrangement produced. Tests carried out on a representative sample of the most onerous PENDA are deemed to verify the performance of similar and less onerous assemblies of the same general construction and rating. ~~For example a temperature rise test carried out on an 800 A PENDA-O with 5 outgoing circuits is deemed to apply to a PENDA-O of the same construction (same general design of enclosure, same design of busbars and same incoming units) with 8 outgoing circuits of the same rating as those included in the PENDA-O that was temperature rise tested. The same approach applies to short circuit verification.~~ Examples of tests carried out on most onerous representative sample include:

- 1) A temperature rise test carried out on 800 A PENDA with 5 outgoing circuits is deemed to apply to an 800 A PENDA of the same construction (same general enclosure, same design of busbars and same incoming unit(s)) with 8 outgoing circuits of the same design.
- 2) Mechanical strength tests carried out on the smallest and largest enclosure in a series are deemed to apply to all other enclosures in the series, when they have the same number of doors and when they are of the same general construction.
- 3) Providing the busbar and connection support arrangements are the same, short-circuit tests carried out on a design of busbar and connections are deemed to apply to higher current ratings of busbars and connections.
- 4) A short circuit test carried out on 800 A PENDA with 8 outgoing circuits is deemed to apply to an 800 A PENDA of the same construction (same general enclosure, same design of busbars and same incoming unit(s)) with 5 outgoing circuits of the same design.
- 5) For a given short circuit rating, a short circuit tests carried out on an incoming or outgoing circuit is deemed to apply to higher current rating circuits of the same design and construction

Additional last paragraph:

Where necessary to suit their particular network parameters, users ~~may~~ can specify more onerous or additional test requirements.

10.2 Strength of materials and parts

10.2.2 Resistance to corrosion

10.2.2.1 ~~Test procedure~~ Verification by test

Replacement of last paragraph with the following:

When the corrosion resistance properties and projected life, as agreed between the manufacturer and the user, ~~can~~ shall be confirmed by reference to ISO 9223:2012, the tests detailed herein need not be performed.

In all other cases the corrosion resistance of each design of assembly shall be verified by severity test A or B, as applicable and as detailed 10.2.2.3 of IEC 61439-1:2020 and 10.2.2.2.

10.2.2.2 Severity test A

Replacement of the test specification (paragraph 2) with the following:

Damp heat cycling test of IEC 60068-2-30:2005: Severity – temperature 55 °C, 6 cycles and variant 1.

At the end of the test, the specimens are removed from the test chamber.

Compliance is checked by visual inspection. The parts tested shall not show rust, cracking or other deterioration. However, surface corrosion of the protective coating is allowed.

10.2.2.4 Results to be obtained

This subclause of IEC 61439-1:2020 is not applicable in respect of tests carried out in accordance with 10.2.2.2.

10.2.3 Properties of insulating materials

Additional subclause:

~~10.2.3.101~~ Dry heat test

~~The complete ASSEMBLY shall be placed in an oven, the internal temperature of which is raised to (100 ± 2) °C over a period of 2 h to 3 h and maintained at this temperature for 5 h.~~

~~Compliance is checked by inspection that there are no visible signs of deterioration. Deformation of protective covers manufactured from insulating materials is acceptable if they are more than 6 mm distant from parts which may have a temperature rise in excess of 40 K and do not support live components.~~

10.2.3.102101 Verification of category of flammability

Representative specimens of each of the materials of enclosures, barriers and other insulating parts shall be subjected to a flammability test in accordance with test method A – horizontal burning test of IEC 60695-11-10:2013.

Compliance is checked by inspection that each set of specimens can be classified to category HB40 criteria a) or b) in accordance with 8.4.3 of IEC 60695-11-10:2013.

10.2.6 Verification of protection against mechanical impact (IK code)

This subclause of IEC 61439-1:2020 is not applicable to assemblies complying with this document.

Additional subclauses:

10.2.101 Verification of mechanical strength

10.2.101.1 General

Where enclosures are manufactured in a series and are of the same design and construction, only the smallest and largest size of enclosure shall be tested.

The tests shall be carried out at an ambient temperature between 10 °C and 40 °C.

With the exception of the test of 10.2.101.7, a new sample assembly ~~may~~ can be used for each of the independent tests. If the same sample assembly is used for more than one test of 10.2.101, the compliance check for the second numeral of the degree of protection (IP code) need only be applied when the tests on that sample have been completed.

All tests shall be carried out with the assembly fixed at its normal service mounting and where appropriate, added support at normal ground level as indicated in Figure 102a, Figure 102b, Figure 103a, and Figure 103b.

Dimensions in millimetres

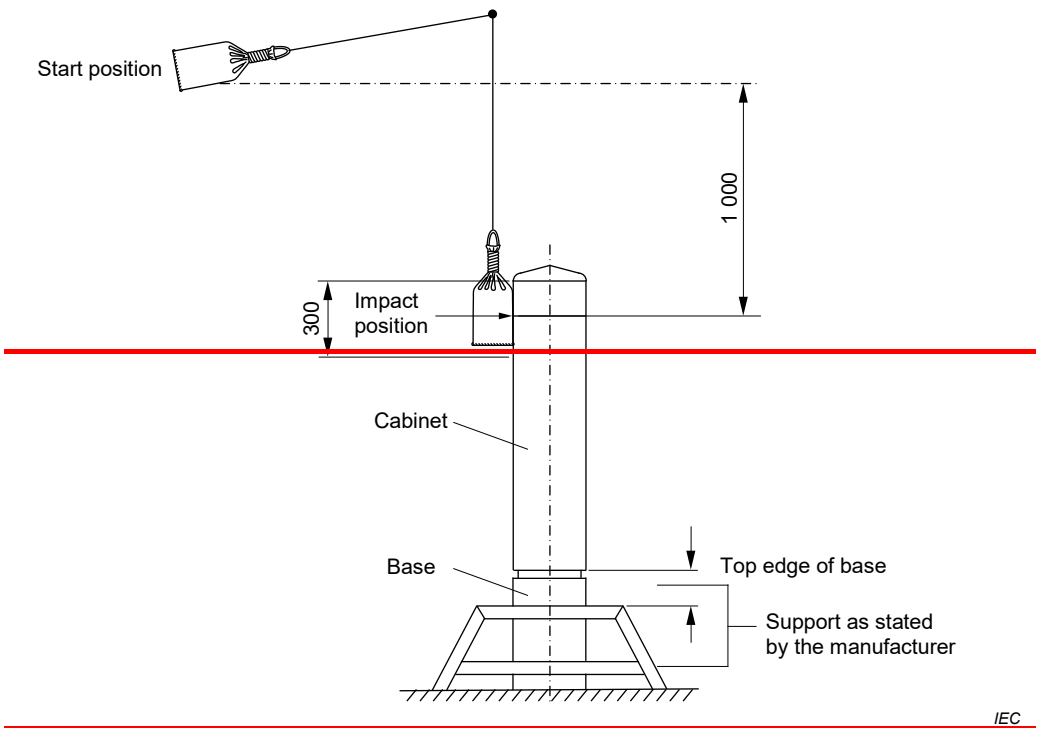


Figure 102a – Diagram of test to verify the resistance to shock load of a ground-mounted PENDA-O with embedded base

Dimensions in millimetres

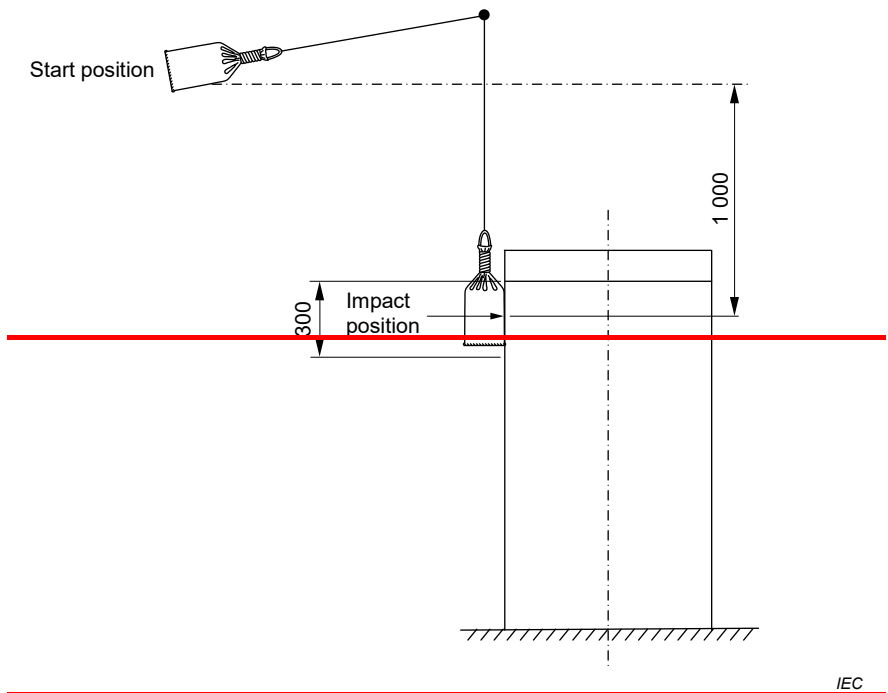


Figure 102b – Diagram of test to verify resistance to shock load of a ground-mounted PENDA-O without embedded base

Dimensions in millimetres

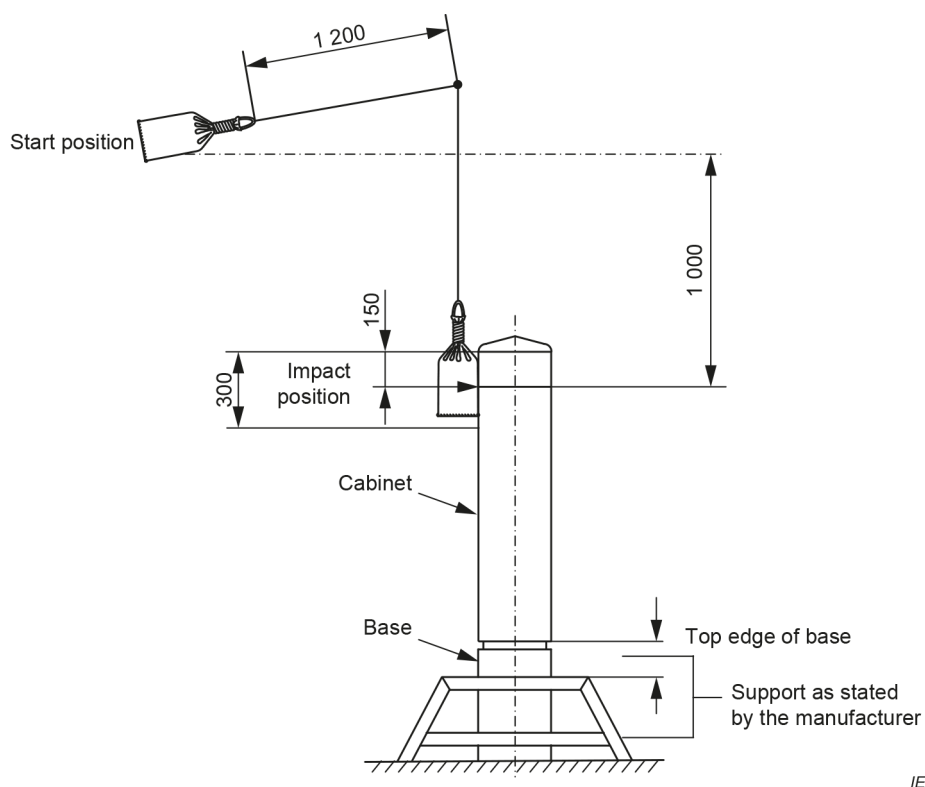


Figure 102a – Diagram of test to verify the resistance to shock load of a ground mounted PENDA-O with embedded base

Dimensions in millimetres

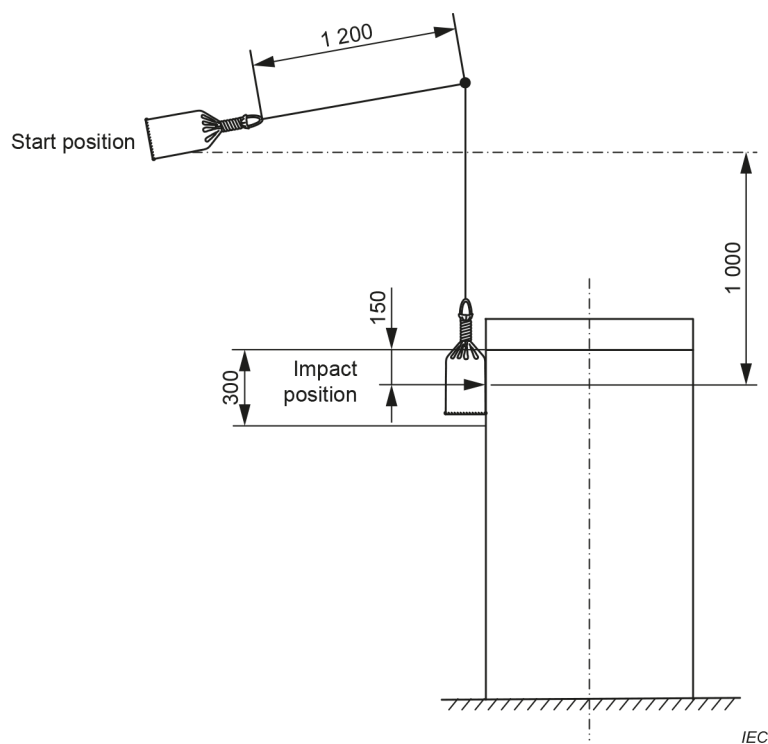
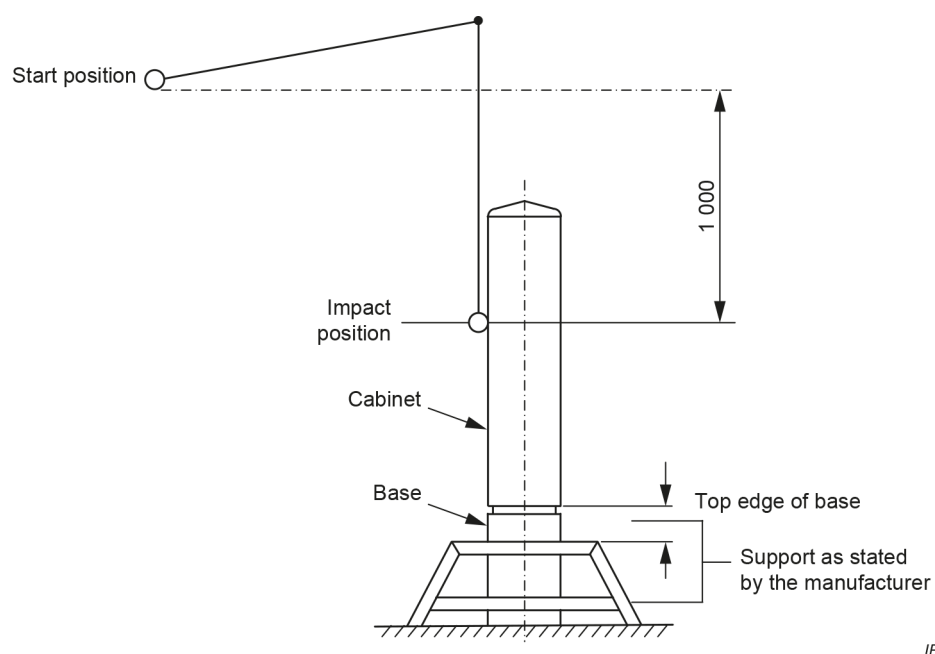


Figure 102b – Diagram of test to verify resistance to shock load of a ground mounted PENDA-O without embedded base

Figure 102 – Diagram of test to verify resistance to shock load of a PENDA-O

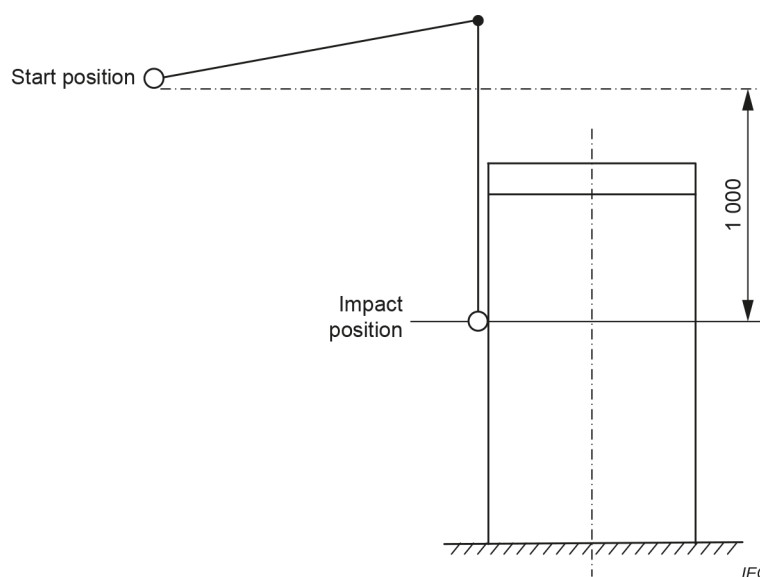
Dimensions in millimetres



IEC

Figure 103a – Diagram of test to verify impact force withstand for a ground mounted PENDA-O with embedded base

Dimensions in millimetres



IEC

Figure 103b – Diagram of test to verify impact force withstand for a ground mounted PENDA-O without embedded base

Figure 103 – Diagram of test to verify impact force withstand of a PENDA-O

With the exception of the test of 10.2.101.8, the door(s) of the assembly, if applicable, shall be locked at the commencement of the test and remain locked for the duration of the test.

10.2.101.2 Verification of resistance to static load

The following tests shall be carried out on all types of PENDA-0:

Test 1 – An evenly distributed load of 8 500 N/m² shall be applied for 5 min to the roof of the enclosure (see Figure 104).

Test 2 – A force of 1 200 N shall be applied for 5 min in turn to the front and back upper edges of the roof of the enclosure (see Figure 104).

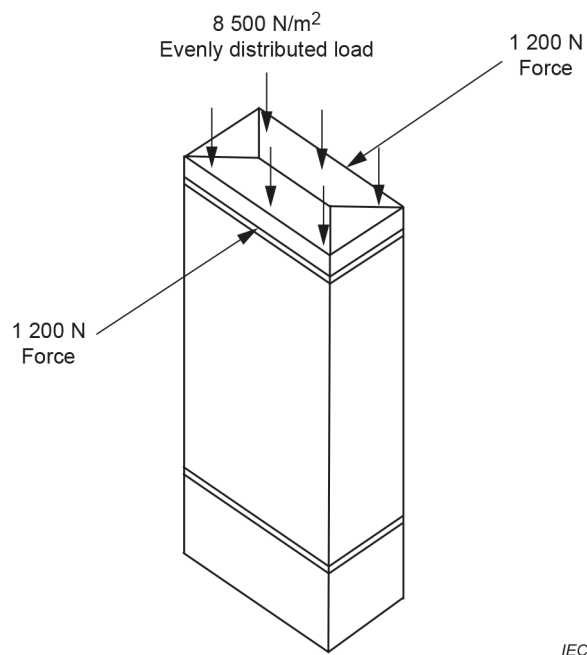


Figure 104 – Diagram of test to verify the resistance to static load

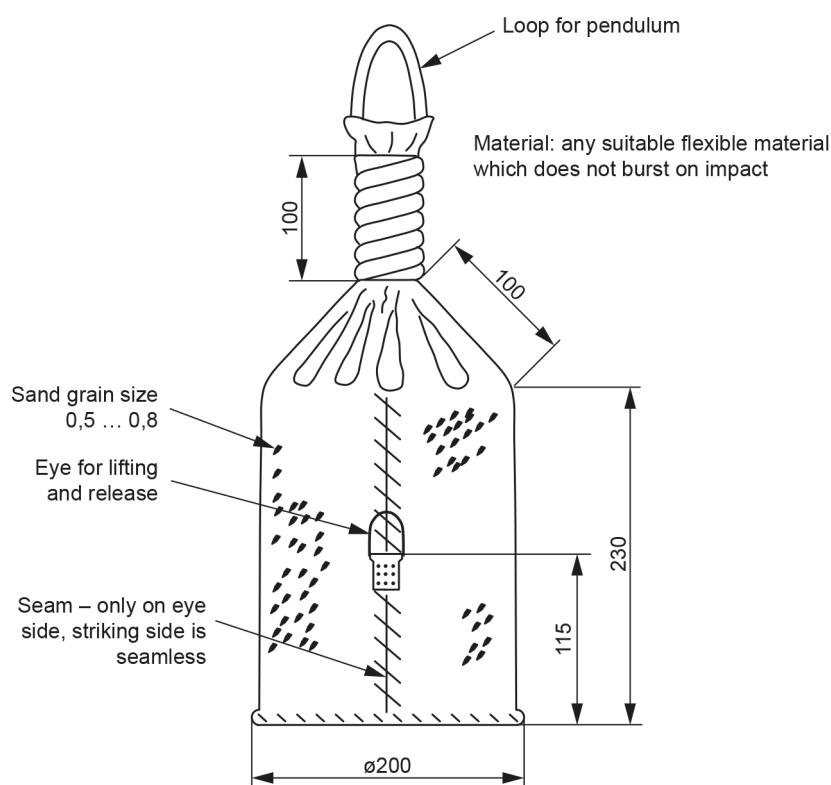
Compliance is checked by verification after the test that the minimum degree of protection is in accordance with 8.2.2, and the operation of the door(s) and locking points are not impaired; also by verification that the electrical clearances have remained satisfactory for the duration of the tests and in the case of an assembly having metallic enclosure, that no contact between live parts and the enclosure has occurred caused by permanent or temporary distortion.

10.2.101.3 Verification of resistance to shock load

The test shall apply to all types of PENDA-O.

A bag in accordance with Figure 105 containing dry sand and having a total mass of 15 kg shall be hung from an overhead support vertically above the surface under test and at ~~least~~ 1 m above the ~~highest~~ strike point of the assembly.

Dimensions in millimetres



IEC

Figure 105 – Sandbag for test to verify the resistance to shock load

Each test shall consist of one blow aimed at the upper part of each of the vertical surfaces of the assembly which are visible when the assembly is installed in its normal service position. Separate enclosures ~~may~~ can be used for each of the test blows.

In the case of an enclosure of cylindrical form, the test shall consist of three blows which are positioned with an angular displacement of 120°.

A test shall consist of raising the lifting eye through a height of 1 m and allowing the sandbag to fall through a vertical arc to impact the approximate centre of the upper part of the surface of the assembly under test (see Figure 102a and Figure 102b).

Compliance is checked by verification after the test that the degree of protection remains in accordance with 8.2.2, and the operation of the door(s) and locking points are not impaired; also by verification that the electrical clearances have remained satisfactory for the duration of the tests and, in the case of an assembly having a metallic enclosure, that no contact between live parts and the enclosure has occurred caused by permanent or temporary distortion. In the case of an assembly having an insulating enclosure, if the appropriate conditions are satisfied, then damage such as small dents or small degrees of surface cracking or flaking are disregarded, provided that there are no associated cracks detrimental to the serviceability of the assembly.

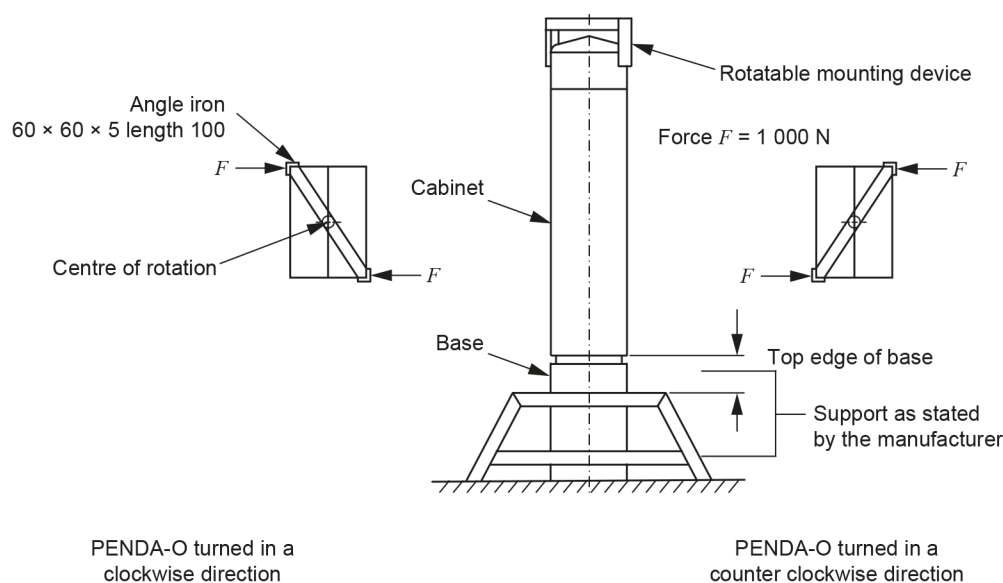
10.2.101.4 Verification of resistance to torsional stress

The test only applies to all types of PENDA-O.

The test is carried out using a horizontally rotatable frame constructed from 60 mm × 60 mm × 5 mm angle iron, having vertical locations 100 mm long at the frame arm's extremities. The assembly under test is rigidly fixed at its base and the frame closely fitted over it, so that the end locations of the frame arm are in contact with the roof and walls of the assembly.

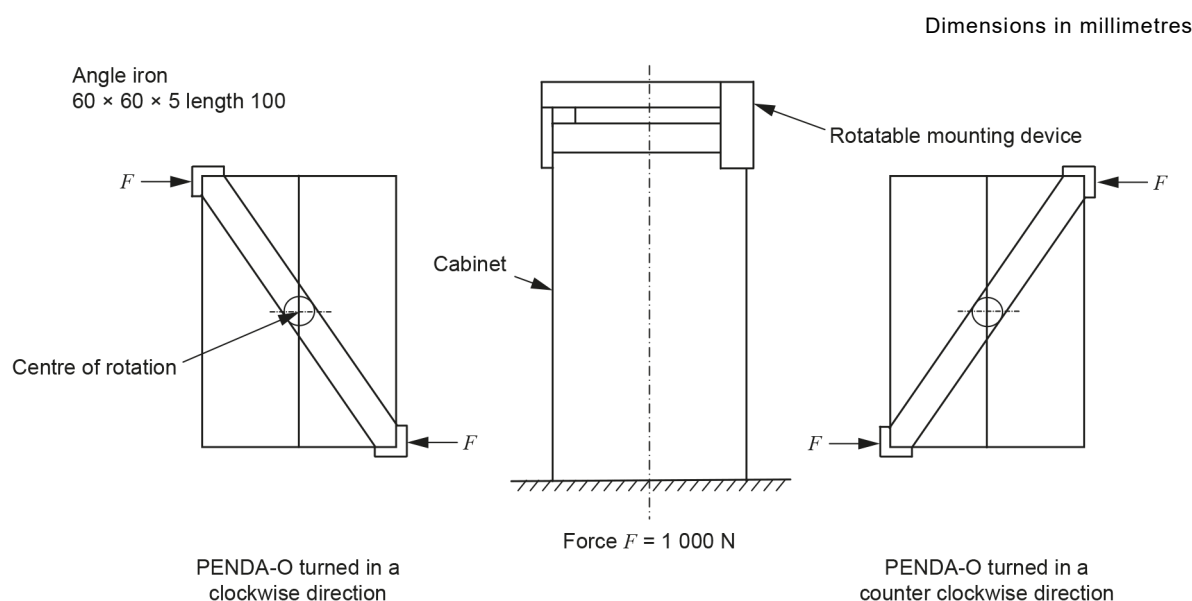
The assembly, with the door(s) closed shall have a torsional force ~~of 2 × 1 000 N~~ applied for 30 s as shown in Figure 106a and Figure 106b. This is comprised two separate tests, first with the two forces each of 1 000 N applied to twist the PENDA in a clockwise direction and the second with the two forces each of 1 000 N applied to rotate the PENDA in an anti-clockwise direction.

Dimensions in millimetres



IEC

Figure 106a – Diagram of test to verify the resistance to torsional stress of a ground mounted PENDA-O with embedded base



IEC

Figure 106b – Diagram of test to verify resistance to torsional stress of a ground mounted PENDA-O without embedded base

Figure 106 – Diagram of test to verify resistance to torsional stress of a PENDA-O

Compliance is checked by verifying that the door(s) remain closed for the duration of the test and by verification after the test that the degree of protection remains in accordance with 8.2.2.

10.2.101.5 Verification of impact force withstand

10.2.101.5.1 Test applicable to ~~PENDAs~~ PENDA-Os designed for operation at ambient temperatures of between 40 °C and –25 °C

The test shall be carried out using impact apparatus in the form of a pendulum incorporating a 9 mm external diameter tube at least 1 m long. The pendulum shall be arranged to swing through a vertical arc.

Attached to one end is a solid steel ball of 2 kg mass, which shall be raised through a height of 1 m and allowed to drop and impact the surface of the assembly under test, thus providing an impact energy of 20 J (see Figure 103a and Figure 103b).

For each of the two tests detailed below, the test shall consist of one blow aimed at the centre of each of the vertical surfaces of the assembly which are visible when it is installed in its normal service position. Separate enclosures ~~may~~ can be used for each of the test blows.

In the case of an enclosure of cylindrical form, the test shall consist of three blows which are positioned with an angular displacement of 120°.

Test 1 shall be carried out at an ambient air temperature of between 10 °C and 40 °C after the assembly has been kept within these temperatures for not less than 12 h.

Test 2 shall be carried out at an ambient air temperature of between 10 °C and 40 °C immediately after the assembly has been kept at a temperature of $-25_{-0,5}^0$ °C for a period of not less than 12 h. ~~Test 2 is not required when a PENDA has an enclosure made of metallic material.~~

Compliance is checked by verification after the test that the degree of protection remains in accordance with 8.2.2, and the operation of the door(s) and locking points are not impaired; also by verification that the electrical clearances have remained satisfactory for the duration of the tests and in the case of an assembly having a metallic enclosure, that no contact between live parts and the enclosure has occurred caused by permanent or temporary distortion. In the case of an assembly having an insulating enclosure, if the appropriate conditions are satisfied, then damage such as small dents or small degrees of surface cracking or flaking are disregarded, provided that there are no associated cracks detrimental to the serviceability of the assembly.

10.2.101.5.2 Test applicable to ~~PENDAs~~ PENDA-Os designed for operation in arctic climate (see 7.1.1.2)

~~Following the ASSEMBLY being kept at a temperature of °C for a period of not less than 12 h, the impact test shall be carried out in an ambient air temperature between 10 °C and 40 °C and at a time when the external temperature of the enclosure has recovered to a temperature not higher than 40 °C. The test sequence shall be as follows:~~

~~Tests 1 and 2 involve applying a force of 1 500 N for 30 s to an earthed metal test body against the enclosure at the considered 10 weakest points. The test body shall be of spherical or hemispherical shape having a radius of 100 mm ± 3 mm and a surface hardness of HB 160 according to ISO 6506-1.~~

The specimen shall be kept at a temperature of -50 °C for a period of not less than 12 h. Tests shall be carried out in an ambient air temperature between 10 °C and 40 °C and at a time when the external temperature of the enclosure has recovered to a temperature not higher than -40 °C.

Test sequence:

For test 1 and test 2 a metal test body of spherical or hemispherical shape having a radius of 100 mm ± 3 mm and a surface hardness of HB 160 according to ISO 6506-1:2014 shall be used. Test 2 requires earthing of the test body.

Test 1 shall be carried out on an empty PENDA-O applying a force of 1 500 N for 30 s at the considerably 10 weakest points of the enclosure.

Compliance is checked by verification after the test that the degree of protection remains in accordance with 8.2.2 and the operation of the door(s) and locking points are not impaired.

Test 2 shall be carried out on an assembly containing equipment that provides the minimum clearances inside the enclosure. A metal enclosure shall be connected to earth, in case of an enclosure made of insulating materials all parts within the assembly that are intended to be connected to earth shall be connected to earth. ~~For the duration of the impact test, an a.c. voltage in accordance with 10.9.2.2 of Part 1 shall be applied between all live parts connected together and the earth.~~

A force of 1 500 N shall be applied for 30 s to earthed metal test body at the considerably 10 weakest points of the enclosure. For the duration of the test, an AC voltage in accordance with 10.9.2.2 of IEC 61439-1:2020, shall be applied between all live parts connected together and the earth.

Compliance is checked by verification that no puncture or flashovers occurs.

Test 3 shall be carried out on an empty enclosure using an impact apparatus as described in 10.2.101.5.1, but having a solid steel ball with an approximate mass of 15 kg. This striker element shall be raised through a height of approximately 1 m allowed to drop and impact the surface of the assembly under test, to provide an impact energy of 150 J (see Figure 103a and Figure 103b).

The test shall consist of one blow aimed at the centre of each of the vertical surfaces of the assembly which are visible when it is installed in its normal service position. Separate enclosures ~~may~~ can be used for each of the test blows.

In the case of an enclosure of cylindrical form, the test shall consist of three blows which are positioned with an angular displacement of 120°.

~~Compliance for Test 1 is checked by verification after the test that the degree of protection remains in accordance with 8.2.2, and the operation of the door(s) and locking points are not impaired.~~

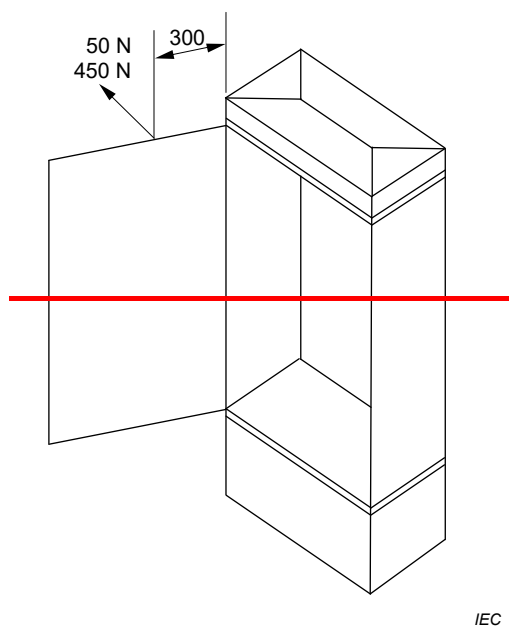
~~Compliance for Test 2 is checked by verification that no puncture or flashover occurs.~~

Compliance for Test 3 is checked by verification after the test that the degree of the protection is at least IP3X.

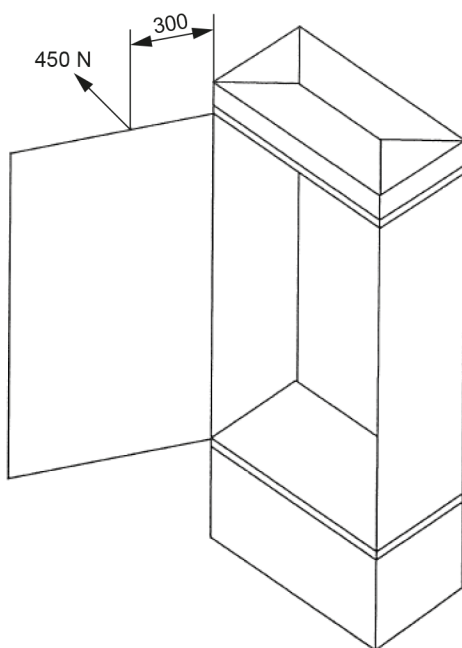
10.2.101.6 Verification of mechanical strength of doors

The test applies to all types of PENDA-O having a door(s) hinged on a vertical edge of the enclosure.

The tests shall be carried out with the door(s) fully open and in contact with the designed restraint. A load of ~~50~~ 450 N shall be applied at the top edge of the door perpendicular to the plane of the door(s) and at a distance 300 mm from the hinged edge and maintained for 3 s (see Figure 107). ~~Unless the door(s) are designed to be unhinged without the use of a tool for maintenance or operational use, the test shall then be repeated with the load increased to 450 N (see Figure 107).~~



Dimensions in millimetres



IEC

Figure 107 – Diagram of test to verify the mechanical strength of doors

Compliance is checked by verification that ~~the door(s) have not become unhinged and the operation of the door(s), hinges and locking points are not impaired by the application of a load of 50 N. In addition, by verification that~~ the degree of protection remains in accordance with 8.2.2 after the door(s) has (have) been closed following tests. If the door(s) become unhinged during the ~~450 N~~ test, this is not regarded as a failure if it is possible to reinstall the same door(s) without the use of a tool and the operation of hinges and locking points are not impaired.

10.2.101.7 Verification of resistance to axial load of metal inserts in synthetic material

The test only applies to all types of assembly when threaded metal inserts are provided to retain the mounting plate or switchgear and controlgear supports in place.

The test shall be carried out on a representative specimen of each type and size of metal insert. Also, if there is a difference in the thickness of the profile of the material surrounding a particular insert, the test shall be repeated for this condition.

During the test the assembly shall be fully supported on a platform.

A screw-eye shall be fitted to each insert under test and an axial force in accordance with Table 102 shall be applied for 10 s in an attempt to extract the insert from its anchorage.

Table 102 – Axial load to be applied to the inserts

Size of insert	Axial load N
M4	350
M5	350
M6	500
M8	500
M10	800
M12	800

Compliance is checked by inspection that the inserts remain undamaged and in their original position; and also, that there is no cracking of the surrounding material forming the anchorage for the insert.

NOTE Small cracks, created by air bubbles that were visible before the test, but not affected by the application of the axial load, are ignored.

10.2.101.8 Verification of resistance to mechanical shock impacts induced by sharp-edged objects

The test applies to all types of PENDA-O.

The test shall be carried out using an impact apparatus as described in 10.2.101.5.1 but having a steel striker element with a mass of 5 kg and having the end profiled as shown in Figure 108.

Dimensions in millimetres

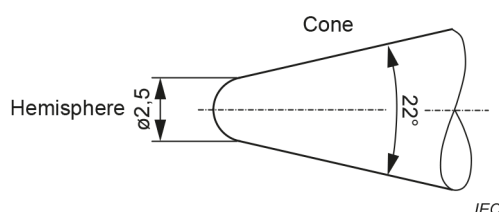


Figure 108 – Striker element for test of resistance to mechanical shock impacts induced by sharp-edged objects

The striker element shall be raised through a height of 0,4 m and allowed to drop and impact the surface of the assembly under test, thus providing an impact energy of 20 J (see Figure 103a and Figure 103b).

Each test shall consist of one blow aimed at the centre of each of the vertical surfaces of the assembly which are visible when the assembly is installed in its normal service position. Separate enclosures ~~may~~ can be used for each of the test blows.

In the case of an enclosure of cylindrical form, the test shall consist of three blows which are positioned with an angular displacement of 120°.

Test 1 shall be carried out at an ambient air temperature between 10 °C and 40 °C after the assembly has been kept within these temperatures for not less than 12 h.

Test 2 shall be carried out at an ambient air temperature between 10 °C and 40 °C immediately after the assembly has been kept at a temperature of -25 ± 5 °C for a period of not less than 12 h. **Test 2 is not required when a PENDA has an enclosure made of metallic material.**

Compliance is checked by inspection that cracks resulting from the blows are contained within a circle of diameter not exceeding 15 mm. If the tip of the striker element has penetrated the enclosure of the assembly, it shall not be possible to insert a gauge of 4 mm diameter having a hemispherical tip, applied to the hole with a force of 5 N.

10.2.101.9 Test of mechanical strength of a base intended to be embedded in the ground

This test is applicable to a PENDA-O only.

The test shall be carried out with the PENDA-O fixed to the base, in accordance with Figure 109 and the manufacturer's installation instructions. A mechanical force is transferred via a thick-walled steel tube and shall be applied to the lowest part of the longest section of the PENDA base that is beneath the ground surface when it is installed.

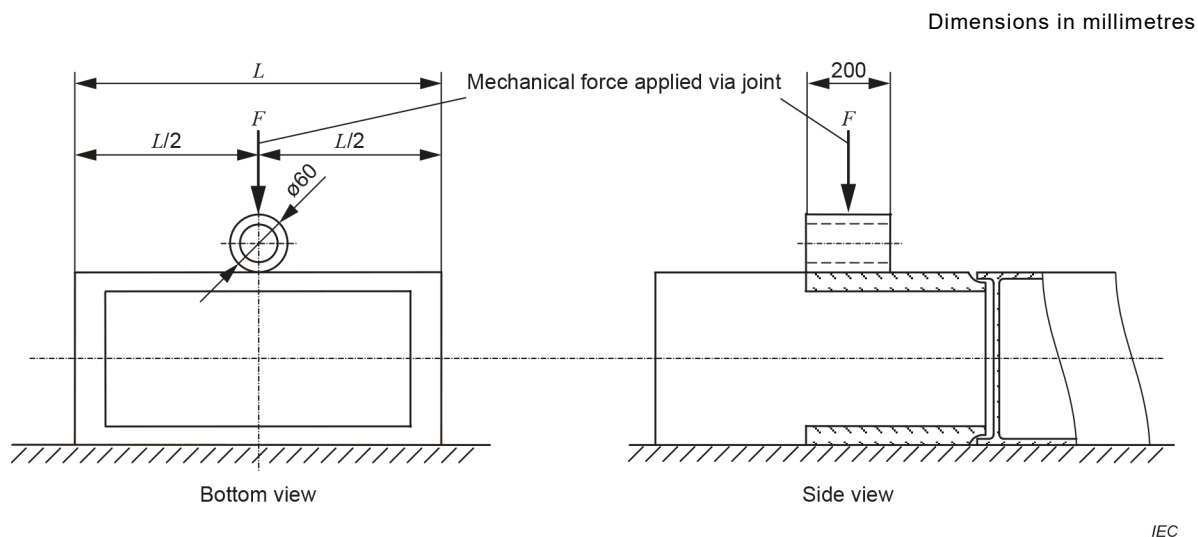


Figure 109 – Typical test arrangement for mechanical strength of base

If the design of the base includes one or more permanent supports (back to front), the force shall be applied by means of a number of steel tubes. One tube shall be placed in the centre of each unsupported length. The individual forces shall simultaneously be applied to each tube and shall be calculated according to the following formula:

$$F = 3,5 \text{ N/mm} \times L$$

where L is the unsupported length in millimetres.

The force(s) shall be applied for 1 min. After this period and whilst the force is still maintained, the degree of protection shall be verified.

If there is another section of the PENDA-O base that is of similar length but has a different profile, the test shall be repeated on this section.

Compliance is checked by inspection that the base has not broken and by verification that the degree of protection of that part of the PENDA-O and base which is normally above ground remains in accordance with 8.2.2.

10.3 Degree of protection of assemblies (IP Code)

Add the following first paragraph:

The declared IP for a PENDA-O does not apply to a base intended to be embedded in the ground. However, the declared IP does apply to the bottom of the assembly.

If a PENDA-O consists of an enclosure and a base (to be embedded in the ground or to be surface mounted) without sealed cable entry, ingress of solid objects through the ground area inside of the base is excluded when determining the degree of protection. When the PENDA is mounted on a base with an open ground area the PENDA and any exposed area of the base shall provide the PENDA with the minimum level of IP protection specified in this document.

NOTE In some instances, additional measure are taken to prohibit the entry of animals/vermin

10.5 Protection against electric shock and integrity of protective circuits

10.5.3.1 General

Replacement of the two paragraphs with the following:

Verification shall be achieved by the application of tests in accordance with 10.5.3.5 of IEC 61439-1:2020.

10.9 Dielectric properties

Additional subclause:

10.9.2.101 Power frequency dielectric test for PENDA's installed in substation with separate HV and LV neutral earths.

When a PENDA is to form part of a distribution substation in which the HV and LV earths are separated an additional dielectric test shall be carried out. A test voltage equal to the HV line to earth voltage shall be applied between the phase plus neutral conductor and the extraneous conductive part of the PENDA for a period of 10 s. Acceptance criteria shall be as given in 10.9.2.4 of IEC 61439-1:2020.

10.9.3 Impulse withstand voltage

10.9.3.1 General

Replacement of the first paragraph with:

Verification shall be achieved by the application of tests in accordance with one of the alternative test methods detailed in 10.9.3.2 to 10.9.3.4, inclusive, of IEC 61439-1:2020.

10.10 ~~Verification of~~ Temperature rise

10.10.1 General

Replacement with:

It shall be verified that the temperature-rise limits specified in 9.2 of IEC 61439-1:2020 for the different parts of the assembly will not be exceeded. Verification shall be by test as specified in 10.10.2 of IEC 61439-1:2020.

Temperature rise verification shall be undertaken on a representative sample(s). It is not necessary to test every arrangement produced, but the sample(s) tested shall, with reasonable accuracy, be thermally the most onerous for the range being considered. Assessing the most onerous arrangement(s) shall take into consideration total power loss, rating of devices (incoming and outgoing), sections of conductors, size of the enclosure and the rated current of the assembly.

10.10.2 Verification by testing

10.10.2.2 Selection of the representative arrangement

10.10.2.2.1 General

Addition of the following paragraph:

When the design of a PENDA is suitable for installation in a recess in wall the temperature rise test shall be carried out with ~~adequate~~ insulation to simulate the presence of the wall.

10.11 Short-circuit withstand strength

10.11.1 General

Replacement with:

With the exception of the circuits of assemblies that are exempt verification in accordance with 10.11.2 of IEC 61439-1:2020, the short-circuit withstand strength specified by the manufacturer shall be verified. Verification shall be by means of test as specified in 10.11.5 of IEC 61439-1:2020.

11 Routine verification

This clause of IEC 61439-1:2020 applies.

Annexes

The annexes of IEC 61439-1:2020 are applicable except as follows:

Annexes A, C, D, H, ~~A~~ K and ~~P~~ M are not applicable.

Modification of Annex ~~Q~~ L.

Addition of Annexes AA, BB, CC and DD.

Annex ~~Q~~ L
(informative)

Guidance on verification of temperature rise ~~verification~~

Modifications:

~~Q.4~~ Calculation

~~This clause of Part 1 is not applicable.~~

~~Q.5~~ Design rules

~~This clause of Part 1 is not applicable.~~

L.4 Verification assessment

This clause of IEC 61439-1:2020 is not applicable.

L.5 Verification by comparison with a reference design

This clause of IEC 61439-1:2020 is not applicable.

Annex AA
(normative)

Cross-section of conductors

Table AA.1 applies for the connection of one cable per terminal.

Table AA.1 – Minimum and maximum cross-section of copper and aluminium conductors, suitable for connection (see 8.8)

Rated current A	Solid or stranded conductors (aluminium or copper) Cross-sections mm ²		Flexible copper conductors Cross sections mm ²	
	Minimum	Maximum	Minimum	Maximum
6	0,75	1,5	0,5	1,5
8	1	2,5	0,75	2,5
10	1	2,5	0,75	2,5
12	1	2,5	0,75	2,5
16	1,5	4	1	4
20	1,5	6	1	4
25	2,5	6	1,5	4
32	2,5	10	1,5	6
40	4	16	2,5	10
63	6	25	6	16
80	10	35	10	25
100	16	50	16	35
125	25	70	25	50
160	35	95	35	70
200	50	150	50	95
250	70	150	70	120
315	70	240	95	185
400	70	240	95	185
500	70	300	95	240
630	70	300	95	240
This table applies for the connection of one conductor per terminal.				
If the external conductors are connected directly to built-in apparatus, the cross-sections indicated in the relevant specification are valid.				
In cases where it is necessary to provide for conductors other than those specified in the table, agreement shall be reached between the manufacturer and the user.				

The approximate relationship between mm² and AWG/kcmil sizes as shown in Table AA.2 is to be used where metric sized round copper conductors are not available.

Table AA.2 – Standard cross-sections of round copper conductors and approximate relationship between mm² and AWG/kcmil sizes (see 8.8 of IEC 61439-1:2020)

Rated cross-section mm ²	AWG/kcmil size	Equivalent metric area mm ²
0,2	24	0,205
0,34	22	0,324
0,5	20	0,519
0,75	18	0,82
1	–	–
1,5	16	1,3
2,5	14	2,1
4	12	3,3
6	10	5,3
10	8	8,4
16	6	13,3
25	4	21,2
35	2	33,6
–	1	42,4
50	0	53,5
70	00	67,4
95	000	85,0
–	0000	107,2
120	250 kcmil	127
150	300 kcmil	152
185	350 kcmil	177
–	400 kcmil	203
240	500 kcmil	253
300	600 kcmil	304
NOTE The dash, when it appears, counts as a size when considering connecting capacity (see 8.8).		

Annex BB (informative)

Items subject to agreement between the assembly manufacturer and the user

The information given in Table BB.1 below is subject to an agreement between the assembly manufacturer and the user. In some cases, information declared by the assembly manufacturer ~~may~~ can take the place of an agreement.

**Table BB.1 – Items subject to agreement between
the ASSEMBLY manufacturer and the user**

Characteristics	Reference clause or subclause ^c	Default arrangement ^b	Options listed in document	User requirement ^a
Electrical system				
Earthing system	5.6, 8.4.3.1, 8.4.3.2.3, 8.6.2, 10.5, 11.4	Manufacturer's standard, selected to suit local requirements	TT / TN-C / TN-C-S / IT, TN-S	
Nominal voltage (V)	3.8.9.1, 5.2.1, 8.5.3	Local, according to installation conditions	max 1 000 V AC	
Transient overvoltages	5.2.4, 8.5.3, 9.1, Annex G	Overvoltage category IV	None	
Temporary overvoltages	9.1	Nominal system voltage + 1 200 V	None	
Rated frequency f_n (Hz)	3.8.12, 5.5, 8.5.3, 10.10.2.3.1, 10.11.5.4	According to local installation conditions	50 Hz/60 Hz	
Additional on-site testing requirements: wiring, operational performance and function	11.10	Manufacturer's standard, according to application	None	
Short-circuit withstand capability				
Prospective short-circuit current at supply terminals I_{cp} (kA)	3.8.7	Determined by the electrical system	None	
Prospective short-circuit current in the neutral	10.11.5.3.5	Max. 60 % of phase values	None	
Prospective short-circuit current in the protective circuit	10.11.5.6	Max. 60 % of phase values	None	
SCPD in the incoming functional unit requirement	9.3.2	According to local installation conditions	Yes / No	
Co-ordination of short-circuit protective devices including external short-circuit protective device details	9.3.4	According to local installation conditions	None	
Data associated with loads likely to contribute to the short-circuit current	9.3.2	No loads likely to make a significant contribution allowed for	None	

Characteristics	Reference clause or subclause ^c	Default arrangement ^b	Options listed in document	User requirement ^a
Protection of persons against electric shock in accordance with IEC 60364-4-41				
Type of protection against electric shock – Basic protection (protection against direct contact)	8.4.2	Basic protection	According to local installation regulations	
Type of protection against electric shock – Fault protection (protection against indirect contact)	8.4.3	According to local installation conditions	Automatic disconnection of supply / Electrical separation / Total insulation	
Installation environment				
Location type	3.5, 8.1.4, 8.2	Manufacturer's standard, according to application	Indoor / outdoor	
Protection against ingress of solid foreign bodies and ingress of water	8.2.2, 8.2.3	Indoor (enclosed): minimum IP2X Outdoor - minimum: IP34D when installed in places accessible to the public. In other locations IP33.		
External mechanical impact NOTE IEC 61439-5 does not nominate specific IK codes.	10.2.101	As Standard (IEC 61439-5)	None	
Resistance to UV radiation (applies for outdoor assemblies only unless specified otherwise)	10.2.4	Indoor: Not applicable. Outdoor: Temperate climate	None	
Resistance to corrosion	10.2.2	Normal Indoor/Outdoor arrangements	None	
Ambient air temperature – Lower limit	7.1.1	Indoor: –5 °C Outdoor: Normal climate –25 °C Arctic climate –50 °C	None	
Ambient air temperature – Upper limit	7.1.1	40 °C	None	
Ambient air temperature – Daily average maximum	7.1.1, 9.2	35 °C	None	
Maximum relative humidity	7.1.21	Indoor: 50 95 % at –5 °C to +30 °C 70 % at +35 °C 57 % at +40 °C Outdoor: 100 % at –25 °C to +27 °C 60 % at 35 °C 46 % at 40 °C	None	

Characteristics	Reference clause or subclause ^c	Default arrangement ^b	Options listed in document	User requirement ^a
Pollution degree (of the installation environment)	7.1.32	Industrial PENDA-O: 4 PENDA-I: 3	2, 3, 4	
Altitude	7.1.41	≤ 2 000 m	None	
EMC environment (A or B)	9.4, 10.12, Annex J	A/B	A/B	
Special service conditions (e.g. vibration, exceptional condensation, heavy pollution, corrosive environment, strong electric or magnetic fields, fungus, small creatures, explosion hazards, heavy vibration and shocks, earthquakes)	7.2, 8.5.4, 9.3.3 Table 7	No special service conditions	Arctic climate	
Installation method				
Type	3.3, 5.6	Manufacturer's standard	Various e.g. ground mounting, transformer mounting, pole mounting, surface wall mounting or mounting within a recess within a wall	
Stationary/Movable	3.5	Stationary	None	
Maximum overall dimensions and weight	5.6, 6.2.1	Manufacturer's standard, according to application	None	
External conductor type(s)	8.8	Cable	Bare or insulated bars.	
Direction(s) of external conductors	8.8	From below	From above	
External conductor material	8.8	Copper/aluminium	None	
External phase conductor, cross sections, and terminations	8.8	As defined within the document	None	
External PE, N, PEN conductors cross sections, and terminations	8.8	As defined within the document	None	
Special terminal identification requirements	8.8	Manufacturer's standard	None	
Storage and handling				
Maximum dimensions and weight of transport units	6.2.2, 10.2.5	Manufacturer's standard	None	
Methods of transport (e.g. forklift, crane)	6.2.2, 8.1.6	Manufacturer's standard	None	
Environmental conditions different from the service conditions	7.3	As service conditions	None	
Packing details	6.2.2	Manufacturer's standard	None	
Operating arrangements				
Access to manually operated devices	8.4	Authorised persons	None	

Characteristics	Reference clause or subclause ^c	Default arrangement ^b	Options listed in document	User requirement ^a
Location of manually operated devices	8.5.5	Easily accessible	None	
Isolation of load installation equipment items	8.4.2, 8.4.3.3, 8.4.6.2	Manufacturer's standard	Individual / groups / all	
Maintenance and upgrade capabilities				
Requirements related to accessibility for inspection and similar operations	8.4.6.2.2	No requirements for accessibility	None	
Requirements related to accessibility for maintenance in service by authorized persons	8.4.6.2.3	No requirements for accessibility	None	
Requirements related to accessibility for extension in service by authorized persons	8.4.6.2.4	No requirements for accessibility	None	
Method of functional units connection	8.5.1, 8.5.2	Manufacturer's standard	None	
Protection against direct contact with hazardous live internal parts during maintenance or upgrade (e.g. functional units, main busbars, distribution busbars)	8.4	No requirements for protection during maintenance or upgrade	None	
Current carrying capability				
Maximum total load current to be supplied by the assembly (from which the rated current of the assembly I_{nA} (A) will be determined)	3.8.9, 10.1, 5.3, 8.4.3.2.3, 8.5.3, 8.8, 10.10.2, 10.10.3, 10.11.5, Annex E	Manufacturer's standard, according to application	None	
Rated current of circuits I_{nG} (A)	5.3.2	Manufacturer's standard, according to application	None	
Rated diversity factor	5.4, 10.10.2.3, Annex E	As defined within the standard	RDF for groups of circuits / RDF for whole ASSEMBLY	
Design current I_B and nature of load for each circuit; alternatively, I_n of the devices and nature of the load (in such cases, the assumed loading factors can be used based on the relevant part of IEC 61439)	3.8.10.8	None	None	
Ratio of cross section of the neutral conductor to phase conductors: phase conductors up to and including 16 mm ²	8.6.1	100 %	None	
Ratio of cross section of the neutral conductor to phase conductors: phase conductors above 16 mm ²	8.6.1	50 % (min. 16 mm ²)	None	
^a For exceptionally onerous applications, the user may need to can specify more stringent requirements to those in the standard. ^b In some cases information declared by the assembly manufacturer may can take the place of an agreement. ^c When a reference does not exist in this document, IEC 61439-1:2020 applies.				

Annex CC (informative)

Design verification

Table CC.1 – List of design verifications to be performed

No.	Characteristic to be verified	Clauses or subclauses
1	Strength of material and parts: Resistance to corrosion Properties of insulating materials: Thermal stability Resistance to abnormal heat and fire due to internal electric effects Dry heat Category of flammability Resistance to ultra-violet (UV) radiation Lifting Mechanical impact Resistance to static load Resistance to shock load Resistance to torsional stress Impact force withstand Mechanical strength of doors Resistance to axial load of metal inserts in synthetic material Resistance to mechanical shock induced by sharp objects Mechanical strength of a base intended to be embedded in the ground Marking	10.2 10.2.2 10.2.3 10.2.3.1 10.2.3.2 10.2.3.101 10.2.3.102 10.2.4 10.2.5 10.2.101 10.2.101.2 10.2.101.3 10.2.101.4 10.2.101.5 10.2.101.6 10.2.101.7 10.2.101.8 10.2.101.9 10.2.7
2	Degree of protection of enclosures	10.3
3	Clearances	10.4
4	Creepage distances	10.4
5	Protection against electric shock and integrity of protective circuits: Effective continuity between the exposed conductive parts of the ASSEMBLY and the protective circuit Short-circuit withstand strength of the protective circuit	10.5 10.5.2 10.5.3
6	Incorporation of switching devices and components	10.6
7	Internal electrical circuits and connections	10.7
8	Terminals for external conductors	10.8
9	Dielectric properties: Power-frequency withstand voltage Impulse withstand voltage	10.9 10.9.2 10.9.3
10	Temperature-rise limits	10.10
11	Short-circuit withstand strength	10.11
12	Electromagnetic compatibility (EMC)	10.12
13	Mechanical operation	10.13

No.	Characteristic to be verified	Clauses or subclauses ^a
1	Strength of material and parts:	10.2
	Resistance to corrosion	10.2.2
	Properties of insulating materials:	10.2.3
	Thermal stability	10.2.3.1
	Resistance to abnormal heat and fire due to internal electric effects	10.2.3.2
	Category of flammability	10.2.3.101
	Resistance to ultra-violet (UV) radiation	10.2.4
	Lifting	10.2.5
	Mechanical impact	10.2.101
	Resistance to static load	10.2.101.2
	Resistance to shock load	10.2.101.3
	Resistance to torsional stress	10.2.101.4
	Impact force withstand	10.2.101.5
	Mechanical strength of doors	10.2.101.6
	Resistance to axial load of metal inserts in synthetic material	10.2.101.7
	Resistance to mechanical shock induced by sharp objects	10.2.101.8
	Mechanical strength of a base intended to be embedded in the ground	10.2.101.9
	Marking	10.2.7
	Mechanical operation	10.2.8
2	Degree of protection of enclosures	10.3
3	Clearances	10.4
4	Creepage distances	10.4
5	Protection against electric shock and integrity of protective circuits:	10.5
	Effective continuity between the exposed conductive parts of the assembly and the protective circuit	10.5.2
	Short-circuit withstand strength of the protective circuit	10.5.3
6	Incorporation of switching devices and components	10.6
7	Internal electrical circuits and connections	10.7
8	Terminals for external conductors	10.8
9	Dielectric properties:	10.9
	Power-frequency withstand voltage	10.9.2
	Impulse withstand voltage	10.9.3
10	Temperature-rise limits	10.10
11	Short-circuit withstand strength	10.11
12	Electromagnetic compatibility (EMC)	10.12
^a When a reference does not exist in this document, IEC 61439-1:2020 applies.		

Annex DD (informative)

List of notes concerning certain countries

Subclause	Text
8.8	Add the following note after the last paragraph: NOTE In the United States of America (USA) the conductor size requirement is dependent on the current rating, conductor insulation temperature rating, ambient temperature and configuration, along with conductor insulation type. The specific requirements are contained in the US National Electrical Code (NEC) NFPA 70 Chapter 3.

Bibliography

The bibliography of IEC 61439-1:2020 is applicable except as follows:

Addition:

~~ISO 9223, Corrosion of metals and alloys — Corrosivity of atmospheres — Classification, determination and estimation~~

IEC 60050 (all parts), *International Electrotechnical Vocabulary (IEV)*, available at www.electropedia.org



IEC 61439-5

Edition 3.0 2023-05

INTERNATIONAL STANDARD

NORME INTERNATIONALE

**Low-voltage switchgear and controlgear assemblies –
Part 5: Assemblies for power distribution in public networks**

**Ensembles d'appareillage à basse tension –
Partie 5: Ensembles pour réseaux de distribution publique**



CONTENTS

FOREWORD	3
1 Scope	5
2 Normative references	6
3 Terms and definitions	7
4 Symbols and abbreviations	8
5 Interface characteristics	8
6 Information	8
7 Service conditions	9
8 Constructional requirements	10
9 Performance requirements	12
10 Design verification	12
11 Routine verification	27
Annexes	28
Annex L (informative) Guidance on verification of temperature rise	29
Annex AA (normative) Cross-section of conductors	30
Annex BB (informative) Items subject to agreement between the assembly manufacturer and the user	32
Annex CC (informative) Design verification	36
Annex DD (informative) List of notes concerning certain countries	37
Bibliography	38
Figure 101 – Typical distribution network	6
Figure 102 – Diagram of test to verify resistance to shock load of a PENDA-O	15
Figure 103 – Diagram of test to verify impact force withstand of a PENDA-O	16
Figure 104 – Diagram of test to verify the resistance to static load	17
Figure 105 – Sandbag for test to verify the resistance to shock load	18
Figure 106 – Diagram of test to verify resistance to torsional stress of a PENDA-O	20
Figure 107 – Diagram of test to verify the mechanical strength of doors	23
Figure 108 – Striker element for test of resistance to mechanical shock impacts induced by sharp-edged objects	24
Figure 109 – Typical test arrangement for mechanical strength of base	25
Table 101 – Values of assumed loading	8
Table 102 – Axial load to be applied to the inserts	24
Table AA.1 – Minimum and maximum cross-section of copper and aluminium conductors, suitable for connection (see 8.8)	30
Table AA.2 – Standard cross-sections of round copper conductors and approximate relationship between mm ² and AWG/kcmil sizes (see 8.8 of IEC 61439-1:2020)	31
Table BB.1 – Items subject to agreement between the ASSEMBLY manufacturer and the user	32
Table CC.1 – List of design verifications to be performed	36

INTERNATIONAL ELECTROTECHNICAL COMMISSION

LOW-VOLTAGE SWITCHGEAR AND CONTROLGEAR ASSEMBLIES –**Part 5: Assemblies for power distribution in public networks**

FOREWORD

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
- 2) The formal decisions or agreements of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees.
- 3) IEC Publications have the form of recommendations for international use and are accepted by IEC National Committees in that sense. While all reasonable efforts are made to ensure that the technical content of IEC Publications is accurate, IEC cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.
- 4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC Publication and the corresponding national or regional publication shall be clearly indicated in the latter.
- 5) IEC itself does not provide any attestation of conformity. Independent certification bodies provide conformity assessment services and, in some areas, access to IEC marks of conformity. IEC is not responsible for any services carried out by independent certification bodies.
- 6) All users should ensure that they have the latest edition of this publication.
- 7) No liability shall attach to IEC or its directors, employees, servants or agents including individual experts and members of its technical committees and IEC National Committees for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC Publication or any other IEC Publications.
- 8) Attention is drawn to the Normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.
- 9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

IEC 61439-5 has been prepared by subcommittee 121B: Low-voltage switchgear and controlgear assemblies, of IEC technical committee 121: Switchgear and controlgear and their assemblies for low voltage. It is an International Standard.

This third edition cancels and replaces the second edition published in 2014. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- a) omission of the requirement to conduct mechanical tests at -25 °C when enclosures are made of a metallic material;
- b) addition of assumed loading factors generation supplies and electric vehicle charging applications;
- c) additional dielectric tests when a PENDA is used in a distribution substation with separate HV and LV earths;
- d) further clarification of representative samples for design verification.

The text of this International Standard is based on the following documents:

Draft	Report on voting
121B/173/FDIS	121B/178/RVD

Full information on the voting for its approval can be found in the report on voting indicated in the above table.

The language used for the development of this International Standard is English.

This document was drafted in accordance with ISO/IEC Directives, Part 2, and developed in accordance with ISO/IEC Directives, Part 1 and ISO/IEC Directives, IEC Supplement, available at www.iec.ch/members_experts/refdocs. The main document types developed by IEC are described in greater detail at www.iec.ch/publications.

This document is to be read in conjunction with IEC 61439-1. The provisions of the general rules dealt with in IEC 61439-1 are only applicable to this document insofar as they are specifically cited. When this document states "addition", "modification" or "replacement", the relevant text in IEC 61439-1:2020 is to be adapted accordingly. Subclauses that are numbered with a 101 (102, 103 etc.) suffix are additional to the same subclause in IEC 61439-1:2020.

Tables and figures in IEC 61439-5:2023 that are new are numbered starting with 101.

New annexes in IEC 61439-5:2023 are lettered AA, BB, etc.

The reader's attention is drawn to the fact that Annex DD lists all of the "in-some-country" clauses on differing practices of a less permanent nature relating to the subject of this document.

A list of all parts of the IEC 61439 series, under the general title *Low-voltage switchgear and controlgear assemblies* can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under webstore.iec.ch in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

LOW-VOLTAGE SWITCHGEAR AND CONTROLGEAR ASSEMBLIES –

Part 5: Assemblies for power distribution in public networks

1 Scope

This document defines the specific requirements for public electricity network distribution assemblies (PENDAs).

PENDAs have the following criteria:

- used for the distribution of electrical energy in three phase systems for which the rated voltage does not exceed 1 000 V AC (see Figure 101 for a typical distribution network) and DC systems not exceeding 1 500 V DC;
- stationary;
- open type assemblies are not covered by this document;
- suitable for installation in places where only skilled persons have access for their use, however, outdoor types can be installed in situations that are accessible to ordinary persons
 - intended for use in energy distribution in public power grids;
 - indoor use: assemblies for installation inside of electric power substations;
 - outdoor use: assemblies containing an enclosure suitable for open air installation.

The object of this document is to state the definitions and to specify the service conditions, construction requirements, technical characteristics and tests for PENDAs. Tests at higher performance level can be applicable with some network parameters.

PENDAs can also include control and or signalling devices associated with the distribution of electrical energy.

NOTE 1 Control and monitoring devices can be used in smart grid applications or the transmission of smart grid data.

This document applies to all PENDAs whether they are designed, manufactured on a one-off basis or fully standardised and manufactured in quantity.

The manufacture and/or assembly can be carried out other than by the original manufacturer (see 3.10.1 of IEC 61439-1:2020).

This document does not apply to individual devices and self-contained components, such as motor starters, fuse switches, electronic equipment, etc. which comply with the relevant product standards.

If the substation is owned or operated by a public distribution system operator (DSO), PENDA's which are used as LV distribution panels in transformer substations are within the scope of this document,

This document does not apply to specific types of assemblies covered by other parts of IEC 61439 series.

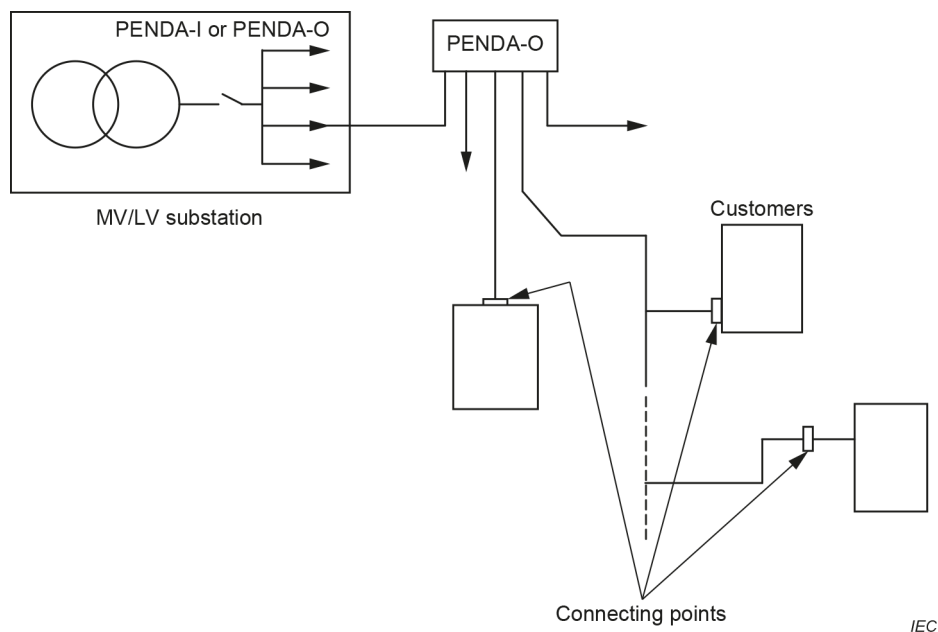


Figure 101 – Typical distribution network

NOTE 2 If a PENDA is equipped with additional equipment (for example meters), in such a way that the main function is changed considerably, then other standards can also apply as agreed between user and manufacturer (see 8.5 of IEC 61439-1:2020).

NOTE 3 Where local regulations and practices permit, a PENDA according to this document can be used in other than public networks.

NOTE 4 DSO's can define additional requirements for their PENDA's.

2 Normative references

This clause of IEC 61439-1:2020 applies except as follows.

Addition:

IEC 60695-11-10:2013, *Fire hazard testing – Part 11-10: Test flames – 50 W horizontal and vertical flame test methods*

IEC 61439-1:2020, *Low-voltage switchgear and controlgear assemblies – Part 1: General rules*

IEC 62262, *Degrees of protection provided by enclosures for electrical equipment against external mechanical impacts (IK code)*

ISO 9223:2012, *Corrosion of metals and alloys – Corrosivity of atmospheres – Classification, determination and estimation*

ISO 6506-1:2014, *Metallic materials – Brinell hardness test – Part 1: Test method*

3 Terms and definitions

This clause of IEC 61439-1:2020 applies except as follows.

3.1 General terms

Additional terms:

3.1.101

public electricity network distribution assembly PENDA

assembly, generally for installation in a public electricity network which in use, receives electrical energy from one or more supplies and distributes that energy through one or more cables to other equipment

Note 1 to entry: A PENDA is installed, operated and maintained solely by skilled persons.

Note 2 to entry: Some types of a PENDA were previously known as a cable distribution cabinet (CDC).

3.1.101.1

outdoor public electricity network distribution assembly PENDA-O

cubicle type public electricity network distribution assembly that is suitable for outdoor installation in places that can, or cannot, be accessible to the public

3.1.101.2

indoor public electricity network distribution assembly PENDA-I

public electricity network distribution assembly suitable for installation indoors, generally without an enclosure, but including all structural parts necessary to support busbars, functional units and other ancillary devices, necessary to complete the assembly

3.1.102

design life

minimum duration for which specified performance characteristics of equipment are expected when the equipment is operated as intended and regularly maintained by instructed persons in accordance with the manufacturer's instructions

[SOURCE: IEC 60050-395:2014, 395-07-93, modified – Note to entry omitted and 'granted' replaced by 'expected when instructions'.]

3.3 External design of assemblies

3.3.1

open-type assembly

This term of IEC 61439-1:2020 does not apply.

3.9 Verification

Modifications:

3.9.1

design verification

Delete the note.

3.9.1.2

verification comparison

This term of IEC 61439-1:2020 does not apply.

3.9.1.3

verification assessment

This term of IEC 61439-1:2020 does not apply.

4 Symbols and abbreviations

This clause of IEC 61439-1:2020 applies.

5 Interface characteristics

This clause of IEC 61439-1:2020 applies, except as follows.

5.4 Rated diversity factor (RDF)

Addition:

In the absence of an agreement between the assembly manufacturer and user concerning the actual load currents, the assumed loading of the outgoing circuits of the assembly or group of outgoing circuits can be based on the values given in Table 101.

For distribution and final circuits, it is assumed that the load current is the rated current of the protective device, I_n , as required by the user, multiplied with the loading factor of Table 101.

Table 101 – Values of assumed loading

Application	Assumed loading factor
Regular distribution grid connections 2 to 3 circuits	0,9
Regular distribution grid connections 4 to 5 circuits	0,8
Regular distribution grid connections 6 to 9 circuits	0,7
Regular distribution grid connections ≥ 10 circuits	0,6
Generation supply (e.g. PV, wind farm, biomass)	1,0
Charging infrastructure for EV	1,0

6 Information

This clause of IEC 61439-1:2020 applies except as follows.

6.1 Assembly designation marking

Addition to first paragraph:

Designation plates can be placed inside an enclosure of an assembly provided their intended place ensures good legibility and visibility when the door(s) is(are) open or the cover is removed.

Replacement of item g):

g) IEC 61439-5.

6.3 Device and/or component identification

Additional paragraph:

In the case of removable fuse-carriers which are specific to a fuseway, a label shall be placed on the fuse carrier as well as on the fuse base, to avoid incorrect interchangeability of the fuse-carrier.

Additional subclause:

6.101 Circuit identification

It shall be possible to identify each functional unit in a clearly visible manner.

7 Service conditions

This clause of IEC 61439-1:2020 applies except as follows.

7.1 Normal service conditions

7.1.1 Climatic conditions

Addition to the first paragraph:

Unless the user specifies that a PENDA shall be suitable for use in an arctic climate, the lower limit of ambient air temperature is -25 °C as specified in Table 15 of IEC 61439-1:2020. For an arctic climate the lower limit of ambient temperature is -50 °C .

7.1.2 Pollution degree

Replace the first paragraph with the following:

The pollution degree referred to in Annex BB is the macro-environmental condition for which the assembly is intended.

7.2 Special service conditions

Additional paragraph:

Additional requirements for a PENDA-O, to be installed where heavy snowfalls occur and where they are adjacent to areas where there is snow clearance by ploughing, are subject to agreement between manufacturer and user.

8 Constructional requirements

This clause of IEC 61439-1:2020 applies except as follows.

8.1 Strength of materials and parts

8.1.1 General

Change the reference to Annex C to Annex BB in paragraph 5 of 8.1.1 of IEC 61439-1:2020.

Addition:

A PENDA-O shall be arranged for ground mounting, transformer mounting, pole mounting, surface wall mounting or mounting within a recess within a wall, as agreed between user and manufacturer.

A PENDA can be directly coupled to a transformer by means of a flange coupling or it can connect to its supply by means of cable or via busbars as agreed between user and manufacturer. Outgoing circuits shall be suitable for connection by means of cables.

A locking device shall be provided on outdoor enclosures which prevents access by unauthorized persons. The fixings of any covers etc. which are removable for installation or maintenance operations shall only be accessible while the door(s) is (are) open.

Any auxiliary equipment, e.g. meters, relays, instruments, circuit breaker trip units, communications equipment, that can be readily replaced, are excluded from the minimum design life of a PENDA.

NOTE When applicable, a design life can be agreed between user and manufacturer, assuming it is operated as intended and regularly maintained by instructed personnel in accordance with the manufacturer's instructions.

8.1.3.2 Resistance of insulating materials to heat and fire

Additional subclause:

8.1.3.2.101 Verification of category of flammability

The insulating materials used for enclosures, barriers and other insulating parts shall have flammability properties in accordance with 10.2.3.101.

8.1.5 Mechanical strength

Additional subclause:

8.1.5.101 Verification of mechanical strength

The mechanical properties of a PENDA-O shall comply with 10.2.101.

Parts of the PENDA-O intended to be embedded in the ground shall withstand the stresses imposed on them during installation and normal service and comply with 10.2.101.9.

8.2 Degree of protection provided by an assembly enclosure

8.2.1 Protection against mechanical impact (IK code)

Subclause 8.2.1 of IEC 61439-1:2020 does not apply.

The mechanical impact tests required by this document are at least equal to IK10 in accordance with IEC 62262 (see 8.1.5.101).

8.2.2 Protection against contact with live parts, ingress of solid foreign bodies and water (IP code)

Addition:

Open type assemblies (IP00) are not covered by this document.

When a PENDA-O is intended to be installed in places accessible to the public, its enclosure shall, when fully-installed in accordance with the manufacturer's instructions, provide a degree of protection of at least IP34D according to IEC 60529. In other locations, the minimum level of protection shall be at least IP33.

PENDA-O's that are intended to be installed in places accessible to the public shall, unless otherwise stated by the user, be designed such that when any temporary cables are connected, the enclosure shall provide a degree of protection of at least IP23C according to IEC 60529. See 8.8.

8.4.2.1 General

The third paragraph does not apply.

Additional subclause:

8.4.2.101 Earthing and short-circuiting means

When specified by the user, the outgoing units in an assembly shall be so constructed that they can be earthed and short-circuited in a secure manner by means of a device(s) recommended by the manufacturer, which ensures the required degree of protection (IP code) is maintained for all parts of the assembly.

8.4.3.1 Installation conditions

Additional paragraph:

When a user specifies the assembly is to feed overhead cable lines, outgoing units shall be designed in such a way that an attached cable(s) can be earthed at the termination(s).

8.8 Terminals for external cables

Replacement of the first three paragraphs with the following:

In the absence of a special agreement between user and manufacturer, terminals shall be capable of accommodating cables having copper or aluminium conductors from the smallest to the largest cross-sectional area corresponding to the appropriate rated current (see Table AA.1).

The terminations for outgoing circuits shall be located so that adequate spacing is provided and to facilitate terminating the phase conductors of a cable irrespective of their lay.

When specified by the user the phase cable terminals of each outgoing circuit shall be separated from all other hazardous live parts. When the terminals of an outgoing circuit are exposed protection shall be provided against accidental contact with other hazardous live parts. Separation and protection shall be from the normal direction of access and in accordance with IPXXB of IEC 60529.

Where specified by the user, the incoming circuit shall be suitable for connection by means of either bare or insulated bars.

Additional subclauses:

8.101 Marking as an obstacle to snow clearance

When specified by the user, a PENDA-O that is intended for use in regions where heavy snowfalls occur in accordance with 7.2, shall be marked as an obstacle to snow clearance. Holders shall be provided, attached to the PENDA-O, to accommodate marking rods and it shall be possible to install and make adjustments to the position of the marking rod from outside the PENDA-O.

8.102 Ease of operation and maintenance

All parts of the assembly shall, as far as practicable, be readily accessible and replaceable without excessive dismantling.

The design shall be such that the cables can be readily connected from the front.

When a PENDA does not have a means of measurement incorporated, it shall be possible, by the use of a portable instrument, to readily and safely measure voltages in all phases of incoming units and on both sides of all current breaking and/or switch devices of outgoing units, also the current in one phase of all outgoing units. During this operation all live parts of the PENDA shall be protected sufficiently to retain the required degree of protection in accordance with 8.2. Instructions concerning the procedure to be adopted shall be provided by the manufacturer.

If the assembly is intended to be connected to a live reserve power, for example a standby generator, the switchgear connecting device shall be designed so that connection can be made with the live parts having a degree of protection of IPXXB according to IEC 60529.

9 Performance requirements

This clause of IEC 61439-1:2020 applies.

10 Design verification

This clause of IEC 61439-1:2020 applies except as follows.

10.1 General

Replace 6th, 7th, and 8th paragraph with:

Design verification shall only be achieved by the application of tests in accordance with Clause 10. The alternatives methods of verification by assessment or comparison with a reference design shall not be used (see Table CC.1).

It is not necessary to test every arrangement produced. Tests carried out on a representative sample of the most onerous PENDA are deemed to verify the performance of similar and less onerous assemblies of the same general construction and rating. Examples of tests carried out on most onerous representative sample include:

- 1) A temperature rise test carried out on 800 A PENDA with 5 outgoing circuits is deemed to apply to an 800 A PENDA of the same construction (same general enclosure, same design of busbars and same incoming unit(s)) with 8 outgoing circuits of the same design.
- 2) Mechanical strength tests carried out on the smallest and largest enclosure in a series are deemed to apply to all other enclosures in the series, when they have the same number of doors and when they are of the same general construction.

- 3) Providing the busbar and connection support arrangements are the same, short-circuit tests carried out on a design of busbar and connections are deemed to apply to higher current ratings of busbars and connections.
- 4) A short circuit test carried out on 800 A PENDA with 8 outgoing circuits is deemed to apply to an 800 A PENDA of the same construction (same general enclosure, same design of busbars and same incoming unit(s)) with 5 outgoing circuits of the same design.
- 5) For a given short circuit rating, a short circuit tests carried out on an incoming or outgoing circuit is deemed to apply to higher current rating circuits of the same design and construction

Additional last paragraph:

Where necessary to suit their particular network parameters, users can specify more onerous or additional test requirements.

10.2 Strength of materials and parts

10.2.2 Resistance to corrosion

10.2.2.1 Verification by test

Replacement of last paragraph with the following:

When the corrosion resistance properties and projected life, as agreed between the manufacturer and the user, shall be confirmed by reference to ISO 9223:2012, the tests detailed herein need not be performed.

In all other cases the corrosion resistance of each design of assembly shall be verified by severity test A or B, as applicable and as detailed 10.2.2.3 of IEC 61439-1:2020 and 10.2.2.2.

10.2.2.2 Severity test A

Replacement of the test specification (paragraph 2) with the following:

Damp heat cycling test of IEC 60068-2-30:2005: Severity – temperature 55 °C, 6 cycles and variant 1.

At the end of the test, the specimens are removed from the test chamber.

Compliance is checked by visual inspection. The parts tested shall not show rust, cracking or other deterioration. However, surface corrosion of the protective coating is allowed.

10.2.2.4 Results to be obtained

This subclause of IEC 61439-1:2020 is not applicable in respect of tests carried out in accordance with 10.2.2.2.

10.2.3 Properties of insulating materials

Additional subclause:

10.2.3.101 Verification of category of flammability

Representative specimens of each of the materials of enclosures, barriers and other insulating parts shall be subjected to a flammability test in accordance with test method A – horizontal burning test of IEC 60695-11-10:2013.

Compliance is checked by inspection that each set of specimens can be classified to category HB40 criteria a) or b) in accordance with 8.4.3 of IEC 60695-11-10:2013.

10.2.6 Verification of protection against mechanical impact (IK code)

This subclause of IEC 61439-1:2020 is not applicable to assemblies complying with this document.

Additional subclauses:

10.2.101 Verification of mechanical strength

10.2.101.1 General

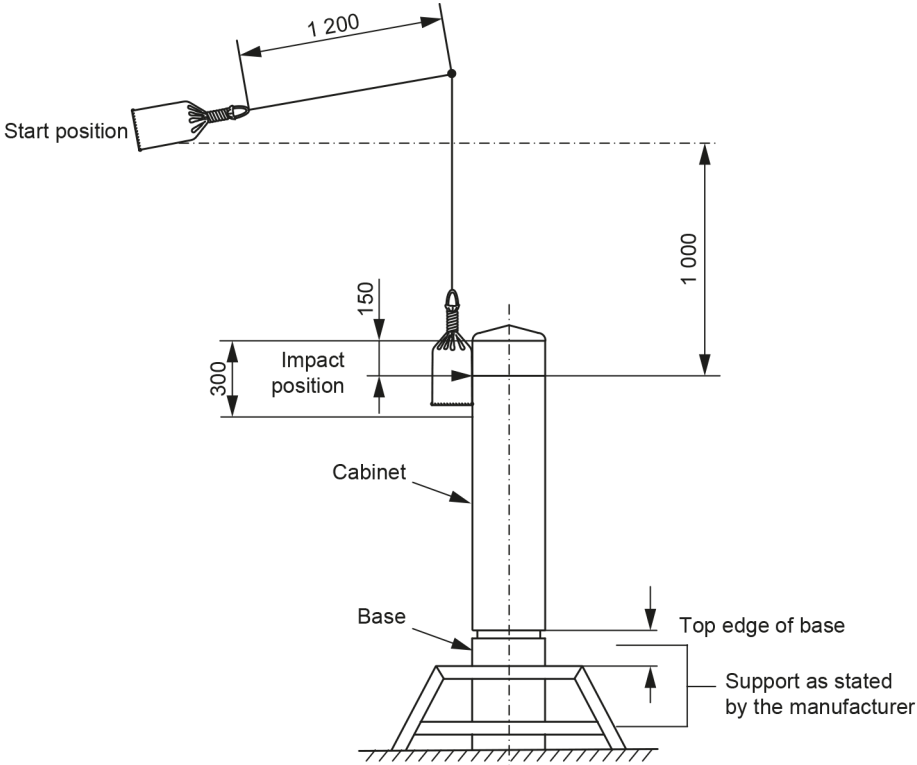
Where enclosures are manufactured in a series and are of the same design and construction, only the smallest and largest size of enclosure shall be tested.

The tests shall be carried out at an ambient temperature between 10 °C and 40 °C.

With the exception of the test of 10.2.101.7, a new sample assembly can be used for each of the independent tests. If the same sample assembly is used for more than one test of 10.2.101, the compliance check for the second numeral of the degree of protection (IP code) need only be applied when the tests on that sample have been completed.

All tests shall be carried out with the assembly fixed at its normal service mounting and where appropriate, added support at normal ground level as indicated in Figure 102a, Figure 102b, Figure 103a, and Figure 103b.

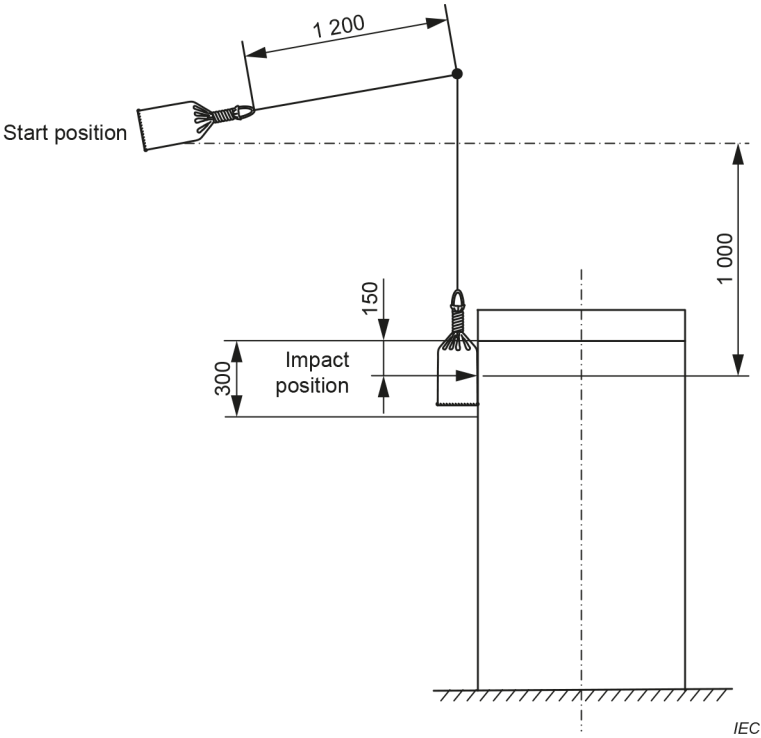
Dimensions in millimetres



IEC

Figure 102a – Diagram of test to verify the resistance to shock load of a ground mounted PENDA-O with embedded base

Dimensions in millimetres

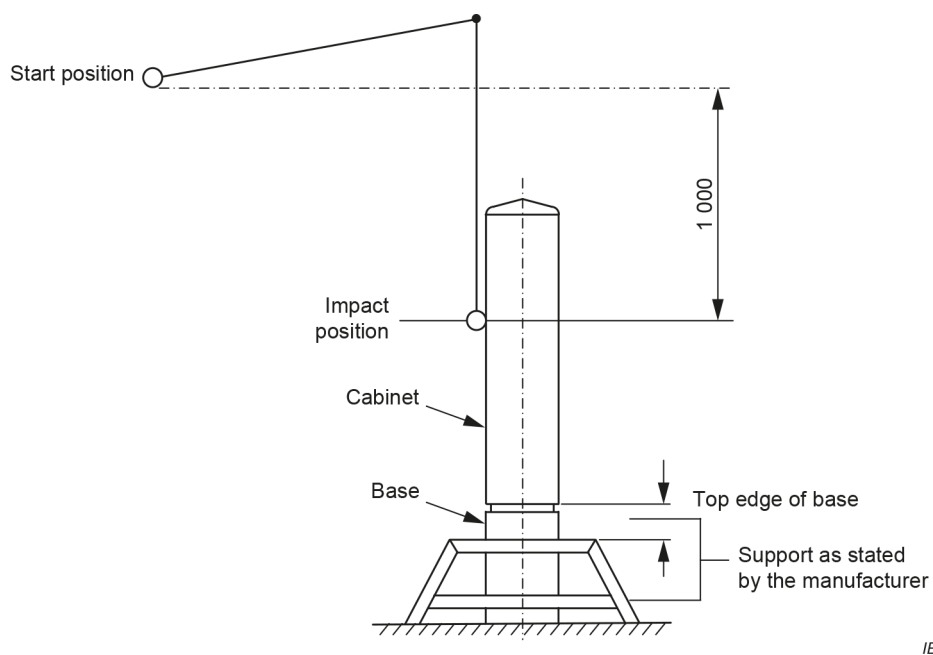


IEC

Figure 102b – Diagram of test to verify resistance to shock load of a ground mounted PENDA-O without embedded base

Figure 102 – Diagram of test to verify resistance to shock load of a PENDA-O

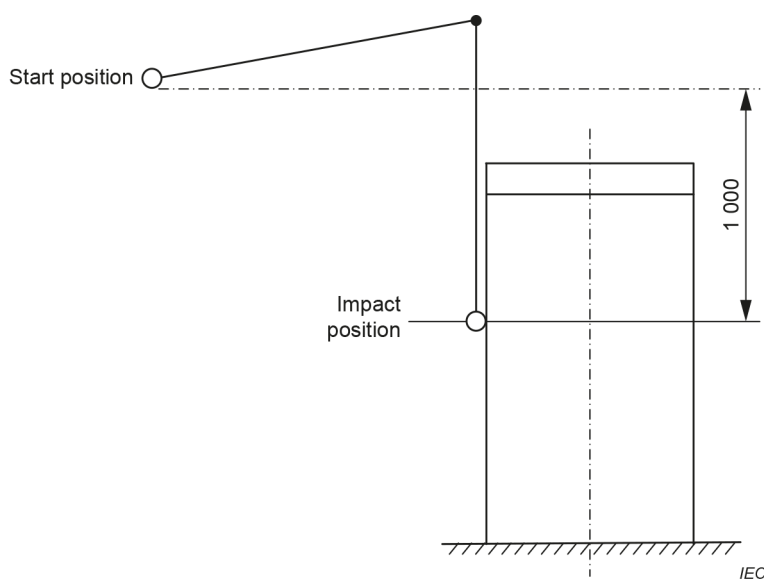
Dimensions in millimetres



IEC

Figure 103a – Diagram of test to verify impact force withstand for a ground mounted PENDA-O with embedded base

Dimensions in millimetres



IEC

Figure 103b – Diagram of test to verify impact force withstand for a ground mounted PENDA-O without embedded base

Figure 103 – Diagram of test to verify impact force withstand of a PENDA-O

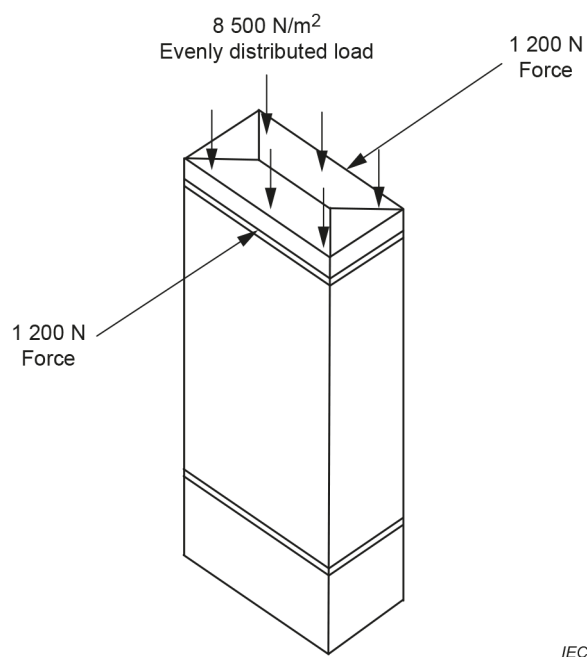
With the exception of the test of 10.2.101.8, the door(s) of the assembly, if applicable, shall be locked at the commencement of the test and remain locked for the duration of the test.

10.2.101.2 Verification of resistance to static load

The following tests shall be carried out on all types of PENDA-0:

Test 1 – An evenly distributed load of 8 500 N/m² shall be applied for 5 min to the roof of the enclosure (see Figure 104).

Test 2 – A force of 1 200 N shall be applied for 5 min in turn to the front and back upper edges of the roof of the enclosure (see Figure 104).



IEC

Figure 104 – Diagram of test to verify the resistance to static load

Compliance is checked by verification after the test that the minimum degree of protection is in accordance with 8.2.2, and the operation of the door(s) and locking points are not impaired; also by verification that the electrical clearances have remained satisfactory for the duration of the tests and in the case of an assembly having metallic enclosure, that no contact between live parts and the enclosure has occurred caused by permanent or temporary distortion.

10.2.101.3 Verification of resistance to shock load

The test shall apply to all types of PENDA-O.

A bag in accordance with Figure 105 containing dry sand and having a total mass of 15 kg shall be hung from an overhead support vertically above the surface under test and at 1 m above the strike point of the assembly.

Dimensions in millimetres

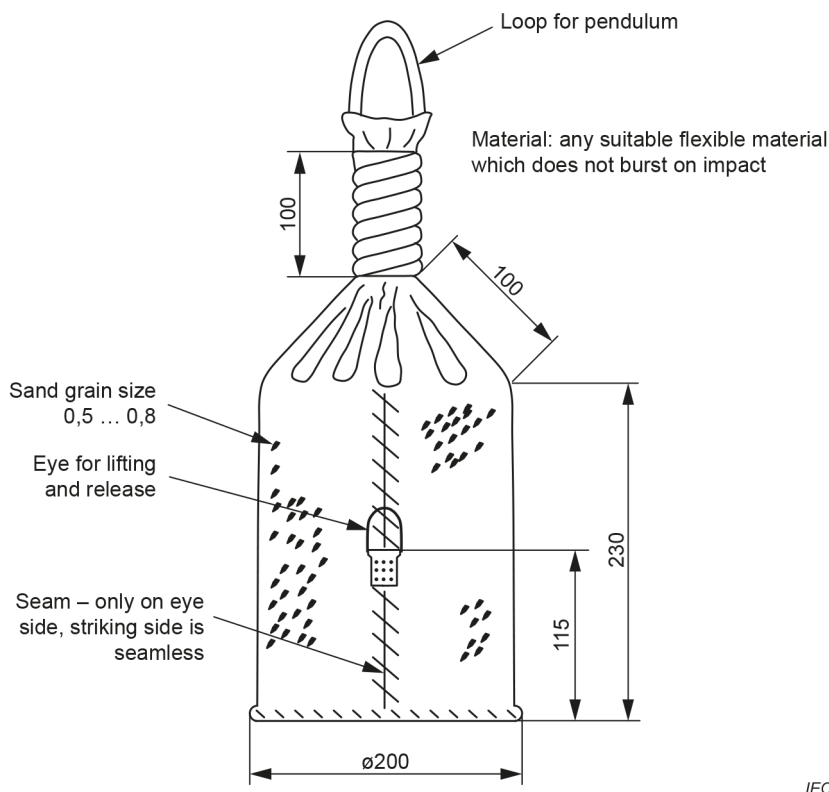


Figure 105 – Sandbag for test to verify the resistance to shock load

Each test shall consist of one blow aimed at the upper part of each of the vertical surfaces of the assembly which are visible when the assembly is installed in its normal service position. Separate enclosures can be used for each of the test blows.

In the case of an enclosure of cylindrical form, the test shall consist of three blows which are positioned with an angular displacement of 120°.

A test shall consist of raising the lifting eye through a height of 1 m and allowing the sandbag to fall through a vertical arc to impact the approximate centre of the upper part of the surface of the assembly under test (see Figure 102a and Figure 102b).

Compliance is checked by verification after the test that the degree of protection remains in accordance with 8.2.2, and the operation of the door(s) and locking points are not impaired; also by verification that the electrical clearances have remained satisfactory for the duration of the tests and, in the case of an assembly having a metallic enclosure, that no contact between live parts and the enclosure has occurred caused by permanent or temporary distortion. In the case of an assembly having an insulating enclosure, if the appropriate conditions are satisfied, then damage such as small dents or small degrees of surface cracking or flaking are disregarded, provided that there are no associated cracks detrimental to the serviceability of the assembly.

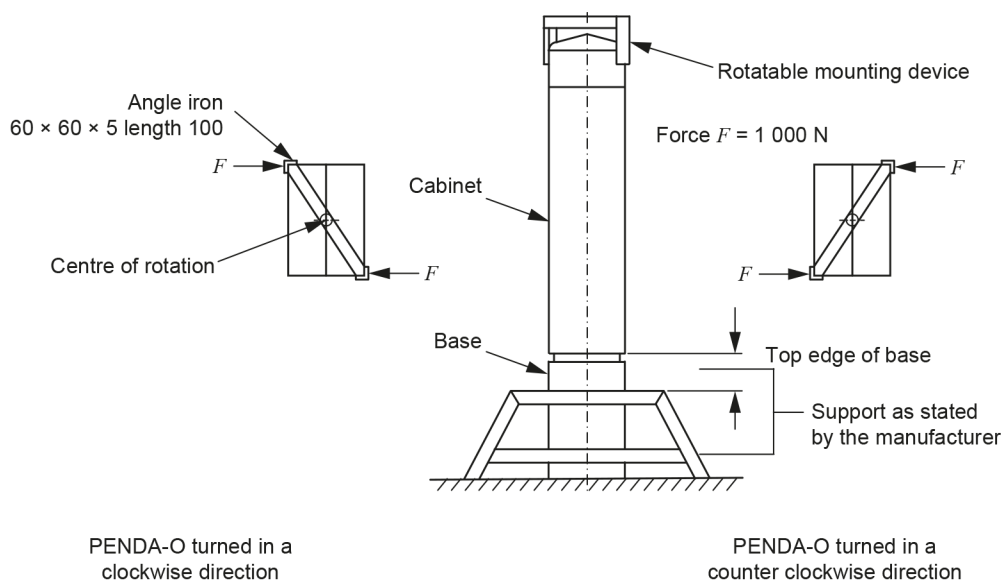
10.2.101.4 Verification of resistance to torsional stress

The test only applies to all types of PENDA-O.

The test is carried out using a horizontally rotatable frame constructed from 60 mm × 60 mm × 5 mm angle iron, having vertical locations 100 mm long at the frame arm's extremities. The assembly under test is rigidly fixed at its base and the frame closely fitted over it, so that the end locations of the frame arm are in contact with the roof and walls of the assembly.

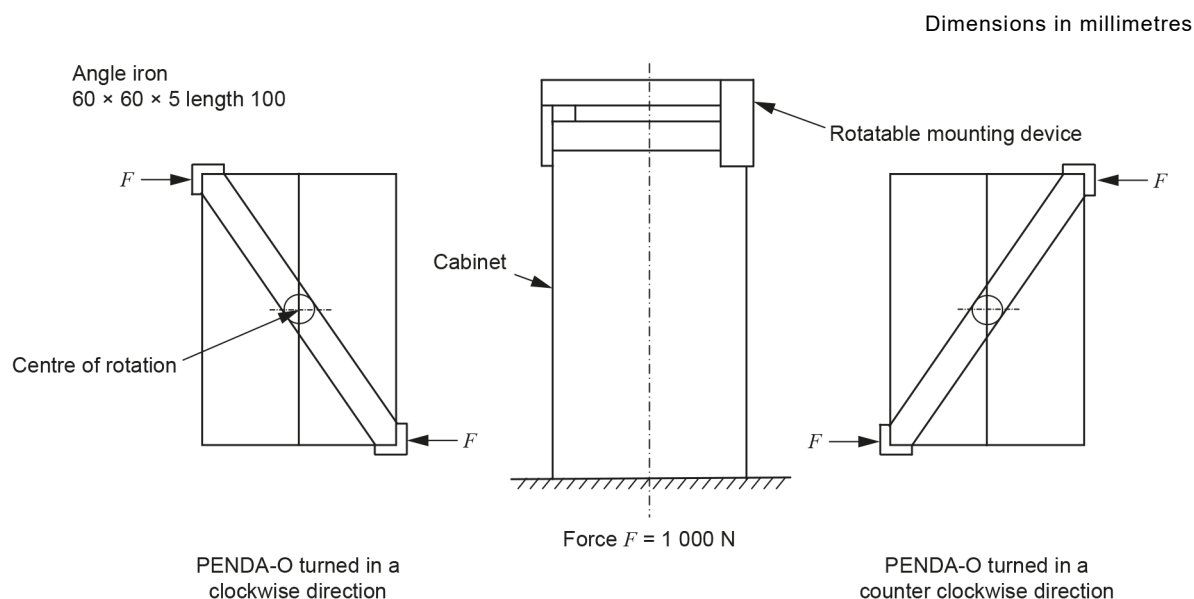
The assembly, with the door(s) closed shall have a torsional force applied for 30 s as shown in Figure 106a and Figure 106b. This is comprised two separate tests, first with the two forces each of 1 000 N applied to twist the PENDA in a clockwise direction and the second with the two forces each of 1 000 N applied to rotate the PENDA in an anti-clockwise direction.

Dimensions in millimetres



IEC

Figure 106a – Diagram of test to verify the resistance to torsional stress of a ground mounted PENDA-O with embedded base



IEC

Figure 106b – Diagram of test to verify resistance to torsional stress of a ground mounted PENDA-O without embedded base

Figure 106 – Diagram of test to verify resistance to torsional stress of a PENDA-O

Compliance is checked by verifying that the door(s) remain closed for the duration of the test and by verification after the test that the degree of protection remains in accordance with 8.2.2.

10.2.101.5 Verification of impact force withstand

10.2.101.5.1 Test applicable to PENDA-Os designed for operation at ambient temperatures of between 40 °C and –25 °C

The test shall be carried out using impact apparatus in the form of a pendulum incorporating a 9 mm external diameter tube at least 1 m long. The pendulum shall be arranged to swing through a vertical arc.

Attached to one end is a solid steel ball of 2 kg mass, which shall be raised through a height of 1 m and allowed to drop and impact the surface of the assembly under test, thus providing an impact energy of 20 J (see Figure 103a and Figure 103b).

For each of the two tests detailed below, the test shall consist of one blow aimed at the centre of each of the vertical surfaces of the assembly which are visible when it is installed in its normal service position. Separate enclosures can be used for each of the test blows.

In the case of an enclosure of cylindrical form, the test shall consist of three blows which are positioned with an angular displacement of 120°.

Test 1 shall be carried out at an ambient air temperature of between 10 °C and 40 °C after the assembly has been kept within these temperatures for not less than 12 h.

Test 2 shall be carried out at an ambient air temperature of between 10 °C and 40 °C immediately after the assembly has been kept at a temperature of $-25_{-0,5}^0$ °C for a period of not less than 12 h. Test 2 is not required when a PENDA has an enclosure made of metallic material.

Compliance is checked by verification after the test that the degree of protection remains in accordance with 8.2.2, and the operation of the door(s) and locking points are not impaired; also by verification that the electrical clearances have remained satisfactory for the duration of the tests and in the case of an assembly having a metallic enclosure, that no contact between live parts and the enclosure has occurred caused by permanent or temporary distortion. In the case of an assembly having an insulating enclosure, if the appropriate conditions are satisfied, then damage such as small dents or small degrees of surface cracking or flaking are disregarded, provided that there are no associated cracks detrimental to the serviceability of the assembly.

10.2.101.5.2 Test applicable to PENDA-Os designed for operation in arctic climate (see 7.1.1)

The specimen shall be kept at a temperature of -50 °C for a period of not less than 12 h. Tests shall be carried out in an ambient air temperature between 10 °C and 40 °C and at a time when the external temperature of the enclosure has recovered to a temperature not higher than -40 °C.

Test sequence:

For test 1 and test 2 a metal test body of spherical or hemispherical shape having a radius of 100 mm ± 3 mm and a surface hardness of HB 160 according to ISO 6506-1:2014 shall be used. Test 2 requires earthing of the test body.

Test 1 shall be carried out on an empty PENDA-O applying a force of 1 500 N for 30 s at the considerably 10 weakest points of the enclosure.

Compliance is checked by verification after the test that the degree of protection remains in accordance with 8.2.2 and the operation of the door(s) and locking points are not impaired.

Test 2 shall be carried out on an assembly containing equipment that provides the minimum clearances inside the enclosure. A metal enclosure shall be connected to earth, in case of an enclosure made of insulating materials all parts within the assembly that are intended to be connected to earth shall be connected to earth.

A force of 1 500 N shall be applied for 30 s to earthed metal test body at the considerably 10 weakest points of the enclosure. For the duration of the test, an AC voltage in accordance with 10.9.2.2 of IEC 61439-1:2020, shall be applied between all live parts connected together and the earth.

Compliance is checked by verification that no puncture or flashovers occurs.

Test 3 shall be carried out on an empty enclosure using an impact apparatus as described in 10.2.101.5.1, but having a solid steel ball with an approximate mass of 15 kg. This striker element shall be raised through a height of approximately 1 m allowed to drop and impact the surface of the assembly under test, to provide an impact energy of 150 J (see Figure 103a and Figure 103b).

The test shall consist of one blow aimed at the centre of each of the vertical surfaces of the assembly which are visible when it is installed in its normal service position. Separate enclosures can be used for each of the test blows.

In the case of an enclosure of cylindrical form, the test shall consist of three blows which are positioned with an angular displacement of 120°.

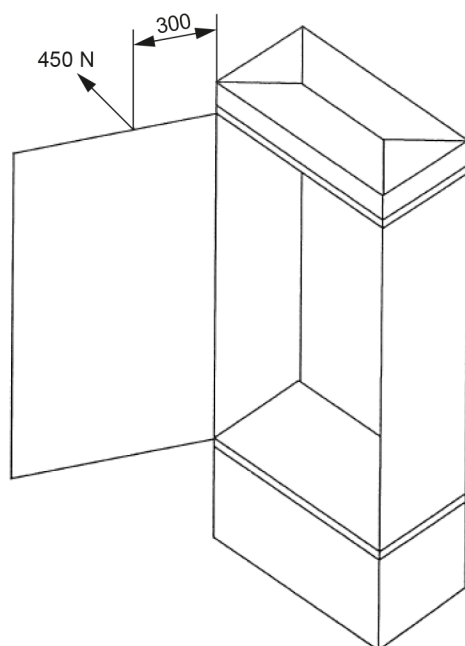
Compliance for Test 3 is checked by verification after the test that the degree of the protection is at least IP3X.

10.2.101.6 Verification of mechanical strength of doors

The test applies to all types of PENDA-O having a door(s) hinged on a vertical edge of the enclosure.

The tests shall be carried out with the door(s) fully open and in contact with the designed restraint. A load of 450 N shall be applied at the top edge of the door perpendicular to the plane of the door(s) and at a distance 300 mm from the hinged edge and maintained for 3 s (see Figure 107).

Dimensions in millimetres



IEC

Figure 107 – Diagram of test to verify the mechanical strength of doors

Compliance is checked by verification that the degree of protection remains in accordance with 8.2.2 after the door(s) has (have) been closed following tests. If the door(s) become unhinged during the test, this is not regarded as a failure if it is possible to reinstall the same door(s) without the use of a tool and the operation of hinges and locking points are not impaired.

10.2.101.7 Verification of resistance to axial load of metal inserts in synthetic material

The test only applies to all types of assembly when threaded metal inserts are provided to retain the mounting plate or switchgear and controlgear supports in place.

The test shall be carried out on a representative specimen of each type and size of metal insert. Also, if there is a difference in the thickness of the profile of the material surrounding a particular insert, the test shall be repeated for this condition.

During the test the assembly shall be fully supported on a platform.

A screw-eye shall be fitted to each insert under test and an axial force in accordance with Table 102 shall be applied for 10 s in an attempt to extract the insert from its anchorage.

Table 102 – Axial load to be applied to the inserts

Size of insert	Axial load N
M4	350
M5	350
M6	500
M8	500
M10	800
M12	800

Compliance is checked by inspection that the inserts remain undamaged and in their original position; and also, that there is no cracking of the surrounding material forming the anchorage for the insert.

NOTE Small cracks, created by air bubbles that were visible before the test, but not affected by the application of the axial load, are ignored.

10.2.101.8 Verification of resistance to mechanical shock impacts induced by sharp-edged objects

The test applies to all types of PENDA-O.

The test shall be carried out using an impact apparatus as described in 10.2.101.5.1 but having a steel striker element with a mass of 5 kg and having the end profiled as shown in Figure 108.

Dimensions in millimetres

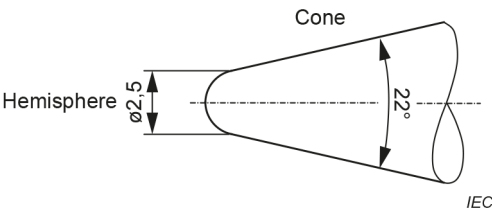


Figure 108 – Striker element for test of resistance to mechanical shock impacts induced by sharp-edged objects

The striker element shall be raised through a height of 0,4 m and allowed to drop and impact the surface of the assembly under test, thus providing an impact energy of 20 J (see Figure 103a and Figure 103b).

Each test shall consist of one blow aimed at the centre of each of the vertical surfaces of the assembly which are visible when the assembly is installed in its normal service position. Separate enclosures can be used for each of the test blows.

In the case of an enclosure of cylindrical form, the test shall consist of three blows which are positioned with an angular displacement of 120°.

Test 1 shall be carried out at an ambient air temperature between 10 °C and 40 °C after the assembly has been kept within these temperatures for not less than 12 h.

Test 2 shall be carried out at an ambient air temperature between 10 °C and 40 °C immediately after the assembly has been kept at a temperature of -25 ± 5 °C for a period of not less than 12 h. Test 2 is not required when a PENDA has an enclosure made of metallic material.

Compliance is checked by inspection that cracks resulting from the blows are contained within a circle of diameter not exceeding 15 mm. If the tip of the striker element has penetrated the enclosure of the assembly, it shall not be possible to insert a gauge of 4 mm diameter having a hemispherical tip, applied to the hole with a force of 5 N.

10.2.101.9 Test of mechanical strength of a base intended to be embedded in the ground

This test is applicable to a PENDA-O only.

The test shall be carried out with the PENDA-O fixed to the base, in accordance with Figure 109 and the manufacturer's installation instructions. A mechanical force is transferred via a thick-walled steel tube and shall be applied to the lowest part of the longest section of the PENDA base that is beneath the ground surface when it is installed.

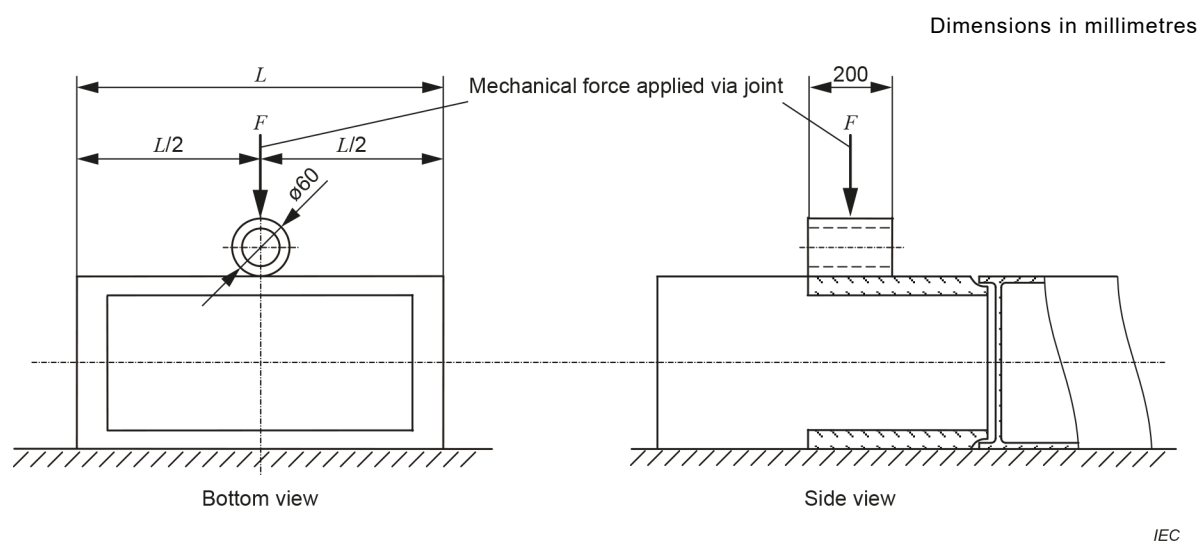


Figure 109 – Typical test arrangement for mechanical strength of base

If the design of the base includes one or more permanent supports (back to front), the force shall be applied by means of a number of steel tubes. One tube shall be placed in the centre of each unsupported length. The individual forces shall simultaneously be applied to each tube and shall be calculated according to the following formula:

$$F = 3,5 \text{ N/mm} \times L$$

where L is the unsupported length in millimetres.

The force(s) shall be applied for 1 min. After this period and whilst the force is still maintained, the degree of protection shall be verified.

If there is another section of the PENDA-O base that is of similar length but has a different profile, the test shall be repeated on this section.

Compliance is checked by inspection that the base has not broken and by verification that the degree of protection of that part of the PENDA-O and base which is normally above ground remains in accordance with 8.2.2.

10.3 Degree of protection of assemblies (IP Code)

Add the following first paragraph:

The declared IP for a PENDA-O does not apply to a base intended to be embedded in the ground. However, the declared IP does apply to the bottom of the assembly.

If a PENDA-O consists of an enclosure and a base (to be embedded in the ground or to be surface mounted) without sealed cable entry, ingress of solid objects through the ground area inside of the base is excluded when determining the degree of protection. When the PENDA is mounted on a base with an open ground area the PENDA and any exposed area of the base shall provide the PENDA with the minimum level of IP protection specified in this document.

NOTE In some instances, additional measure are taken to prohibit the entry of animals/vermin

10.5 Protection against electric shock and integrity of protective circuits

10.5.3.1 General

Replacement of the two paragraphs with the following:

Verification shall be achieved by the application of tests in accordance with 10.5.3.5 of IEC 61439-1:2020.

10.9 Dielectric properties

Additional subclause:

10.9.2.101 Power frequency dielectric test for PENDA's installed in substation with separate HV and LV neutral earths.

When a PENDA is to form part of a distribution substation in which the HV and LV earths are separated an additional dielectric test shall be carried out. A test voltage equal to the HV line to earth voltage shall be applied between the phase plus neutral conductor and the extraneous conductive part of the PENDA for a period of 10 s. Acceptance criteria shall be as given in 10.9.2.4 of IEC 61439-1:2020.

10.9.3 Impulse withstand voltage

10.9.3.1 General

Replacement of the first paragraph with:

Verification shall be achieved by the application of tests in accordance with one of the alternative test methods detailed in 10.9.3.2 to 10.9.3.4, inclusive, of IEC 61439-1:2020.

10.10 Temperature rise

10.10.1 General

Replacement with:

It shall be verified that the temperature-rise limits specified in 9.2 of IEC 61439-1:2020 for the different parts of the assembly will not be exceeded. Verification shall be by test as specified in 10.10.2 of IEC 61439-1:2020.

Temperature rise verification shall be undertaken on a representative sample(s). It is not necessary to test every arrangement produced, but the sample(s) tested shall, with reasonable accuracy, be thermally the most onerous for the range being considered. Assessing the most onerous arrangement(s) shall take into consideration total power loss, rating of devices (incoming and outgoing), sections of conductors, size of the enclosure and the rated current of the assembly.

10.10.2 Verification by testing

10.10.2.2 Selection of the representative arrangement

10.10.2.2.1 General

Addition of the following paragraph:

When the design of a PENDA is suitable for installation in a recess in wall the temperature rise test shall be carried out with insulation to simulate the presence of the wall.

10.11 Short-circuit withstand strength

10.11.1 General

Replacement with:

With the exception of the circuits of assemblies that are exempt verification in accordance with 10.11.2 of IEC 61439-1:2020, the short-circuit withstand strength specified by the manufacturer shall be verified. Verification shall be by means of test as specified in 10.11.5 of IEC 61439-1:2020.

11 Routine verification

This clause of IEC 61439-1:2020 applies.

Annexes

The annexes of IEC 61439-1:2020 are applicable except as follows:

Annexes A, C, D, H, K and M are not applicable.

Modification of Annex L.

Addition of Annexes AA, BB, CC and DD.

Annex L (informative)

Guidance on verification of temperature rise

Modifications:

L.4 Verification assessment

This clause of IEC 61439-1:2020 is not applicable.

L.5 Verification by comparison with a reference design

This clause of IEC 61439-1:2020 is not applicable.

Annex AA (normative)

Cross-section of conductors

Table AA.1 applies for the connection of one cable per terminal.

Table AA.1 – Minimum and maximum cross-section of copper and aluminium conductors, suitable for connection (see 8.8)

Rated current A	Solid or stranded conductors (aluminium or copper) Cross-sections mm ²		Flexible copper conductors Cross sections mm ²	
	Minimum	Maximum	Minimum	Maximum
6	0,75	1,5	0,5	1,5
8	1	2,5	0,75	2,5
10	1	2,5	0,75	2,5
12	1	2,5	0,75	2,5
16	1,5	4	1	4
20	1,5	6	1	4
25	2,5	6	1,5	4
32	2,5	10	1,5	6
40	4	16	2,5	10
63	6	25	6	16
80	10	35	10	25
100	16	50	16	35
125	25	70	25	50
160	35	95	35	70
200	50	150	50	95
250	70	150	70	120
315	70	240	95	185
400	70	240	95	185
500	70	300	95	240
630	70	300	95	240

This table applies for the connection of one conductor per terminal.

If the external conductors are connected directly to built-in apparatus, the cross-sections indicated in the relevant specification are valid.

In cases where it is necessary to provide for conductors other than those specified in the table, agreement shall be reached between the manufacturer and the user.

The approximate relationship between mm² and AWG/kcmil sizes as shown in Table AA.2 is to be used where metric sized round copper conductors are not available.

Table AA.2 – Standard cross-sections of round copper conductors and approximate relationship between mm² and AWG/kcmil sizes (see 8.8 of IEC 61439-1:2020)

Rated cross-section mm ²	AWG/kcmil size	Equivalent metric area mm ²
0,2	24	0,205
0,34	22	0,324
0,5	20	0,519
0,75	18	0,82
1	–	–
1,5	16	1,3
2,5	14	2,1
4	12	3,3
6	10	5,3
10	8	8,4
16	6	13,3
25	4	21,2
35	2	33,6
–	1	42,4
50	0	53,5
70	00	67,4
95	000	85,0
–	0000	107,2
120	250 kcmil	127
150	300 kcmil	152
185	350 kcmil	177
–	400 kcmil	203
240	500 kcmil	253
300	600 kcmil	304
NOTE The dash, when it appears, counts as a size when considering connecting capacity (see 8.8).		

Annex BB (informative)

Items subject to agreement between the assembly manufacturer and the user

The information given in Table BB.1 below is subject to an agreement between the assembly manufacturer and the user. In some cases, information declared by the assembly manufacturer can take the place of an agreement.

**Table BB.1 – Items subject to agreement between
the ASSEMBLY manufacturer and the user**

Characteristics	Reference clause or subclause ^c	Default arrangement ^b	Options listed in document	User requirement ^a
Electrical system				
Earthing system	5.6, 8.4.3.1, 8.4.3.2.3, 8.6.2, 10.5, 11.4	Manufacturer's standard, selected to suit local requirements	TT / TN-C / TN-C-S / IT, TN-S	
Nominal voltage (V)	3.8.9.1, 5.2.1, 8.5.3	Local, according to installation conditions	max 1 000 V AC	
Transient overvoltages	5.2.4, 8.5.3, 9.1, Annex G	Overvoltage category IV	None	
Temporary overvoltages	9.1	Nominal system voltage + 1 200 V	None	
Rated frequency f_n (Hz)	3.8.12, 5.5, 8.5.3, 10.10.3.1, 10.11.5.4	According to local installation conditions	50 Hz/60 Hz	
Additional on-site testing requirements: wiring, operational performance and function	11.10	Manufacturer's standard, according to application	None	
Short-circuit withstand capability				
Prospective short-circuit current at supply terminals I_{cp} (kA)	3.8.7	Determined by the electrical system	None	
Prospective short-circuit current in the neutral	10.11.5.3.5	Max. 60 % of phase values	None	
Prospective short-circuit current in the protective circuit	10.11.5.6	Max. 60 % of phase values	None	
SCPD in the incoming functional unit requirement	9.3.2	According to local installation conditions	Yes / No	
Co-ordination of short-circuit protective devices including external short-circuit protective device details	9.3.4	According to local installation conditions	None	
Data associated with loads likely to contribute to the short-circuit current	9.3.2	No loads likely to make a significant contribution allowed for	None	

Characteristics	Reference clause or subclause ^c	Default arrangement ^b	Options listed in document	User requirement ^a
Protection of persons against electric shock in accordance with IEC 60364-4-41				
Type of protection against electric shock – Basic protection (protection against direct contact)	8.4.2	Basic protection	According to local installation regulations	
Type of protection against electric shock – Fault protection (protection against indirect contact)	8.4.3	According to local installation conditions	Automatic disconnection of supply / Electrical separation / Total insulation	
Installation environment				
Location type	3.5, 8.1.4, 8.2	Manufacturer's standard, according to application	Indoor / outdoor	
Protection against ingress of solid foreign bodies and ingress of water	8.2.2, 8.2.3	Indoor (enclosed): minimum IP2X Outdoor - minimum: IP34D when installed in places accessible to the public. In other locations IP33.		
External mechanical impact NOTE IEC 61439-5 does not nominate specific IK codes.	10.2.101	As Standard (IEC 61439-5)	None	
Resistance to UV radiation (applies for outdoor assemblies only unless specified otherwise)	10.2.4	Indoor: Not applicable. Outdoor: Temperate climate	None	
Resistance to corrosion	10.2.2	Normal Indoor/Outdoor arrangements	None	
Ambient air temperature – Lower limit	7.1.1	Indoor: –5 °C Outdoor: Normal climate –25 °C Arctic climate –50 °C	None	
Ambient air temperature – Upper limit	7.1.1	40 °C	None	
Ambient air temperature – Daily average maximum	7.1.1, 9.2	35 °C	None	
Maximum relative humidity	7.1.1	Indoor: 95 % at –5 °C to +30 °C 70 % at +35 °C 57 % at +40 °C Outdoor: 100 % at –25 °C to +27 °C 60 % at 35 °C 46 % at 40 °C	None	
Pollution degree (of the installation environment)	7.1.2	PENDA-O: 4 PENDA-I: 3	2, 3, 4	

Characteristics	Reference clause or subclause ^c	Default arrangement ^b	Options listed in document	User requirement ^a
Altitude	7.1.1	≤ 2 000 m	None	
EMC environment (A or B)	9.4, 10.12, Annex J	A/B	A/B	
Special service conditions (e.g. vibration, exceptional condensation, heavy pollution, corrosive environment, strong electric or magnetic fields, fungus, small creatures, explosion hazards, heavy vibration and shocks, earthquakes)	7.2, 8.5.4, 9.3.3 Table 7	No special service conditions	Arctic climate	
Installation method				
Type	3.3, 5.6	Manufacturer's standard	Various e.g. ground mounting, transformer mounting, pole mounting, surface wall mounting or mounting within a recess within a wall	
Stationary/Movable	3.5	Stationary	None	
Maximum overall dimensions and weight	5.6, 6.2.1	Manufacturer's standard, according to application	None	
External conductor type(s)	8.8	Cable	Bare or insulated bars.	
Direction(s) of external conductors	8.8	From below	From above	
External conductor material	8.8	Copper/ aluminium	None	
External phase conductor, cross sections, and terminations	8.8	As defined within the document	None	
External PE, N, PEN conductors cross sections, and terminations	8.8	As defined within the document	None	
Special terminal identification requirements	8.8	Manufacturer's standard	None	
Storage and handling				
Maximum dimensions and weight of transport units	6.2.2, 10.2.5	Manufacturer's standard	None	
Methods of transport (e.g. forklift, crane)	6.2.2, 8.1.6	Manufacturer's standard	None	
Environmental conditions different from the service conditions	7.3	As service conditions	None	
Packing details	6.2.2	Manufacturer's standard	None	
Operating arrangements				
Access to manually operated devices	8.4	Authorised persons	None	
Location of manually operated devices	8.5.5	Easily accessible	None	
Isolation of load installation equipment items	8.4.2, 8.4.3.3, 8.4.6.2	Manufacturer's standard	Individual / groups / all	

Characteristics	Reference clause or subclause ^c	Default arrangement ^b	Options listed in document	User requirement ^a
Maintenance and upgrade capabilities				
Requirements related to accessibility for inspection and similar operations	8.4.6.2.2	No requirements for accessibility	None	
Requirements related to accessibility for maintenance in service by authorized persons	8.4.6.2.3	No requirements for accessibility	None	
Requirements related to accessibility for extension in service by authorized persons	8.4.6.2.4	No requirements for accessibility	None	
Method of functional units connection	8.5.1, 8.5.2	Manufacturer's standard	None	
Protection against direct contact with hazardous live internal parts during maintenance or upgrade (e.g. functional units, main busbars, distribution busbars)	8.4	No requirements for protection during maintenance or upgrade	None	
Current carrying capability				
Maximum total load current to be supplied by the assembly (from which the rated current of the assembly I_{nA} (A) will be determined)	3.8.10.1, 5.3, 8.4.3.2.3, 8.5.3, 8.8, 10.10.2, 10.10.3, 10.11.5, Annex E	Manufacturer's standard, according to application	None	
Design current I_B and nature of load for each circuit; alternatively, I_n of the devices and nature of the load (in such cases, the assumed loading factors can be used based on the relevant part of IEC 61439)	3.8.10.8	None	None	
Ratio of cross section of the neutral conductor to phase conductors: phase conductors up to and including 16 mm ²	8.6.1	100 %	None	
Ratio of cross section of the neutral conductor to phase conductors: phase conductors above 16 mm ²	8.6.1	50 % (min. 16 mm ²)	None	
^a For exceptionally onerous applications, the user can specify more stringent requirements to those in the standard. ^b In some cases information declared by the assembly manufacturer can take the place of an agreement. ^c When a reference does not exist in this document, IEC 61439-1:2020 applies.				

Annex CC (informative)

Design verification

Table CC.1 – List of design verifications to be performed

No.	Characteristic to be verified	Clauses or subclauses ^a
1	Strength of material and parts: Resistance to corrosion Properties of insulating materials: Thermal stability Resistance to abnormal heat and fire due to internal electric effects Category of flammability Resistance to ultra-violet (UV) radiation Lifting Mechanical impact Resistance to static load Resistance to shock load Resistance to torsional stress Impact force withstand Mechanical strength of doors Resistance to axial load of metal inserts in synthetic material Resistance to mechanical shock induced by sharp objects Mechanical strength of a base intended to be embedded in the ground Marking	10.2 10.2.2 10.2.3 10.2.3.1 10.2.3.2 10.2.3.101 10.2.4 10.2.5 10.2.101 10.2.101.2 10.2.101.3 10.2.101.4 10.2.101.5 10.2.101.6 10.2.101.7 10.2.101.8 10.2.101.9 10.2.7
	Mechanical operation	10.2.8
2	Degree of protection of enclosures	10.3
3	Clearances	10.4
4	Creepage distances	10.4
5	Protection against electric shock and integrity of protective circuits: Effective continuity between the exposed conductive parts of the assembly and the protective circuit Short-circuit withstand strength of the protective circuit	10.5 10.5.2 10.5.3
6	Incorporation of switching devices and components	10.6
7	Internal electrical circuits and connections	10.7
8	Terminals for external conductors	10.8
9	Dielectric properties: Power-frequency withstand voltage Impulse withstand voltage	10.9 10.9.2 10.9.3
10	Temperature-rise limits	10.10
11	Short-circuit withstand strength	10.11
12	Electromagnetic compatibility (EMC)	10.12
^a When a reference does not exist in this document, IEC 61439-1:2020 applies.		

Annex DD
(informative)

List of notes concerning certain countries

Subclause	Text
8.8	Add the following note after the last paragraph: NOTE In the United States of America (USA) the conductor size requirement is dependent on the current rating, conductor insulation temperature rating, ambient temperature and configuration, along with conductor insulation type. The specific requirements are contained in the US National Electrical Code (NEC) NFPA 70 Chapter 3.

Bibliography

The bibliography of IEC 61439-1:2020 is applicable except as follows:

Addition:

IEC 60050 (all parts), *International Electrotechnical Vocabulary (IEV)*, available at www.electropedia.org

SOMMAIRE

AVANT-PROPOS	41
1 Domaine d'application	43
2 Références normatives	44
3 Termes et définitions	45
4 Symboles et abréviations	46
5 Caractéristiques d'interface	46
6 Informations	46
7 Conditions d'emploi	47
8 Exigences de construction	48
9 Exigences de performance	50
10 Vérification de conception.....	50
11 Vérification individuelle de série	65
Annexes	66
Annexe L (informative) Recommandations concernant la vérification de l'échauffement	67
Annexe AA (normative) Section des conducteurs	68
Annexe BB (informative) Sujets soumis à accord entre le fabricant d'ensembles et l'utilisateur	70
Annexe CC (informative) Vérification de conception.....	75
Annexe DD (informative) Liste des notes concernant certains pays.....	76
Bibliographie.....	77
Figure 101 – Réseau de distribution classique	44
Figure 102 – Schéma de l'essai de vérification de la résistance au choc réparti d'un ERD-E	53
Figure 103 – Schéma de l'essai de vérification de la résistance à l'impact d'un ERD-E	54
Figure 104 – Schéma de l'essai de vérification de la résistance à la charge statique	55
Figure 105 – Sac de sable pour l'essai de vérification de la résistance au choc réparti	56
Figure 106 – Schéma de l'essai de vérification de la résistance à la torsion d'un ERD-E.....	58
Figure 107 – Schéma de l'essai de vérification de la tenue mécanique des portes	61
Figure 108 – Pièce de frappe pour l'essai de résistance aux impacts mécaniques d'objets pointus.....	62
Figure 109 – Disposition d'essai type pour la résistance mécanique du socle	63
Tableau 101 – Valeurs de charge présumée	46
Tableau 102 – Charge axiale à appliquer aux inserts	62
Tableau AA.1 – Sections minimales et maximales des conducteurs en cuivre et en aluminium, adaptées aux raccordements (voir 8.8)	68
Tableau AA.2 – Sections normalisées de conducteurs de cuivre circulaires et relation approximative entre les dimensions en mm ² et AWG/kcmil (voir 8.8 de l'IEC 61439-1:2020)	69
Tableau BB.1 – Sujets soumis à accord entre le fabricant d'ENSEMBLES et l'utilisateur.....	70
Tableau CC.1 – Liste de vérifications de conception à réaliser.....	75

COMMISSION ÉLECTROTECHNIQUE INTERNATIONALE

ENSEMBLES D'APPAREILLAGE À BASSE TENSION –

Partie 5: Ensembles pour réseaux de distribution publique

AVANT-PROPOS

- 1) La Commission Électrotechnique Internationale (IEC) est une organisation mondiale de normalisation composée de l'ensemble des comités électrotechniques nationaux (Comités nationaux de l'IEC). L'IEC a pour objet de favoriser la coopération internationale pour toutes les questions de normalisation dans les domaines de l'électricité et de l'électronique. À cet effet, l'IEC – entre autres activités – publie des Normes internationales, des Spécifications techniques, des Rapports techniques, des Spécifications accessibles au public (PAS) et des Guides (ci-après dénommés "Publication(s) de l'IEC"). Leur élaboration est confiée à des comités d'études, aux travaux desquels tout Comité national intéressé par le sujet traité peut participer. Les organisations internationales, gouvernementales et non gouvernementales, en liaison avec l'IEC, participent également aux travaux. L'IEC collabore étroitement avec l'Organisation Internationale de Normalisation (ISO), selon des conditions fixées par accord entre les deux organisations.
- 2) Les décisions ou accords officiels de l'IEC concernant les questions techniques représentent, dans la mesure du possible, un accord international sur les sujets étudiés, étant donné que les Comités nationaux de l'IEC intéressés sont représentés dans chaque comité d'études.
- 3) Les Publications de l'IEC se présentent sous la forme de recommandations internationales et sont agréées comme telles par les Comités nationaux de l'IEC. Tous les efforts raisonnables sont entrepris afin que l'IEC s'assure de l'exactitude du contenu technique de ses publications; l'IEC ne peut pas être tenue responsable de l'éventuelle mauvaise utilisation ou interprétation qui en est faite par un quelconque utilisateur final.
- 4) Dans le but d'encourager l'uniformité internationale, les Comités nationaux de l'IEC s'engagent, dans toute la mesure possible, à appliquer de façon transparente les Publications de l'IEC dans leurs publications nationales et régionales. Toutes divergences entre toutes Publications de l'IEC et toutes publications nationales ou régionales correspondantes doivent être indiquées en termes clairs dans ces dernières.
- 5) L'IEC elle-même ne fournit aucune attestation de conformité. Des organismes de certification indépendants fournissent des services d'évaluation de conformité et, dans certains secteurs, accèdent aux marques de conformité de l'IEC. L'IEC n'est responsable d'aucun des services effectués par les organismes de certification indépendants.
- 6) Tous les utilisateurs doivent s'assurer qu'ils sont en possession de la dernière édition de cette publication.
- 7) Aucune responsabilité ne doit être imputée à l'IEC, à ses administrateurs, employés, auxiliaires ou mandataires, y compris ses experts particuliers et les membres de ses comités d'études et des Comités nationaux de l'IEC, pour tout préjudice causé en cas de dommages corporels et matériels, ou de tout autre dommage de quelque nature que ce soit, directe ou indirecte, ou pour supporter les coûts (y compris les frais de justice) et les dépenses découlant de la publication ou de l'utilisation de cette Publication de l'IEC ou de toute autre Publication de l'IEC, ou au crédit qui lui est accordé.
- 8) L'attention est attirée sur les références normatives citées dans cette publication. L'utilisation de publications référencées est obligatoire pour une application correcte de la présente publication.
- 9) L'attention est attirée sur le fait que certains des éléments de la présente Publication de l'IEC peuvent faire l'objet de droits de brevet. L'IEC ne saurait être tenue pour responsable de ne pas avoir identifié de tels droits de brevets.

L'IEC 61439-5 a été établie par le sous-comité 121B: Ensembles d'appareillages à basse tension, du comité d'études 121 de l'IEC: Appareillages et ensembles d'appareillages basse tension. Il s'agit d'une Norme internationale.

Cette troisième édition annule et remplace la deuxième édition parue en 2014. Cette édition constitue une révision technique.

Cette édition inclut les modifications techniques majeures suivantes par rapport à l'édition précédente:

- a) omission de l'exigence de réalisation d'essais mécaniques à -25 °C lorsque les enveloppes sont en matériau métallique;
- b) ajout d'application de charge de véhicules électriques et de production d'énergie avec des facteurs de charge présumés;

- c) ajout d'essais diélectriques lorsqu'un ERD est utilisé dans un poste avec terres HT et BT séparées;
- d) description plus détaillée des échantillons représentatifs pour la vérification de la conception.

Le texte de cette Norme internationale est issu des documents suivants:

Projet	Rapport de vote
121B/173/FDIS	121B/178/RVD

Le rapport de vote indiqué dans le tableau ci-dessus donne toute information sur le vote ayant abouti à son approbation.

La langue employée pour l'élaboration de cette Norme internationale est l'anglais.

Ce document a été rédigé selon les Directives ISO/IEC, Partie 2, il a été développé selon les Directives ISO/IEC, Partie 1 et les Directives ISO/IEC, Supplément IEC, disponibles sous www.iec.ch/members_experts/refdocs. Les principaux types de documents développés par l'IEC sont décrits plus en détail sous www.iec.ch/publications.

Le présent document doit être lu conjointement avec l'IEC 61439-1. Les dispositions des règles générales traitées dans l'IEC 61439-1 s'appliquent uniquement au présent document lorsqu'elles sont spécifiquement citées. Lorsque le présent document mentionne "addition", "modification" ou "remplacement", le texte correspondant de l'IEC 61439-1:2020 doit être adapté en conséquence. Les paragraphes qui sont numérotés avec un suffixe 101 (102, 103, etc.) sont ajoutés au même paragraphe de l'IEC 61439-1:2020.

Les nouveaux tableaux et nouvelles figures de l'IEC 61439-5:2023 sont numérotés à partir de 101.

Les nouvelles annexes de l'IEC 61439-5:2023 sont désignées AA, BB, etc.

L'attention du lecteur est attirée sur le fait que l'Annexe DD énumère tous les articles qui traitent des différences à caractère moins permanent inhérentes à certains pays, concernant le sujet du présent document.

Une liste de toutes les parties de la série IEC 61439, publiées sous le titre général *Ensembles d'appareillage à basse tension*, se trouve sur le site web de l'IEC.

Le comité a décidé que le contenu de ce document ne sera pas modifié avant la date de stabilité indiquée sur le site web de l'IEC sous webstore.iec.ch dans les données relatives au document recherché. À cette date, le document sera

- reconduit,
- supprimé,
- remplacé par une édition révisée, ou
- amendé.

ENSEMBLES D'APPAREILLAGE À BASSE TENSION –

Partie 5: Ensembles pour réseaux de distribution publique

1 Domaine d'application

Le présent document définit les exigences spécifiques aux ensembles pour réseaux de distribution publique d'électricité (ERD).

Les ERD remplissent les critères suivants:

- ils sont utilisés pour la distribution de l'énergie électrique dans des systèmes triphasés pour lesquels la tension assignée ne dépasse pas 1 000 V en courant alternatif (se reporter à la Figure 101 qui représente un réseau de distribution classique) et des systèmes à courant continu qui ne dépassent pas 1 500 V en courant continu;
- ils sont fixes;
- les ensembles ouverts ne sont pas couverts par le présent document;
- ils sont adaptés à une installation dans des emplacements où seules des personnes qualifiées ont accès pour leur utilisation; cependant, les types pour l'extérieur peuvent être installés dans des endroits accessibles à des personnes ordinaires;
 - ils sont destinés à être utilisés dans les réseaux de distribution publique;
 - à l'intérieur: ensembles installés à l'intérieur de postes d'alimentation électrique;
 - à l'extérieur: ensembles avec une enveloppe adaptée à une installation en plein air.

L'objet du présent document est de fournir les définitions et de spécifier les conditions d'emploi, les exigences de construction, les caractéristiques techniques et les essais pour les ERD. Des essais à des niveaux de performance supérieurs peuvent s'appliquer pour certains paramètres de réseau.

Les ERD peuvent également inclure des dispositifs de commande et de signalisation associés à la distribution de l'énergie électrique.

NOTE 1 Des dispositifs de commande et de surveillance peuvent être utilisés dans les applications de réseau intelligent ou pour la transmission de données de réseau intelligent.

Le présent document s'applique à tous les ERD qu'ils soient conçus, fabriqués et vérifiés à l'unité ou qu'ils soient complètement normalisés et fabriqués en quantité.

La fabrication et/ou l'assemblage peuvent être réalisés par un tiers qui n'est pas le fabricant d'origine (voir 3.10.1 de l'IEC 61439-1:2020).

Le présent document ne s'applique pas aux appareils individuels et aux composants indépendants, tels que démarreurs de moteurs, fusibles-interrupteurs, matériels électroniques, etc. qui sont conformes aux normes de produit applicables.

Si le poste appartient à (ou est exploité par) un opérateur de réseau de distribution public, les ERD utilisés comme tableaux de distribution basse tension dans les postes de transformation sont inclus dans le domaine d'application du présent document.

Le présent document ne s'applique pas aux types d'ensembles spécifiques qui sont couverts par d'autres parties de la série IEC 61439.

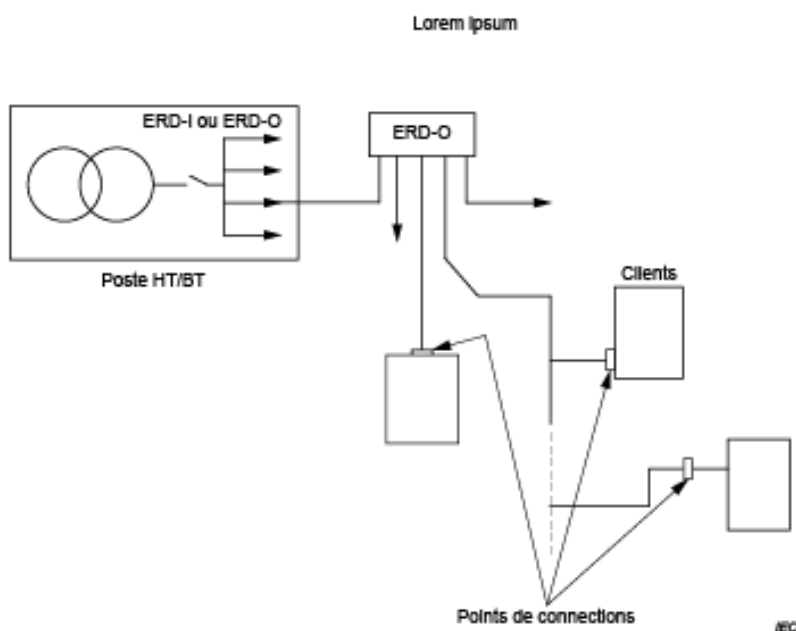


Figure 101 – Réseau de distribution classique

NOTE 2 Si un ERD comporte des matériels complémentaires (par exemple des compteurs), de telle sorte que sa fonction principale est considérablement modifiée, d'autres normes peuvent alors également s'appliquer selon accord entre utilisateur et fabricant (voir 8.5 de l'IEC 61439-1:2020).

NOTE 3 Si les pratiques et règlements locaux le permettent, un ERD conforme au présent document peut être utilisé sur des réseaux autres que publics.

NOTE 4 Les opérateurs de réseau de distribution peuvent définir des exigences supplémentaires pour leurs ERD.

2 Références normatives

L'article de l'IEC 61439-1:2020 s'applique, avec les exceptions suivantes.

Addition:

IEC 60695-11-10:2013, *Essais relatifs aux risques du feu – Partie 11-10: Flammes d'essai – Méthodes d'essai horizontal et vertical à la flamme de 50 W*

IEC 61439-1:2020, *Ensembles d'appareillage à basse tension – Partie 1: Règles générales*

IEC 62262, *Degrés de protection procurés par les enveloppes de matériels électriques contre les impacts mécaniques externes (code IK)*

ISO 9223:2012, *Corrosion des métaux et alliages – Corrosivité des atmosphères – Classification, détermination et estimation*

ISO 6506-1:2014, *Matériaux métalliques – Essai de dureté Brinell – Partie 1: Méthode d'essai*

3 Termes et définitions

L'article de l'IEC 61439-1:2020 s'applique, avec les exceptions suivantes.

3.1 Termes généraux

Termes supplémentaires:

3.1.101

ensemble pour réseaux de distribution publique

ERD

ensemble, généralement destiné à être installé dans un réseau électrique public, qui en exploitation reçoit de l'énergie électrique d'une ou de plusieurs sources et distribue cette énergie par l'intermédiaire d'un ou de plusieurs câbles vers un autre matériel

Note 1 à l'article: Un ERD est installé, mis en fonctionnement et entretenu uniquement par des personnes qualifiées.

Note 2 à l'article: Certains types d'ERD étaient précédemment désignés sous le nom de tableaux de distribution par câbles pour postes (SCDB).

3.1.101.1

ensemble pour réseaux de distribution publique pour l'extérieur

ERD-E

ensemble pour réseaux de distribution publique en armoire adapté à une installation à l'extérieur dans des endroits qui peuvent ou non être accessibles au public

3.1.101.2

ensemble pour réseaux de distribution publique pour l'intérieur

ERD-I

ensemble pour réseaux de distribution publique adapté à une installation à l'intérieur, généralement sans enveloppe, mais comprenant toutes les parties structurelles nécessaires pour supporter les jeux de barres, les unités fonctionnelles et autres dispositifs auxiliaires, nécessaires pour réaliser l'ensemble

3.1.102

durée de vie prévue à la conception

durée de vie spécifiée

durée minimale pendant laquelle les caractéristiques de performance spécifiées pour un matériel sont attendues lorsque le matériel est mis en fonctionnement comme prévu et régulièrement entretenu par des personnes averties conformément aux instructions du fabricant

[SOURCE: IEC 60050-395:2014, 395-07-93, modifié – Note à l'article omise et "garanties" remplacé par "attendues lorsque instructions du fabricant".]

3.3 Présentation extérieure des ensembles

3.3.1

ensemble ouvert

Le terme de l'IEC 61439-1:2020 ne s'applique pas.

3.9 Vérification

Modifications:

3.9.1

vérification de la conception

Supprimer la note.

3.9.1.2

comparaison de vérification

Le terme de l'IEC 61439-1:2020 ne s'applique pas.

3.9.1.3

évaluation de vérification

Le terme de l'IEC 61439-1:2020 ne s'applique pas.

4 Symboles et abréviations

L'article de l'IEC 61439-1:2020 s'applique.

5 Caractéristiques d'interface

L'article de l'IEC 61439-1:2020 s'applique, avec l'exception suivante.

5.4 Facteur de diversité assigné (RDF)

Addition:

En l'absence d'un accord entre le fabricant d'ensembles et l'utilisateur concernant les courants de charge réels, la charge supposée des circuits de départ de l'ensemble ou du groupe des circuits de départ peut être fondée sur les valeurs du Tableau 101.

Pour les circuits de distribution et les circuits terminaux, il est admis par hypothèse que le courant de charge est le courant assigné du dispositif de protection, I_n , exigé par l'utilisateur, multiplié par le facteur de charge du Tableau 101.

Tableau 101 – Valeurs de charge présumée

Application	Facteur de charge présumé
Connexions normales au réseau de distribution 2 à 3 circuits	0,9
Connexions normales au réseau de distribution 4 à 5 circuits	0,8
Connexions normales au réseau de distribution 6 à 9 circuits	0,7
Connexions normales au réseau de distribution ≥ 10 circuits	0,6
Production d'énergie (par exemple photovoltaïque, centrale éolienne, biomasse)	1,0
Infrastructure de charge pour VE	1,0

6 Informations

L'article de l'IEC 61439-1:2020 s'applique, avec les exceptions suivantes.

6.1 Marquage pour l'identification des ensembles

Addition au premier alinéa:

Les plaques signalétiques peuvent être placées à l'intérieur de l'enveloppe d'un ensemble à condition que leur place prévue assure une bonne lisibilité et une bonne visibilité lorsque la ou les portes sont ouvertes ou que le capot est enlevé.

Remplacement du point g):

g) IEC 61439-5.

6.3 Identification des appareils et/ou des composants

Alinéa supplémentaire:

Dans le cas de fusibles dont le porte-fusible spécifique est amovible, une étiquette doit être placée sur le porte-fusible ainsi que sur le socle pour éviter l'échange incorrect du porte-fusible.

Paragraphe supplémentaire:

6.101 Identification de circuit

Il doit être possible d'identifier chaque unité fonctionnelle d'une manière clairement visible.

7 Conditions d'emploi

L'article de l'IEC 61439-1:2020 s'applique, avec les exceptions suivantes.

7.1 Conditions normales d'emploi

7.1.1 Conditions climatiques

Addition au premier alinéa:

La limite inférieure de la température de l'air ambiant est de -25 °C , comme cela est spécifié dans le Tableau 15 de l'IEC 61439-1:2020, à moins que l'utilisateur ne spécifie qu'un ERD doit être adapté pour être utilisé sous un climat arctique. Pour un climat arctique, la limite inférieure de la température ambiante est de -50 °C .

7.1.2 Degré de pollution

Remplacer le premier alinéa par le texte suivant:

Le degré de pollution dont il est question à l'Annexe BB est la condition de macroenvironnement pour laquelle l'ensemble est prévu.

7.2 Conditions spéciales d'emploi

Alinéa supplémentaire:

Des exigences complémentaires pour l'installation d'un ERD-E dans un endroit soumis à de fortes chutes de neige et avoisinant des zones où la neige est dégagée par chasse-neige font l'objet d'un accord entre le fabricant et l'utilisateur.

8 Exigences de construction

L'article de l'IEC 61439-1:2020 s'applique, avec les exceptions suivantes.

8.1 Résistance des matériaux et des parties

8.1.1 Généralités

Remplacer la référence à l'Annexe C par l'Annexe BB au 5^e alinéa du 8.1.1 de l'IEC 61439-1:2020.

Addition:

Un ERD-E doit convenir pour une fixation au sol, un montage sur transformateur, un montage sur poteau, une fixation en saillie sur mur ou un encastrement dans un mur selon l'accord fixé entre l'utilisateur et le fabricant.

Un ERD peut être couplé directement à un transformateur au moyen d'un couplage à brides ou il peut être connecté à son alimentation au moyen d'un câble ou de jeux de barres selon l'accord fixé entre l'utilisateur et le fabricant. Les circuits de départ doivent être adaptés à une connexion au moyen de câbles.

Un système de verrouillage doit être prévu sur les enveloppes pour installation à l'extérieur afin d'interdire l'accès aux personnes non autorisées. Les fixations de tous les capots, etc., qui sont amovibles pour les opérations d'installation ou d'entretien doivent être accessibles seulement lorsque la ou les portes sont ouvertes.

Tous les matériels auxiliaires (par exemple compteurs, relais, instruments, unités de déclenchement de disjoncteur, matériels de communication) qui peuvent être facilement remplacés sont exclus de la durée de vie minimale prévue à la conception d'un ERD.

NOTE Le cas échéant, la durée de vie prévue à la conception peut être fixée par accord entre l'utilisateur et le fabricant, en prenant pour hypothèse qu'il est mis en fonctionnement comme prévu et régulièrement entretenu par des personnes averties conformément aux instructions du fabricant.

8.1.3.2 Résistance des matériaux isolants à la chaleur et au feu

Paragraphe supplémentaire:

8.1.3.2.101 Vérification du classement au feu

Les matériaux isolants utilisés pour les enveloppes, les écrans et autres parties isolantes doivent avoir des propriétés d'inflammabilité conformes au 10.2.3.101.

8.1.5 Résistance mécanique

Paragraphe supplémentaire:

8.1.5.101 Vérification de la résistance mécanique

Les propriétés mécaniques d'un ERD-E doivent être conformes au 10.2.101.

Les parties de l'ERD-E prévues pour être encastrées dans le sol doivent résister aux contraintes qui leur sont imposées pendant l'installation et en usage normal, et satisfaire aux exigences du 10.2.101.9.

8.2 Degré de protection procuré par l'enveloppe d'un ensemble

8.2.1 Protection contre les impacts mécaniques (code IK)

Le 8.2.1 de l'IEC 61439-1:2020 ne s'applique pas.

Les essais d'impacts mécaniques exigés par le présent document équivalent au moins au code IK10 selon l'IEC 62262 (voir 8.1.5.101).

8.2.2 Protection contre les contacts avec des parties actives, contre la pénétration de corps étrangers solides et d'eau (code IP)

Addition:

Les ensembles ouverts (IP00) ne sont pas couverts par le présent document.

Lorsqu'un ERD-E est destiné à être installé dans des endroits accessibles au public, son enveloppe doit, après installation complète conformément aux instructions du fabricant, procurer un degré de protection au moins égal à IP34D, conformément à l'IEC 60529. Dans d'autres endroits, le niveau minimal de protection doit être au moins égal à IP33.

Les ERD-E destinés à être installés dans des endroits accessibles au public doivent, sauf indication contraire de l'utilisateur, être conçus de telle sorte que, lorsque tous les câbles temporaires sont connectés, l'enveloppe doit procurer un degré de protection au moins égal à IP23C, conformément à l'IEC 60529. Voir 8.8.

8.4.2.1 Généralités

Le troisième alinéa ne s'applique pas.

Paragraphe supplémentaire:

8.4.2.101 Moyens de mise à la terre et de mise en court-circuit

Lorsque cela est spécifié par l'utilisateur, les unités de départ d'un ensemble doivent être construites de telle sorte qu'elles puissent être mises à la terre et mises en court-circuit de façon sûre, à l'aide d'un ou de plusieurs dispositifs recommandés par le fabricant, qui assurent la conservation du degré de protection (code IP) exigé pour toutes les parties de l'ensemble.

8.4.3.1 Conditions d'installation

Alinéa supplémentaire:

Lorsqu'un utilisateur spécifie que l'ensemble doit alimenter un ou plusieurs câbles de lignes aériennes, les unités de départ doivent être conçues de telle manière que chaque câble fixé puisse être mis à la terre à ses terminaisons.

8.8 Bornes pour câbles externes

Remplacement des trois premiers alinéas par les suivants:

En l'absence d'un accord particulier entre l'utilisateur et le fabricant, les bornes doivent être capables de recevoir les câbles à âme en cuivre ou en aluminium de la plus petite à la plus grande section qui correspondent au courant assigné approprié (voir Tableau AA.1).

Les terminaisons pour les circuits de départ doivent être situées de sorte qu'un espacement approprié soit prévu et de manière à faciliter la terminaison des conducteurs de phase d'un câble quel que soit leur pas.

Lorsque cela est spécifié par l'utilisateur, les bornes des câbles de phase de chaque circuit de départ doivent être séparées de toutes les autres parties actives dangereuses. Si les bornes d'un circuit de départ sont exposées, la protection contre les contacts accidentels avec d'autres parties actives dangereuses doit être assurée. La séparation et la protection doivent être assurées dans le sens d'accès normal et conformément au code IPXXB de l'IEC 60529.

Si l'utilisateur le spécifie, le circuit d'arrivée doit convenir pour la connexion au moyen de barres nues ou isolées.

Paragraphes supplémentaires:

8.101 Marquage comme obstacle au dégagement de la neige

Lorsque cela est spécifié par l'utilisateur, un ERD-E destiné à être utilisé dans des régions soumises à de fortes chutes de neige selon 7.2 doit être signalé comme obstacle au dégagement de la neige. Des fixations doivent être prévues, solidaires de l'ERD-E, pour disposer des perches de repérage; il doit être possible d'installer et de régler la position des perches de repérage de l'extérieur de l'ERD-E.

8.102 Facilité de fonctionnement et maintenance

Toutes les parties de l'ensemble doivent, autant que possible en pratique, être facilement accessibles et remplaçables sans démontage excessif.

La conception doit être telle que les câbles puissent être facilement raccordés par l'avant.

Lorsqu'un ERD ne comporte pas de dispositif de mesure incorporé, il doit être possible, en utilisant un instrument portatif, de mesurer facilement et en toute sécurité les tensions dans toutes les phases des unités d'arrivée et des deux côtés de tous les appareils de coupure de courant et/ou de connexion des unités de départ, ainsi que le courant dans une phase de toutes les unités de départ. Pendant cette opération, toutes les parties actives de l'ERD doivent être suffisamment protégées pour conserver le degré de protection exigé conformément au 8.2. Des instructions concernant la procédure à adopter doivent être fournies par le fabricant.

Si l'ensemble est prévu pour être raccordé à une source de puissance de réserve sous tension, par exemple un générateur de secours, le dispositif de raccordement de l'appareillage de connexion doit être conçu de sorte que la connexion puisse être réalisée avec les parties actives qui procurent un degré de protection IPXXB conformément à l'IEC 60529.

9 Exigences de performance

L'article de l'IEC 61439-1:2020 s'applique.

10 Vérification de conception

L'article de l'IEC 61439-1:2020 s'applique, avec les exceptions suivantes.

10.1 Généralités

Remplacer les 6^e, 7^e et 8^e alinéas par:

La vérification de conception doit être réalisée uniquement par le biais d'essais conformément à l'Article 10. Les autres méthodes de vérification par évaluation ou comparaison à une conception de référence ne doivent pas être utilisées (voir Tableau CC.1).

Il n'est pas nécessaire de soumettre à l'essai toutes les configurations produites. Les essais effectués sur un échantillon représentatif des ERD les plus exigeants sont présumés vérifier la performance des ensembles analogues et moins exigeants de mêmes constructions générales et valeurs assignées. Exemples d'essais effectués sur l'échantillon représentatif le plus exigeant:

- 1) un essai d'échauffement effectué sur un ERD 800 A avec 5 circuits de départ est présumé s'appliquer à un ERD 800 A de même construction (même enveloppe générale, même conception des jeux de barres et même(s) unité(s) d'arrivée) avec 8 circuits de départ de même conception;
- 2) des essais de résistance mécanique effectués sur la plus petite et la plus grande enveloppe d'une série sont présumés s'appliquer à toutes les autres enveloppes de la série, si elles présentent le même nombre de portes et si leur construction générale est identique;
- 3) si la disposition des jeux de barres et des supports de connexion est identique, des essais de court-circuit effectués sur une conception de jeux de barres et de connexions sont présumés s'appliquer aux jeux de barres et connexions de courants assignés plus élevés;
- 4) un essai de court-circuit effectué sur un ERD 800 A avec 8 circuits de départ est présumé s'appliquer à un ERD 800 A de même construction (même enveloppe générale, même conception des jeux de barres et même(s) unité(s) d'arrivée) avec 5 circuits de départ de même conception;
- 5) pour un courant assigné de court-circuit donné, des essais de court-circuit effectués sur un circuit d'arrivée ou de départ sont présumés s'appliquer aux circuits de courants assignés plus élevés de mêmes conception et construction.

Alinéa final supplémentaire:

Si nécessaire, pour s'adapter à leurs paramètres particuliers de réseau, les utilisateurs peuvent spécifier des exigences d'essais plus sévères ou complémentaires.

10.2 Résistance des matériaux et des parties

10.2.2 Résistance à la corrosion

10.2.2.1 Vérification par essai

Remplacement du dernier alinéa par le suivant:

Si les propriétés de résistance à la corrosion et de durée de vie, selon accord entre fabricant et utilisateur, peuvent être confirmées par référence à l'ISO 9223:2012, il n'est pas nécessaire de réaliser les essais ci-dessous.

Dans tous les autres cas, la résistance à la corrosion de chaque conception d'ensemble doit être vérifiée par l'essai de sévérité A ou B, selon le cas, décrit en 10.2.2.3 de l'IEC 61439-1:2020 et en 10.2.2.2.

10.2.2.2 Essai de sévérité A

Remplacement des spécifications d'essai (alinéa 2) par les suivantes:

Essai cyclique de chaleur humide de l'IEC 60068-2-30:2005: Sévérité – température 55 °C, 6 cycles et variante 1.

À la fin de l'essai, les échantillons sont retirés de l'enceinte d'essai.

La conformité est vérifiée par examen visuel. Les parties soumises à l'essai ne doivent pas présenter de rouille, de fissures ni autres détériorations. Toutefois, une corrosion de surface de la couche de protection est admise.

10.2.2.4 Résultats à obtenir

Le paragraphe de l'IEC 61439-1:2020 ne s'applique pas en ce qui concerne les essais effectués conformément au 10.2.2.2.

10.2.3 Propriétés des matériaux isolants

Paragraphe supplémentaire:

10.2.3.101 Vérification du classement au feu

Des échantillons représentatifs de chacun des matériaux des enveloppes, écrans et autres parties isolantes doivent être soumis à l'essai d'inflammabilité selon la méthode d'essai A (essai de combustion horizontale) de l'IEC 60695-11-10:2013.

La conformité est vérifiée par un examen qui montre que chaque lot d'échantillons peut être classé en catégorie HB40, critère a) ou b), conformément au 8.4.3 de l'IEC 60695-11-10:2013.

10.2.6 Vérification de la protection contre les impacts mécaniques (code IK)

Le paragraphe de l'IEC 61439-1:2020 ne s'applique pas aux ensembles conformes au présent document.

Paragraphe supplémentaire:

10.2.101 Vérification de la résistance mécanique

10.2.101.1 Généralités

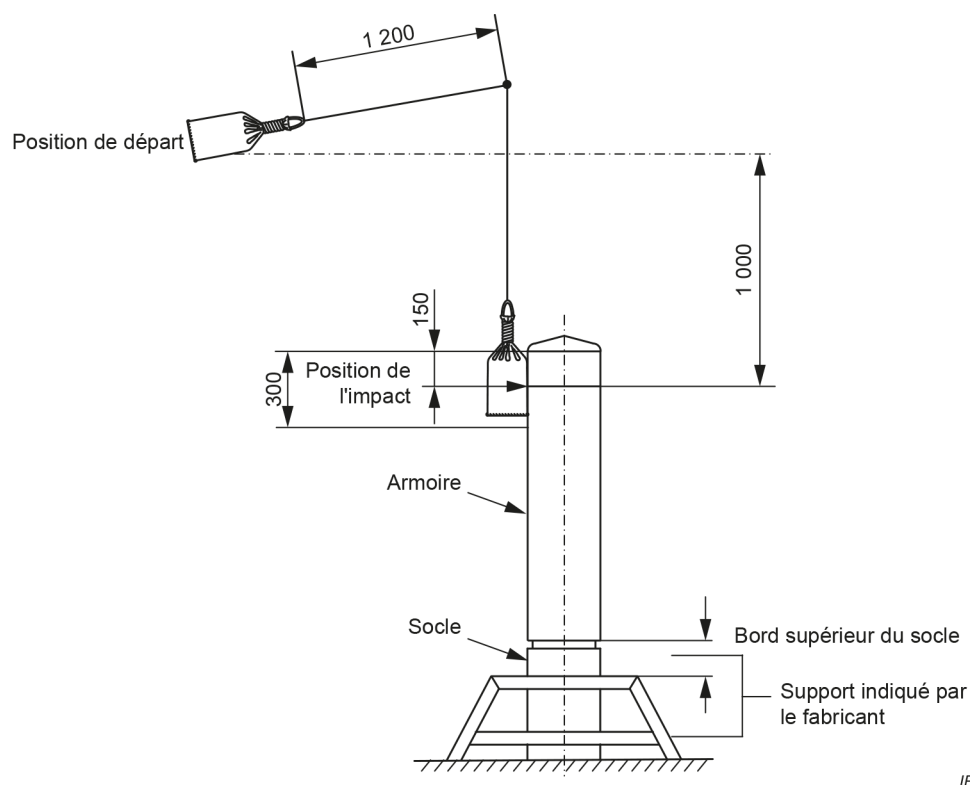
Lorsque des enveloppes sont fabriquées en série et sont de mêmes conception et construction, seules la plus petite et la plus grande taille d'enveloppe doivent être soumises aux essais.

Les essais doivent être effectués à une température ambiante comprise entre 10 °C et 40 °C.

À l'exception de l'essai du 10.2.101.7, un nouvel échantillon d'ensemble peut être utilisé pour chacun des essais indépendants. Si le même échantillon d'ensemble est utilisé pour plusieurs essais du 10.2.101, il n'est nécessaire de vérifier la conformité pour le deuxième chiffre du degré de protection (code IP) qu'au terme des essais réalisés sur l'échantillon concerné.

Tous les essais doivent être effectués avec l'ensemble fixé selon ses conditions normales d'emploi et, le cas échéant, avec un support complémentaire au niveau normal du sol, comme cela est indiqué sur la Figure 102a, la Figure 102b, la Figure 103a et la Figure 103b.

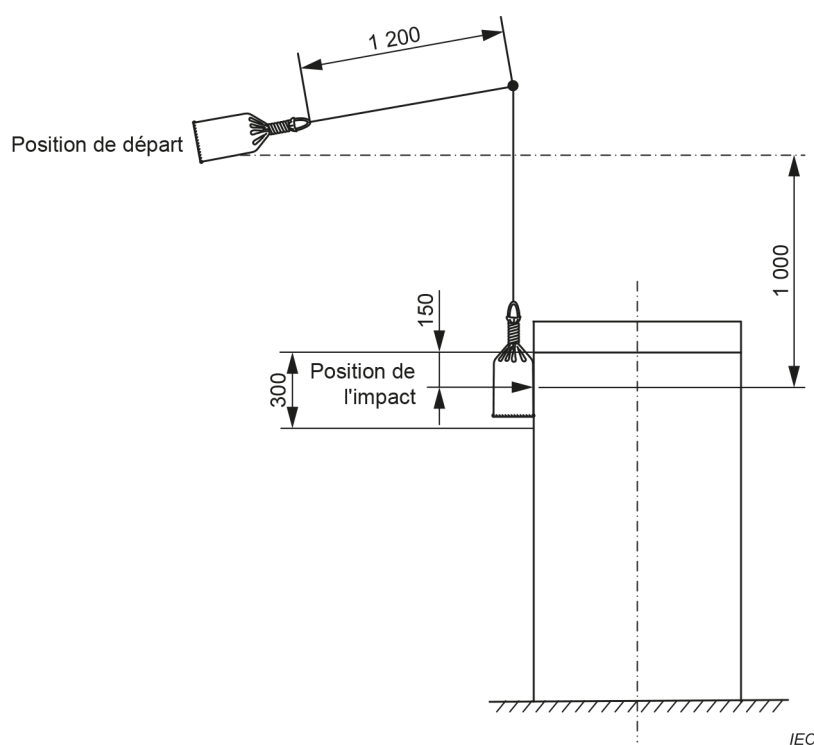
Dimensions en millimètres



IEC

Figure 102a – Schéma de l'essai de vérification de la résistance au choc réparti d'un ERD-E fixé au sol avec socle encastré

Dimensions en millimètres

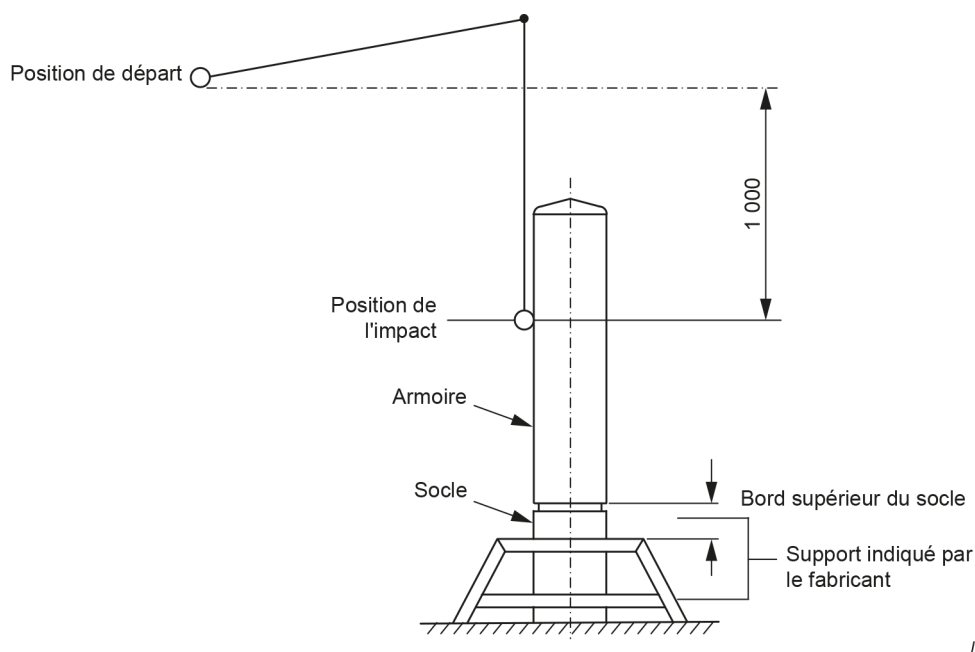


IEC

Figure 102b – Schéma de l'essai de vérification de la résistance au choc réparti d'un ERD-E fixé au sol sans socle encastré

Figure 102 – Schéma de l'essai de vérification de la résistance au choc réparti d'un ERD-E

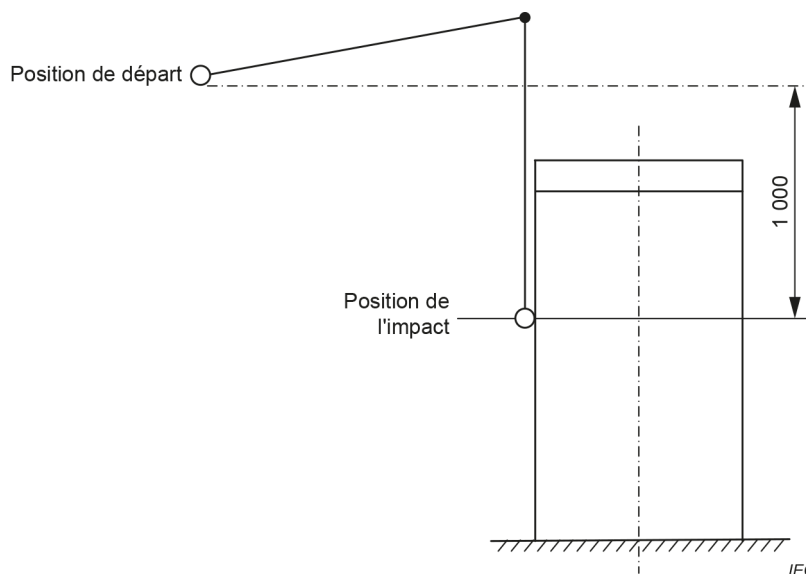
Dimensions en millimètres



IEC

Figure 103a – Schéma de l'essai de vérification de la résistance à l'impact d'un ERD-E fixé au sol avec socle encastré

Dimensions en millimètres



IEC

Figure 103b – Schéma de l'essai de vérification de la résistance à l'impact d'un ERD-E fixé au sol sans socle encastré

Figure 103 – Schéma de l'essai de vérification de la résistance à l'impact d'un ERD-E

À l'exception de l'essai du 10.2.101.8, la ou les portes de l'ensemble, le cas échéant, doivent être verrouillées au début de l'essai et le rester pendant la durée de l'essai.

10.2.101.2 Vérification de la résistance à la charge statique

Les essais suivants doivent être effectués sur tous les types d'ERD-E:

Essai 1 – Une charge uniformément répartie de 8 500 N/m² doit être appliquée pendant 5 min sur le toit de l'enveloppe (voir Figure 104).

Essai 2 – Une force de 1 200 N doit être appliquée successivement pendant 5 min sur les bords supérieurs à l'avant et à l'arrière du toit de l'enveloppe (voir Figure 104).

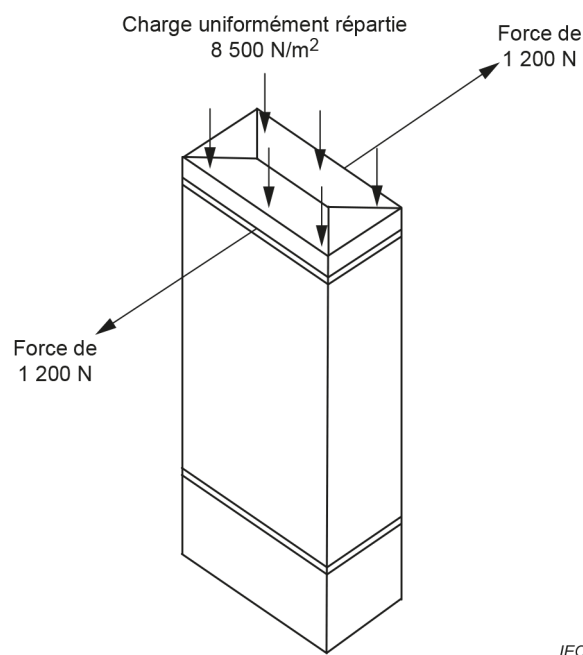


Figure 104 – Schéma de l'essai de vérification de la résistance à la charge statique

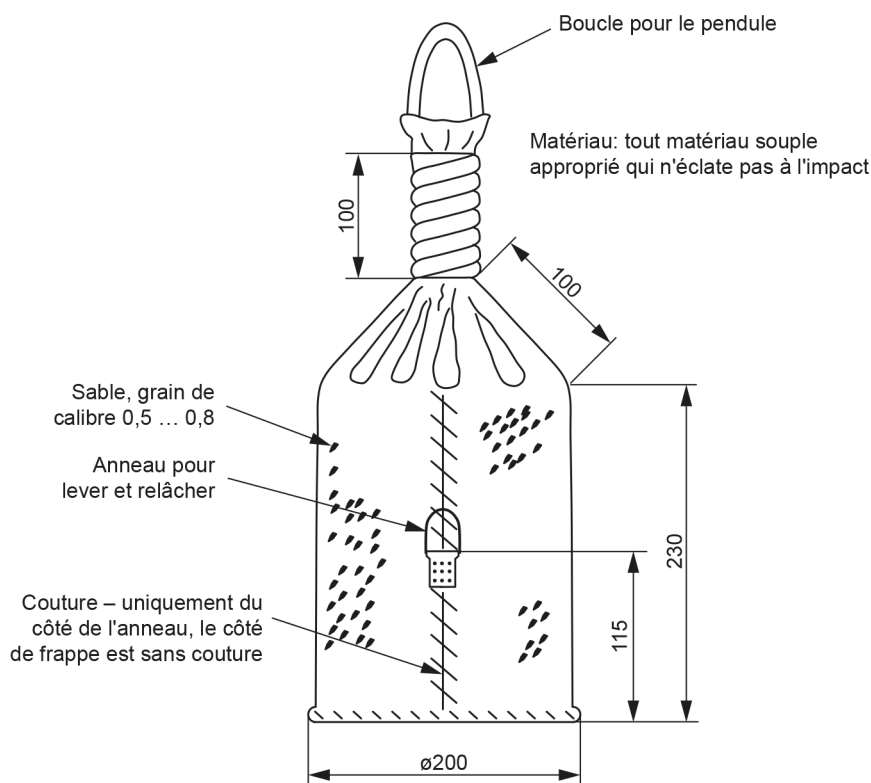
La conformité est vérifiée en contrôlant après l'essai que le degré de protection minimal est conforme au 8.2.2 et que le fonctionnement de la ou des portes ainsi que les points de verrouillage ne sont pas altérés; la conformité est également vérifiée en s'assurant que les distances d'isolement électrique restent satisfaisantes pendant la durée des essais et, dans le cas d'un ensemble à enveloppe métallique, qu'aucun contact ne s'est produit entre les parties actives et l'enveloppe par suite d'une déformation permanente ou temporaire.

10.2.101.3 Vérification de la résistance au choc réparti

L'essai doit s'appliquer à tous les types d'ERD-E.

Un sac conforme à la Figure 105 qui contient du sable sec et d'une masse totale de 15 kg doit être suspendu verticalement à un support au-dessus de la surface à l'essai et à 1 m au-dessus du point d'impact de l'ensemble.

Dimensions en millimètres



IEC

Figure 105 – Sac de sable pour l'essai de vérification de la résistance au choc réparti

Chaque essai doit consister en un choc appliqué à la partie supérieure de chacune des surfaces verticales de l'ensemble qui sont visibles quand l'ensemble est installé dans sa position normale de service. Des enveloppes séparées peuvent être utilisées pour chacun des coups d'essai.

Dans le cas d'une enveloppe de forme cylindrique, l'essai doit comporter trois chocs suivant un déplacement angulaire de 120°.

Un essai doit consister à soulever l'anneau de levage sur une hauteur de 1 m et à laisser tomber le sac de sable suivant un arc vertical pour venir heurter le centre approximatif de la partie supérieure de la surface de l'ensemble à l'essai (voir Figure 102a et Figure 102b).

La conformité est vérifiée en contrôlant après l'essai que le degré de protection reste conforme au 8.2.2 et que le fonctionnement de la ou des portes ainsi que les points de verrouillage ne sont pas altérés; la conformité est également vérifiée en s'assurant que les distances d'isolement électrique restent satisfaisantes pendant la durée des essais et, dans le cas d'un ensemble à enveloppe à métallique, qu'aucun contact ne s'est produit entre les parties actives et l'enveloppe par suite d'une déformation permanente ou temporaire. Dans le cas d'un ensemble à enveloppe isolante, si les conditions appropriées sont remplies, les dommages tels que de petites ébréchures ou de faibles zones de fissure ou d'écaillage de la surface ne sont pas pris en compte, sous réserve qu'aucune fissure associée ne compromette l'aptitude à l'emploi de l'ensemble.

10.2.101.4 Vérification de la résistance à la torsion

L'essai s'applique uniquement à tous les types d'ERD-E.

L'essai est effectué en utilisant un châssis horizontal rotatif, construit avec des cornières d'acier de 60 mm × 60 mm × 5 mm, avec des logements verticaux de 100 mm de longueur aux extrémités des bras. L'ensemble à l'essai est fixé rigidement sur son socle et le châssis est placé solidement dessus, de manière que les logements des extrémités des bras du châssis soient en contact avec le toit et les parois de l'ensemble.

L'ensemble, porte(s) fermé(e)s, doit supporter une force de torsion appliquée pendant 30 s comme cela est indiqué sur la Figure 106a et la Figure 106b. Il s'agit de deux essais séparés, le premier étant réalisé en appliquant les deux forces de 1 000 N chacune pour tordre l'ERD dans le sens des aiguilles d'une montre, le second étant réalisé en appliquant les deux forces de 1 000 N chacune pour faire tourner l'ERD dans le sens inverse des aiguilles d'une montre.

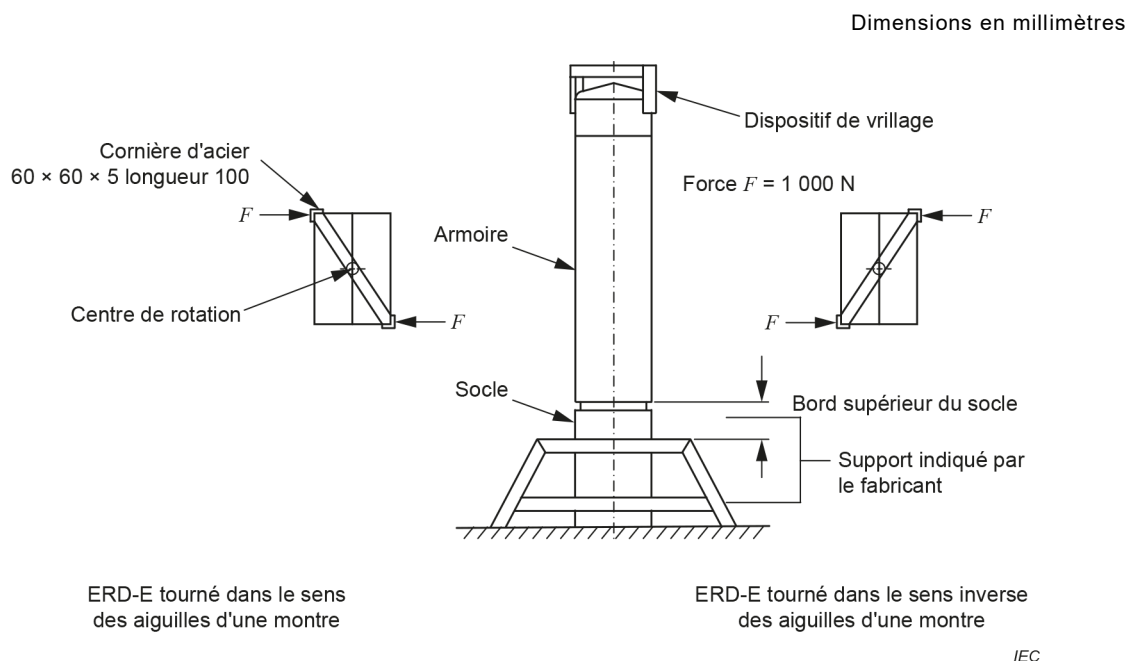


Figure 106a – Schéma de l'essai de vérification de la résistance à la torsion d'un ERD-E fixé au sol avec socle encastré

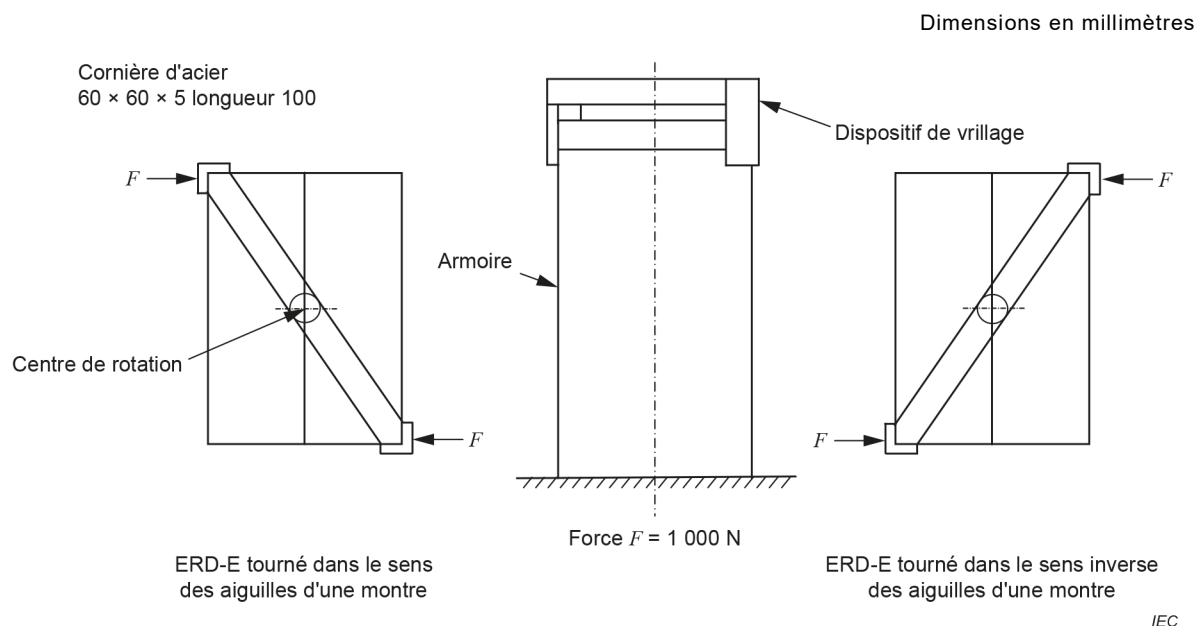


Figure 106b – Schéma de l'essai de vérification de la résistance à la torsion d'un ERD-E fixé au sol sans socle encastré

Figure 106 – Schéma de l'essai de vérification de la résistance à la torsion d'un ERD-E

La conformité est vérifiée en s'assurant que la ou les portes restent fermées pendant la durée de l'essai et, après l'essai, en contrôlant que le degré de protection reste conforme au 8.2.2.

10.2.101.5 Vérification de la résistance à l'impact

10.2.101.5.1 Essai applicable aux ERD-E conçus pour fonctionner à des températures ambiantes comprises entre -25 °C et 40 °C

L'essai doit être effectué à l'aide d'un appareil d'essai de chocs pendulaire qui comporte un tube de 9 mm de diamètre externe et d'au moins 1 m de longueur. Le pendule doit être disposé de façon à parcourir un arc vertical.

Une sphère d'acier d'une masse de 2 kg est fixée à une extrémité; elle doit être soulevée d'une hauteur de 1 m puis relâchée pour venir heurter la surface de l'ensemble à l'essai, ce qui produit une énergie d'impact de 20 J (voir Figure 103a et Figure 103b).

Pour chacun des deux essais décrits ci-dessous, l'essai doit consister en un choc appliqué au centre de chacune des surfaces verticales de l'ensemble qui sont visibles quand celui-ci est installé dans sa position normale de service. Des enveloppes séparées peuvent être utilisées pour chacun des coups d'essai.

Dans le cas d'une enveloppe de forme cylindrique, l'essai doit comporter trois chocs suivant un déplacement angulaire de 120°.

L'essai 1 doit être effectué à une température de l'air ambiant comprise entre 10 °C et 40 °C, après le maintien de l'ensemble dans cette plage de températures pendant au 12 h.

L'essai 2 doit être effectué à une température de l'air ambiant comprise entre 10 °C et 40 °C immédiatement après avoir maintenu l'ensemble à une température de $-25_{-0,5}^0$ °C pendant au moins 12 h. L'essai 2 n'est pas exigé si l'enveloppe de l'ERD est en métal.

La conformité est vérifiée en contrôlant après l'essai que le degré de protection reste conforme au 8.2.2 et que le fonctionnement de la ou des portes ainsi que les points de verrouillage ne sont pas altérés; la conformité est également vérifiée en s'assurant que les distances d'isolement électrique restent satisfaisantes pendant la durée des essais et, dans le cas d'un ensemble à enveloppe a métallique, qu'aucun contact ne s'est produit entre les parties actives et l'enveloppe par suite d'une déformation permanente ou temporaire. Dans le cas d'un ensemble à enveloppe isolante, si les conditions appropriées sont remplies, les dommages tels que de petites ébréchures ou de faibles zones de fissure ou d'écaillage de la surface ne sont pas pris en compte, sous réserve qu'aucune fissure associée ne compromette l'aptitude à l'emploi de l'ensemble.

10.2.101.5.2 Essai applicable aux ERD-E conçus pour une utilisation dans un climat arctique (voir 7.1.1)

L'échantillon doit être maintenu à une température de -50 °C pendant au moins 12 h. Les essais doivent être effectués à une température de l'air ambiant comprise entre 10 °C et 40 °C et au moment où la température externe de l'enveloppe est revenue à une valeur inférieure ou égale à -40 °C.

Séquence d'essai:

Pour l'essai 1 et l'essai 2, une pièce d'essai en métal, sphérique ou hémisphérique, d'un rayon de 100 mm ± 3 mm et d'une dureté de surface 160 HB selon l'ISO 6506-1:2014 doit être utilisée. L'essai 2 exige la mise à la terre la pièce d'essai.

L'essai 1 doit être effectué sur un ERD-E vide en appliquant une force de 1 500 N pendant 30 s aux 10 points considérablement plus faibles de l'enveloppe.

La conformité est vérifiée en contrôlant après l'essai que le degré de protection reste conforme au 8.2.2 et que le fonctionnement de la ou des portes ainsi que les points de verrouillage ne sont pas altérés.

L'essai 2 doit être effectué sur un ensemble qui contient le matériel qui fournit les distances d'isolement minimales dans l'enveloppe. Une enveloppe métallique doit être reliée à la terre. Dans le cas d'une enveloppe en matériaux isolants, toutes les parties de l'ensemble destinées à être reliées à la terre doivent l'être.

Une force de 1 500 N doit être appliquée pendant 30 s à la pièce d'essai métallique mise à la terre, aux 10 points considérablement plus faibles de l'enveloppe. Au cours de l'essai, une tension en courant alternatif conforme au 10.9.2.2 de l'IEC 61439-1:2020 doit être appliquée entre toutes les parties actives reliées entre elles et la terre.

La conformité est vérifiée par l'absence de perforations et d'amorçages.

L'essai 3 doit être effectué sur une enveloppe vide en utilisant l'appareil d'essai de chocs décrit en 10.2.101.5.1, mais avec une sphère d'acier d'une masse d'environ 15 kg. Cette pièce de frappe doit être soulevée d'une hauteur d'environ 1 m puis relâchée pour venir heurter la surface de l'ensemble à l'essai, ce qui produit une énergie d'impact de 150 J (voir Figure 103a et Figure 103b).

L'essai doit consister en un choc appliqué au centre de chacune des surfaces verticales de l'ensemble qui sont visibles quand celui-ci est installé dans sa position normale de service. Des enveloppes séparées peuvent être utilisées pour chacun des coups d'essai.

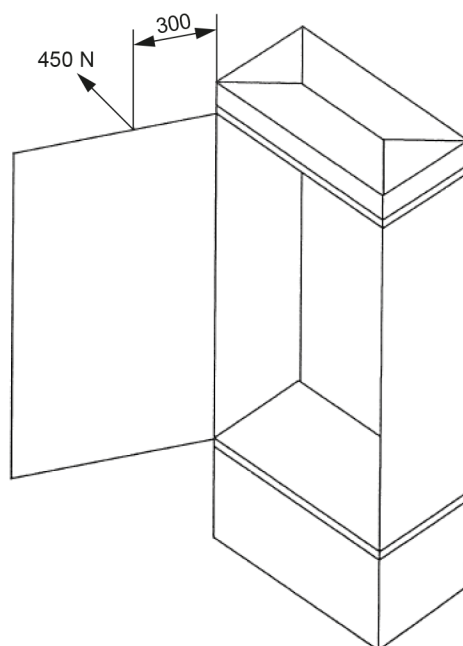
Dans le cas d'une enveloppe de forme cylindrique, l'essai doit comporter trois chocs suivant un déplacement angulaire de 120°.

La conformité à l'essai 3 est vérifiée en contrôlant après l'essai que le degré de protection est au moins égal à IP3X.

10.2.101.6 Vérification de la résistance mécanique des portes

L'essai s'applique à tous les types d'ERD-E qui comportent une ou plusieurs portes articulées sur un bord vertical de l'enveloppe.

Les essais doivent être effectués avec la ou les portes complètement ouvertes et en contact avec les dispositifs de retenue prévus. Une force de 450 N doit être appliquée sur le bord supérieur de la porte perpendiculairement au plan de la ou des portes et à une distance de 300 mm du bord à charnières et maintenue pendant 3 s (voir Figure 107).



IEC

Figure 107 – Schéma de l'essai de vérification de la tenue mécanique des portes

La conformité est vérifiée en s'assurant que le degré de protection reste conforme au 8.2.2 après fermeture de la ou des portes à la suite des essais. Si la ou les portes se défont pendant l'essai, cela n'est pas considéré comme un échec s'il est possible de remettre en place la ou les mêmes portes sans l'aide d'un outil et si le fonctionnement des charnières et les points de verrouillage ne sont pas altérés.

10.2.101.7 Vérification de la résistance axiale des inserts métalliques dans les matériaux synthétiques

L'essai s'applique uniquement à tous les types d'ensembles lorsque des inserts filetés en métal sont prévus pour maintenir la plaque de montage ou les supports d'appareillage en place.

L'essai doit être effectué sur un échantillon représentatif de chaque type et dimension d'insert métallique. Aussi, s'il y a une différence dans l'épaisseur de la forme du matériau qui entoure un insert défini, l'essai doit être renouvelé pour cette disposition.

Pendant l'essai, l'ensemble doit être complètement fixé sur une plateforme.

Un anneau fileté doit être mis en place dans chaque insert à l'essai et une force axiale selon le Tableau 102 doit être appliquée pendant 10 s en vue d'extraire l'insert de son ancrage.

Tableau 102 – Charge axiale à appliquer aux inserts

Taille de l'insert	Charge axiale N
M4	350
M5	350
M6	500
M8	500
M10	800
M12	800

La conformité est vérifiée par un examen qui montre que les inserts n'ont pas subi de dommages et demeurent dans leur position initiale; la conformité est également vérifiée en contrôlant l'absence de fissure dans le matériau qui entoure et constitue l'ancrage de l'insert.

NOTE Les petites fissures créées par des bulles d'air qui étaient visibles avant l'essai, mais qui ne sont pas affectées par l'application de la charge axiale, ne sont pas prises en compte.

10.2.101.8 Vérification de la résistance aux impacts mécaniques d'objets pointus

L'essai s'applique à tous les types d'ERD-E.

L'essai doit être effectué en utilisant l'appareil d'essai de chocs décrit en 10.2.101.5.1, mais avec une pièce de frappe en acier d'une masse de 5 kg et dont la forme d'extrémité est représentée à la Figure 108.

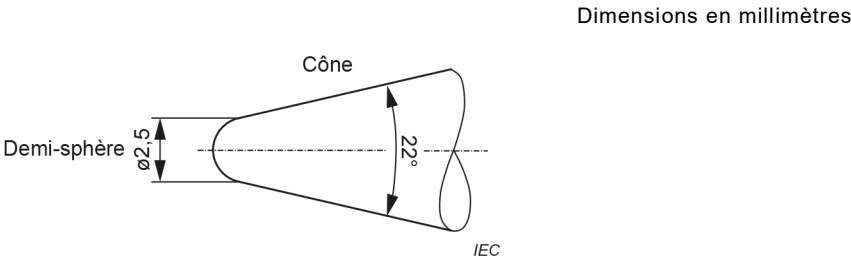


Figure 108 – Pièce de frappe pour l'essai de résistance aux impacts mécaniques d'objets pointus

Cette pièce de frappe doit être soulevée d'une hauteur de 0,4 m puis relâchée pour venir heurter la surface de l'ensemble à l'essai, ce qui produit une énergie d'impact de 20 J (voir Figure 103a et Figure 103b).

Chaque essai doit consister en un choc appliqué au centre de chacune des surfaces verticales de l'ensemble qui sont visibles quand l'ensemble est installé dans sa position normale de service. Des enveloppes séparées peuvent être utilisées pour chacun des coups d'essai.

Dans le cas d'une enveloppe de forme cylindrique, l'essai doit comporter trois chocs suivant un déplacement angulaire de 120°.

L'essai 1 doit être effectué à une température de l'air ambiant comprise entre 10 °C et 40 °C, après le maintien de l'ensemble dans cette plage de températures pendant au 12 h.

L'essai 2 doit être effectué à une température de l'air ambiant comprise entre 10 °C et 40 °C immédiatement après avoir maintenu l'ensemble à une température de -25_{-5}^0 °C pendant au moins 12 h. L'essai 2 n'est pas exigé si l'enveloppe de l'ERD est en métal.

La conformité est vérifiée par un examen qui montre que les fissures dues aux chocs sont contenues dans un cercle dont le diamètre n'excède pas 15 mm. Si l'extrémité de la pièce de frappe a pénétré dans l'enveloppe de l'ensemble, il ne doit pas être possible d'insérer un calibre de 4 mm de diamètre à extrémité hémisphérique, appliqué à la perforation avec une force de 5 N.

10.2.101.9 Essai de résistance mécanique d'un socle prévu pour être encastré dans le sol

Cet essai s'applique uniquement à un ERD-E.

Cet essai doit être effectué avec l'ERD-E fixé sur son socle, conformément à la Figure 109 et aux instructions d'installation du fabricant. Une force mécanique est transmise par l'intermédiaire d'un tube d'acier à paroi épaisse et doit être appliquée à la partie la plus basse du côté le plus long du socle de l'ERD situé sous la surface du sol après installation.

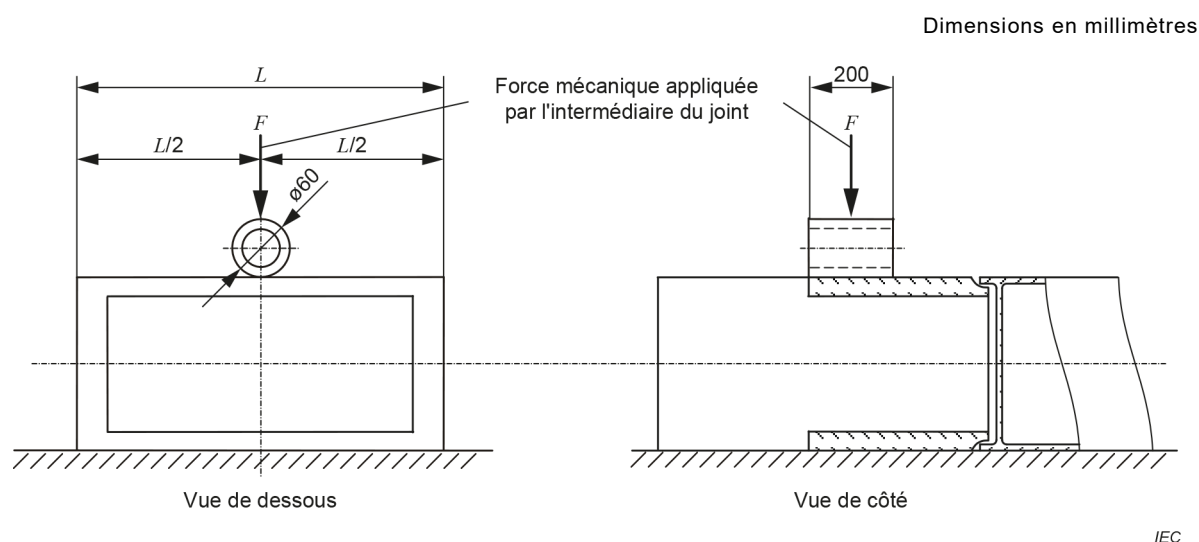


Figure 109 – Disposition d'essai type pour la résistance mécanique du socle

Si, par conception, le socle comprend un ou plusieurs supports permanents (de l'arrière vers l'avant), la force doit être appliquée en utilisant le nombre correspondant de tubes d'acier. Chaque tube doit être placé au milieu de la longueur entre supports. Les forces doivent être appliquées simultanément sur chaque tube et doivent être calculées au moyen de la formule suivante:

$$F = 3,5 \text{ N/mm} \times L$$

où L est la distance entre supports en millimètres.

La ou les forces doivent être appliquées pendant 1 min. À l'issue de cette période, l'effort étant maintenu, le degré de protection doit être vérifié.

S'il existe un autre côté du socle de l'ERD-E qui est de même longueur, mais dont le profil est différent, l'essai doit être répété sur ce côté.

La conformité est vérifiée par un examen qui montre que le socle n'est pas cassé et en contrôlant que le degré de protection de cette partie du socle et de l'ERD-E, qui est normalement au-dessus du sol, reste conforme au 8.2.2.

10.3 Degré de protection des ensembles (code IP)

Ajouter le premier alinéa suivant:

Le code IP déclaré pour un ERD-E ne s'applique pas à un socle prévu pour être encastré dans le sol. Cependant, le code IP déclaré s'applique à la partie inférieure de l'ensemble.

Si un ERD-E est constitué d'une enveloppe et d'un socle (à encastrer dans le sol ou à fixer en saillie) sans entrée de câbles étanche, la pénétration d'objets solides à travers l'espace au sol à l'intérieur du socle n'est pas prise en compte lors de la détermination du degré de protection. Si l'ERD est monté sur un socle avec un espace au sol ouvert, l'ERD et toutes les surfaces exposées du socle doivent procurer à l'ERD le niveau minimal de protection IP spécifié dans le présent document.

NOTE Dans certains cas, des mesures supplémentaires sont prises pour empêcher l'entrée d'animaux/de nuisibles.

10.5 Protection contre les chocs électriques et intégrité des circuits de protection

10.5.3.1 Généralités

Remplacement des deux alinéas par le suivant:

La vérification doit être réalisée par le biais d'essais conformément au 10.5.3.5 de l'IEC 61439-1:2020.

10.9 Propriétés diélectriques

Paragraphe supplémentaire:

10.9.2.101 Essai diélectrique à la fréquence du réseau pour les ERD installés dans des postes avec terres neutres HT et BT séparées

Lorsqu'un ERD doit faire partie intégrante d'un poste de distribution dans lequel les terres HT et BT sont séparées, un essai diélectrique supplémentaire doit être effectué. Une tension d'essai égale à la tension phase-terre HT doit être appliquée pendant 10 s entre le conducteur phase plus neutre et l'élément conducteur étranger de l'ERD. Les critères d'acceptation doivent être conformes au 10.9.2.4 de l'IEC 61439-1:2020.

10.9.3 Tension de tenue aux chocs

10.9.3.1 Généralités

Remplacement du premier alinéa par:

La vérification doit être réalisée par le biais d'essais conformément à l'une des méthodes d'essai alternatives décrites du 10.9.3.2 au 10.9.3.4 (inclus) de l'IEC 61439-1:2020.

10.10 Échauffement

10.10.1 Généralités

Remplacement par:

Il doit être vérifié que les limites d'échauffement spécifiées en 9.2 de l'IEC 61439-1:2020 pour les différentes parties de l'ensemble ne sont pas dépassées. La vérification doit être réalisée par un essai spécifié en 10.10.2 de l'IEC 61439-1:2020.

L'échauffement doit être vérifié sur un ou plusieurs échantillons représentatifs. Il n'est pas nécessaire de soumettre à l'essai toutes les configurations produites, mais les échantillons doivent, avec une exactitude raisonnable, être les plus défavorables d'un point de vue thermique pour les variantes étudiées. L'évaluation des configurations les plus défavorables doit prendre en considération la puissance dissipée totale, les caractéristiques assignées des appareils (de départ et d'arrivée), les sections des conducteurs, la taille de l'enveloppe et le courant assigné de l'ensemble.

10.10.2 Vérification par des essais

10.10.2.2 Choix de la (des) configuration(s) représentative(s)

10.10.2.2.1 Généralités

Addition de l'alinéa suivant:

Lorsque la conception d'un ERD est adaptée à une installation en encastrement dans un mur, l'essai d'échauffement doit être effectué avec une isolation afin de simuler la présence du mur.

10.11 Tenue aux courts-circuits

10.11.1 Généralités

Remplacement par:

À l'exception des circuits des ensembles qui sont exemptés de la vérification conformément au 10.11.2 de l'IEC 61439-1:2020, la tenue aux courts-circuits spécifiée par le fabricant doit être vérifiée. La vérification doit être réalisée par un essai spécifié en 10.11.5 de l'IEC 61439-1:2020.

11 Vérification individuelle de série

L'article de l'IEC 61439-1:2020 s'applique.

Annexes

Les annexes de l'IEC 61439-1:2020 s'appliquent, avec les exceptions suivantes:

Les Annexes A, C, D, H, K et M ne s'appliquent pas.

Modification de l'Annexe L.

Addition des Annexes AA, BB, CC et DD.

Annexe L (informative)

Recommandations concernant la vérification de l'échauffement

Modifications:

L.4 Évaluation de vérification

L'article de l'IEC 61439-1:2020 ne s'applique pas.

L.5 Vérification par comparaison à une conception de référence

L'article de l'IEC 61439-1:2020 ne s'applique pas.

Annexe AA
(normative)

Section des conducteurs

Le Tableau AA.1 s'applique au raccordement d'un câble par borne.

Tableau AA.1 – Sections minimales et maximales des conducteurs
en cuivre et en aluminium, adaptées aux raccordements (voir 8.8)

Courant assigné A	Conducteurs à âmes massives ou toronnées (aluminium ou cuivre) Sections mm ²		Conducteurs souples en cuivre Sections mm ²	
	Minimales	Maximales	Minimales	Maximales
6	0,75	1,5	0,5	1,5
8	1	2,5	0,75	2,5
10	1	2,5	0,75	2,5
12	1	2,5	0,75	2,5
16	1,5	4	1	4
20	1,5	6	1	4
25	2,5	6	1,5	4
32	2,5	10	1,5	6
40	4	16	2,5	10
63	6	25	6	16
80	10	35	10	25
100	16	50	16	35
125	25	70	25	50
160	35	95	35	70
200	50	150	50	95
250	70	150	70	120
315	70	240	95	185
400	70	240	95	185
500	70	300	95	240
630	70	300	95	240
Ce tableau s'applique au raccordement d'un seul conducteur par borne.				
Si les conducteurs externes sont directement raccordés à l'appareillage incorporé, les sections indiquées dans la spécification correspondante s'appliquent.				
Dans les cas où il est nécessaire d'utiliser des conducteurs de sections différentes de celles indiquées dans ce tableau, un accord doit être trouvé entre le fabricant et l'utilisateur.				

La relation approximative entre les dimensions en mm² et AWG/kcmil indiquées dans le Tableau AA.2 doit être utilisée lorsque les conducteurs de cuivre circulaires de dimensions métriques ne sont pas disponibles.

Tableau AA.2 – Sections normalisées de conducteurs de cuivre circulaires et relation approximative entre les dimensions en mm² et AWG/kcmil (voir 8.8 de l'IEC 61439-1:2020)

Section assignée mm ²	Dimension AWG/kcmil	Surface métrique équivalente mm ²
0,2	24	0,205
0,34	22	0,324
0,5	20	0,519
0,75	18	0,82
1	–	–
1,5	16	1,3
2,5	14	2,1
4	12	3,3
6	10	5,3
10	8	8,4
16	6	13,3
25	4	21,2
35	2	33,6
–	1	42,4
50	0	53,5
70	00	67,4
95	000	85,0
–	0000	107,2
120	250 kcmil	127
150	300 kcmil	152
185	350 kcmil	177
–	400 kcmil	203
240	500 kcmil	253
300	600 kcmil	304
NOTE Le tiret, s'il est présent, indique une dimension qui prend en compte la capacité de connexion (voir 8.8).		

Annexe BB (informative)

Sujets soumis à accord entre le fabricant d'ensembles et l'utilisateur

Les informations données dans le Tableau BB.1 ci-après sont soumises à accord entre le fabricant d'ensembles et l'utilisateur. Dans certains cas, les informations déclarées par le fabricant d'ensembles peuvent se substituer à un accord.

Tableau BB.1 – Sujets soumis à accord entre le fabricant d'ENSEMBLES et l'utilisateur

Caractéristiques	Article ou paragraphe de référence ^c	Configuration par défaut ^b	Options énumérées dans le document	Exigence de l'utilisateur ^a
Système électrique				
Système de mise à la terre	5.6, 8.4.3.1, 8.4.3.2.3, 8.6.2, 10.5, 11.4	Norme du fabricant choisie pour répondre aux exigences locales	TT / TN-C / TN-C-S / IT, TN-S	
Tension nominale (V)	3.8.9.1, 5.2.1, 8.5.3	Selon les conditions d'installation locales	max. 1 000 V en courant alternatif	
Surtensions transitoires	5.2.4, 8.5.3, 9.1, Annexe G	Catégorie de surtension IV	Aucune	
Surtensions temporaires	9.1	Tension réseau nominale + 1 200 V	Aucune	
Fréquence assignée f_n (Hz)	3.8.12, 5.5, 8.5.3, 10.10.3.1, 10.11.5.4	Selon les conditions d'installation locales	50 Hz/60 Hz	
Exigences supplémentaires d'essai sur site: câblage, performance de fonctionnement et fonction	11.10	Norme du fabricant, selon l'application	Aucune	
Tenue aux courts-circuits				
Courant de court-circuit présumé aux bornes d'alimentation I_{cp} (kA)	3.8.7	Déterminé par le système électrique	Aucune	
Courant de court-circuit présumé dans le neutre	10.11.5.3.5	Au maximum 60 % des valeurs pour les phases	Aucune	
Courant de court-circuit présumé dans le circuit de protection	10.11.5.6	Au maximum 60 % des valeurs pour les phases	Aucune	
Exigence relative à la présence d'un dispositif de protection contre les courts-circuits (DPCC) dans l'unité fonctionnelle d'arrivée	9.3.2	Selon les conditions d'installation locales	Oui/Non	
Coordination des dispositifs de protection contre les courts-circuits, y compris les informations concernant les dispositifs de protection externes contre les courts-circuits	9.3.4	Selon les conditions d'installation locales	Aucune	

Caractéristiques	Article ou paragraphe de référence ^c	Configuration par défaut ^b	Options énumérées dans le document	Exigence de l'utilisateur ^a
Données associées à des charges susceptibles de contribuer au courant de court-circuit	9.3.2	Aucune charge susceptible de contribuer de manière significative	Aucune	
Protection des personnes contre les chocs électriques conformément à l'IEC 60364-4-41				
Type de protection contre les chocs électriques – Protection principale (protection contre le contact direct)	8.4.2	Protection principale	Selon les règles d'installation locales	
Type de protection contre les chocs électriques – Protection en cas de défaut (protection contre le contact indirect)	8.4.3	Selon les conditions d'installation locales	Coupure automatique de l'alimentation/ Séparation électrique/ Isolation totale	
Environnement de l'installation				
Type d'emplacement	3.5, 8.1.4, 8.2	Norme du fabricant, selon l'application	Intérieur/ Extérieur	
Protection contre la pénétration de corps étrangers solides et l'infiltration d'eau	8.2.2, 8.2.3	Intérieur (sous enveloppe): au minimum IP2X Extérieur - au minimum: IP34D dans le cas d'installations dans des emplacements accessibles au public. IP33 dans d'autres emplacements.		
Impact mécanique externe NOTE L'IEC 61439-5 ne désigne pas de codes IK particuliers.	10.2.101	D'après la présente norme (IEC 61439-5)	Aucune	
Résistance aux rayonnements UV (s'applique uniquement aux ensembles extérieurs, sauf spécification contraire)	10.2.4	Intérieur: Ne s'applique pas. Extérieur: Climat tempéré	Aucune	
Résistance à la corrosion	10.2.2	Configurations intérieures/ extérieures normales	Aucune	
Température de l'air ambiant – Limite inférieure	7.1.1	Intérieur: –5 °C Extérieur: Climat normal –25 °C Climat arctique –50 °C	Aucune	
Température de l'air ambiant – Limite supérieure	7.1.1	40 °C	Aucune	
Température de l'air ambiant – Moyenne journalière maximale	7.1.1, 9.2	35 °C	Aucune	

Caractéristiques	Article ou paragraphe de référence ^c	Configuration par défaut ^b	Options énumérées dans le document	Exigence de l'utilisateur ^a
Humidité relative maximale	7.1.1	Intérieur: 95 % entre –5 °C et +30 °C 70 % à +35 °C 57 % à +40 °C Extérieur: 100 % entre –25 °C et +27 °C 60 % à 35 °C 46 % à 40 °C	Aucune	
Degré de pollution (de l'environnement de l'installation)	7.1.2	ERD-E: 4 ERD-I: 3	2, 3, 4	
Altitude	7.1.1	≤ 2 000 m	Aucune	
Environnement CEM (A ou B)	9.4, 10.12, Annexe J	A/B	A/B	
Conditions spéciales d'emploi (par exemple, les vibrations, une condensation exceptionnelle, une forte pollution, un environnement corrosif, des champs électriques ou magnétiques élevés, des moisissures, de petits animaux, des dangers d'explosion, de forts chocs et vibrations, des séismes)	7.2, 8.5.4, 9.3.3 Tableau 7	Aucune condition spéciale d'emploi	Climat arctique	
Méthode d'installation				
Type	3.3, 5.6	Norme du fabricant	Plusieurs, par exemple fixation au sol, montage sur transformateur, montage sur poteau, fixation en saillie sur mur ou encastrement dans un mur	
Fixe/Mobile	3.5	Fixe	Aucune	
Dimensions hors-tout et masse maximales	5.6, 6.2.1	Norme du fabricant, selon l'application	Aucune	
Type(s) de conducteur(s) externe(s)	8.8	Câble	Barres nues ou isolées.	
Direction(s) des conducteurs externes	8.8	Par en dessous	Par au-dessus	
Matériau des conducteurs externes	8.8	Cuivre/ Aluminium	Aucune	
Sections et terminaisons de conducteurs de phase externes	8.8	Comme cela est défini dans le document	Aucune	
Sections et terminaisons des conducteurs PE, N et PEN externes	8.8	Comme cela est défini dans le document	Aucune	
Exigences spéciales d'identification des bornes	8.8	Norme du fabricant	Aucune	
Stockage et manutention				
Dimensions et poids maximaux des unités de transport	6.2.2, 10.2.5	Norme du fabricant	Aucune	
Méthodes de transport (par exemple, chariot élévateur, grue)	6.2.2, 8.1.6	Norme du fabricant	Aucune	

Caractéristiques	Article ou paragraphe de référence ^c	Configuration par défaut ^b	Options énumérées dans le document	Exigence de l'utilisateur ^a
Conditions d'environnement différentes des conditions d'emploi	7.3	Selon les conditions d'emploi	Aucune	
Informations d'emballage	6.2.2	Norme du fabricant	Aucune	
Configurations de fonctionnement				
Accès aux appareils manœuvrés à la main	8.4	Personnes autorisées	Aucune	
Emplacement des appareils manœuvrés à la main	8.5.5	Facilement accessible	Aucune	
Sectionnement des matériels de l'installation de charge	8.4.2, 8.4.3.3, 8.4.6.2	Norme du fabricant	Individuels/ groupes/ tous types	
Capacités d'entretien et d'évolution				
Exigences relatives à l'accessibilité en vue d'un examen ou d'opérations analogues	8.4.6.2.2	Aucune exigence relative à l'accessibilité	Aucune	
Exigences relatives à l'accessibilité pour entretien en service par des personnes autorisées	8.4.6.2.3	Aucune exigence relative à l'accessibilité	Aucune	
Exigences relatives à l'accessibilité pour extension en service par des personnes autorisées	8.4.6.2.4	Aucune exigence relative à l'accessibilité	Aucune	
Méthode de raccordement des unités fonctionnelles	8.5.1, 8.5.2	Norme du fabricant	Aucune	
Protection contre les contacts directs avec des parties internes actives dangereuses au cours d'un entretien ou d'une évolution (par exemple, les unités fonctionnelles, les jeux de barres principaux, les jeux de barres de distribution)	8.4	Aucune exigence relative à la protection au cours d'un entretien ou d'une évolution	Aucune	
Courant admissible				
Courant de charge total maximal à fournir par l'ensemble (à partir duquel le courant assigné de l'ensemble I_{nA} (A) est déterminé)	3.8.10.1, 5.3, 8.4.3.2.3, 8.5.3, 8.8, 10.10.2, 10.10.3, 10.11.5, Annexe E	Norme du fabricant, selon l'application	Aucune	
Courant d'emploi I_B et nature de la charge pour chaque circuit; à défaut, I_n des appareils et nature de la charge (dans ce cas, les facteurs de charge présumés peuvent être utilisés selon la partie applicable de l'IEC 61439)	3.8.10.8	Aucune	Aucune	
Rapport de la section du conducteur neutre à celle des conducteurs de phase: conducteurs de phase jusqu'à et y compris 16 mm ²	8.6.1	100 %	Aucune	

Caractéristiques	Article ou paragraphe de référence ^c	Configuration par défaut ^b	Options énumérées dans le document	Exigence de l'utilisateur ^a
Rapport de la section du conducteur neutre à celle des conducteurs de phase: conducteurs de phase au-dessus de 16 mm ²	8.6.1	50 % (au minimum 16 mm ²)	Aucune	
^a Pour les applications particulièrement sévères, l'utilisateur peut spécifier des exigences plus rigoureuses que celles établies dans la norme. ^b Dans certains cas, les informations déclarées par le fabricant d'ensembles peuvent se substituer à un accord. ^c Lorsqu'une référence n'existe pas dans le présent document, l'IEC 61439-1:2020 s'applique.				

Annexe CC (informative)

Vérification de conception

Tableau CC.1 – Liste de vérifications de conception à réaliser

N°	Caractéristique à vérifier	Articles ou paragraphes ^a
1	Résistance des matériaux et des parties: Résistance à la corrosion Propriétés des matériaux isolants: Stabilité thermique Résistance à la chaleur anormale et au feu dus aux effets électriques internes Catégorie de classement au feu Résistance aux rayonnements ultraviolets (UV) Levage Impacts mécaniques Résistance à la charge statique Résistance au choc réparti Résistance à la torsion Résistance à l'impact Résistance mécanique des portes Résistance axiale des inserts métalliques dans les matériaux synthétiques Résistance aux chocs mécaniques induits par des objets à angles vifs Résistance mécanique d'un socle prévu pour être encastré dans le sol Marquage	10.2 10.2.2 10.2.3 10.2.3.1 10.2.3.2 10.2.3.101 10.2.4 10.2.5 10.2.101 10.2.101.2 10.2.101.3 10.2.101.4 10.2.101.5 10.2.101.6 10.2.101.7 10.2.101.8 10.2.101.9 10.2.7
	Manœuvre mécanique	10.2.8
2	Degré de protection procuré par les enveloppes	10.3
3	Distances d'isolement	10.4
4	Lignes de fuite	10.4
5	Protection contre les chocs électriques et intégrité des circuits de protection: Continuité effective entre les parties conductrices exposées de l'ensemble et le circuit de protection Tenue du circuit de protection aux courts-circuits	10.5 10.5.2 10.5.3
6	Intégration des appareils de connexion et des composants	10.6
7	Circuits électriques internes et connexions	10.7
8	Bornes pour conducteurs externes	10.8
9	Propriétés diélectriques: Tension de tenue à fréquence industrielle Tension de tenue aux chocs	10.9 10.9.2 10.9.3
10	Limites d'échauffement	10.10
11	Tenue aux courts-circuits	10.11
12	Compatibilité électromagnétique (CEM)	10.12
^a Lorsqu'une référence n'existe pas dans le présent document, l'IEC 61439-1:2020 s'applique.		

Annexe DD
(informative)

Liste des notes concernant certains pays

Paragraphe	Texte
8.8	Ajouter la note suivante après le dernier alinéa: NOTE Aux États-Unis d'Amérique (USA), l'exigence relative à la taille des conducteurs dépend du courant assigné, de la température assignée de l'isolation des conducteurs, de la température ambiante et de la configuration, ainsi que du type d'isolation des conducteurs. Les exigences spécifiques figurent dans le code électrique national américain (NEC, <i>National Electrical Code</i>) NFPA 70, Chapitre 3.

Bibliographie

La bibliographie de l'IEC 61439-1:2020 s'applique, avec l'exception suivante:

Addition:

IEC 60050 (toutes les parties), *Vocabulaire Électrotechnique International (IEV)*, disponible à l'adresse www.electropedia.org

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

3, rue de Varembé
PO Box 131
CH-1211 Geneva 20
Switzerland

Tel: + 41 22 919 02 11
info@iec.ch
www.iec.ch